Land at South Marston and Rowborough
Swindon

Transport Assessment

Swindon EV Consortium

Brookbanks

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Executive Summary

Brookbanks Consulting (BCL) is appointed by the Swindon Eastern Villages Consortium to produce a Transport Assessment and Travel Plan to accompany an outline planning application for a residential-led scheme, together with areas of mixed use comprising: community; employment; education; local retail uses; and formal and informal open space provision on land east of Swindon and north of A420.

The proposed development will comprise up to 2380 new houses, a primary school and supporting uses including a local centre and mixed use area incorporating leisure & community facilities, low scale employment opportunities and open space provision. This represents a comprehensive first phase in the development of the Eastern Villages.

As part of evidence prepared in support of the Swindon Eastern Villages development proposals, BCL has worked in conjunction with Swindon Borough Council (SBC) and Highways England (HE) representatives to work toward agreement on all highways matters, including trip generation forecasts, traffic modelling methodology and results, together with the production of a package of mitigation measures to enhance the highway network and accessibility of the area while mitigating for transport impacts of the development proposals.

The process followed has been used by BCL as the basis for this Transport Assessment, with the key agreements applied to the development proposals.

In relation to transport policy, the proposed development accords with national, regional and local transport and land-use policies. These support Swindon Borough Council's aim to promote sustainable development that seeks to ensure wider travel choices in the form of the most sustainable forms of transport and thus reduce the use of the private car particularly for single person use. The proposals seek to address positively the principles set out in relation to South Marston in the Draft South Marston SPD and take account of the guidance also in the Draft Eastern Villages SPD.

A number of vehicular accesses are proposed into the existing highway network, being from a compact roundabout on Thornhill Road, with access also from several locations off Old Vicarage Lane and from a signalised junction on the A420 towards the eastern extent of the site. These accesses are consistent with the Local Plan evidence base documents that have been tested independently and again within this document. In accordance with the Draft South Marston and Eastern Villages SPD no access is taken off Nightingale or Rowborough Lanes.

The western access point off Thornhill Road will provide access to the South Marston residential development. The proposed accesses off Old Vicarage Lane will provide a secondary point of access serving the expanded South Marston. The development area at Rowborough will be served by accesses from / to Old Vicarage Lane and via the eastern access under the railway. The phasing of the connections back to the A420, including the timing of the delivery of the rail under bridge at the eastern end of the site, will be a matter for discussion with SBC.

Arrangements to enhance public transport provision are provided together with new and enhanced footways and cycleways in the area which will reduce the need to travel by private car.

Pedestrian and cycle connections to the neighbouring South Marston and Sainsbury's development and the public transport interchange at Gablecross are proposed to be enhanced and potential effects of severance caused by the A420 is proposed to be reduced through the provision of upgraded junctions with crossing facilities.

A bespoke bus service will be introduced operating at a 20 minute frequency and will connect the development to the town centre, rail station as well as the bus interchange at Fleming Way.

It is agreed that a key component of the transport mitigation measures is the implementation of a comprehensive Travel Plan (TP), which is of a scale sufficient to warrant a standalone document accompanying this Transport Assessment. The Travel Plan proposes a target for the site which will reduce the mode share by single occupancy vehicles by 10%.

The final components of the package of transport mitigation measures are the provision of highway improvements either directly by the developer or through developer contributions at various locations on the A420 and A419 and in South Marston.

The potential impacts on the highway network have been tested through the Swindon area wide SATURN traffic model together with the A419 / A420 PARAMICS traffic model. This has resulted in the identification of a range of offsite interventions that will delivered by the development. This included:

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road.
- **New Access onto A420:** New signal controlled junction at the east end of phase one to provide access from the A420.
- **Police HQ Signals:** Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire
- **Gablecross Roundabout:** Conversion to a signalled crossroads (this is a revised intervention from the original proposed intervention which was localised widening on approach arms)
- **White Hart interchange:** Signalisation off several arms of the existing interchange layout.
- **Merlin Way / A419 slip road roundabout:** Signalisation of several arms of the existing interchange layout (subsequent traffic modelling of the revised development layout together with anticipated traffic growth for the Swindon Borough raised that mitigation measures were necessary at this junction location).

The assessment process has identified a range of possible interventions, as identified above. However, the HE and SBC have also identified potential alternative schemes at several of these locations. Through the subsequent discussions with the Regulatory Bodies, it may be that a contribution towards these alternative schemes could be agreed, as opposed to the completion of the BCL identified schemes.

This represents a comprehensive package of interventions delivered through, and arising from, the development. The detailed traffic modelling completed for this TA has also considered future development resulting from the wider growth proposals to the east of Swindon, as identified in the Local Plan. As would be expected, this work has identified further interventions that will be necessary to deliver the wider growth. However, the assessments demonstrates that delivery of this Phase I element of the growth does not prejudice the delivery of the subsequent phases. Such further interventions that have been anticipated in the Eastern Villages SPD for instance and include a diamond interchange at White Hart and the signalisation of the Greenbridge roundabout. The SPD envisages the phased implementation of wider transport improvements in accordance with the phased implementation of the eastern villages development.

Overall, the development provides mitigation in relation to the transport networks and aims to minimise travel through the implementation of the Travel Plan to sufficiently minimise the impact of the development on the highway network. Good connectivity is afforded to the surrounding urban area and accessibility to a range of key services, facilities and opportunities has been provided.

Design of the transport and highway proposals is considerate to the specific concerns of the local South Marston community. The access strategy has been considered to specifically limit the increase in traffic through South Marston. The selection of a roundabout junction on Thornhill Road provides a direct link into the development to reduce any need to travel through South Marston. Proposals for Old Vicarage Lane include the alteration of the priority such that traffic generated by the development is discouraged from travelling towards South Marston. This proposed junction arrangement was also considered to be effective in

avoiding traffic passing through South Marston in the technical studies conducted by JMP as background to the draft South Marston SPD.

A connection through South Marston and connections at Rowborough will be provided, the routes of which have been identified by the masterplan to connect to the external public highway. Access from Rowborough onto Old Vicarage Lane is proposed at two points. By this means the traffic modelling work confirms that traffic generated from within Rowborough towards Old Vicarage Lane will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road. As such the transport strategy route provides an equitable and effective route choice. The proposed link between Old Vicarage Lane and Thornhill Road is therefore demonstrated not to perform a southern bypass role for the South Marston area. Indeed the link has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

The potential impact of the proposals for South Marston has been fully appraised, including Pound Corner. This assessment indicates that the delivery of these proposals will not materially increase the flows through the village or impede access to the A420 for existing residents.

Furthermore in relation to the highway assessment, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included. Whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner.

Notwithstanding, the additional traffic generated by the development, with the improvements proposed to the Gablecross roundabout and the junction of Old Vicarage Lane, the level of queuing at Gablecross roundabout on the Thornhill Road arm is forecast to reduce compared to the no development and no improvement position and the queuing on Old Vicarage Lane will be limited and acceptable.

The impact on South Marston will therefore be limited by:

- Discouraging the use of new highway links north of the A420 and south of South Marston being used in preference to the A420.
- Discouraging vehicles using South Marston village.

Consideration has also been given to the wider objectives in relation to transport in South Marston as set out in the Draft SPD. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

To reduce the potential attractiveness for development trips to pass through South Marston, consideration should be given towards measures to influence route choice. This could likely include a signage strategy to highlight alternative routes and a range of traffic calming measures providing both traffic control and a better environmental conditions within South Marston.

While the traffic management improvements may be considered to be beneficial to the network, highway users and the local communities, it should be noted that transportation capacity is adequate with the highway capacity improvements alone. As such, traffic management improvements are betterments rather than measures that are strictly necessary for network operation.

Detailed discussions have occurred with the HE and SBC. This included the submission of the TA at key stages throughput the project. Comments were received from the regulatory bodies at each stage. The TA has been amended accordingly, with green text identifying subsequent changes.

Impacts of the development have not changed substantially from previous revisions. However as highlighted above, some revised mitigation measures are required at the two identified junctions of Gablecross Roundabout (conversion to a signalised junction) and Merlin Way / A419 slip road roundabout (addition of traffic signals on approach arms). This is in light of some updated traffic flows generated by the new development in addition to anticipated traffic growth for the Swindon Borough.

In summary, the proposals demonstrate that a well-considered approach to developing transport and highways proposals for the first phase of development at Swindon Eastern Villages is able to ensure the proposals are able to meet national, regional and local policy and guidance while meeting with the identified aspirations of the growth area.

1 Introduction

1.1 Brookbanks Consulting Limited (BCL) is appointed by the Swindon Eastern Village Consortium (SEVC) to complete a Transportation Assessment (TA) and a Travel Plan (TP) for a proposed mixed use development on land to the east of Swindon. The proposed site is part of the wider Eastern Villages (EV) growth area that is supported by Swindon Borough Council planning policies and the adopted Local Plan.

1.2 Development to the east of Swindon has been mooted over many years, supported through various iterations of the Borough Council's Local Development Framework. Initially, the Plan policies identified that circa 12,000 dwellings could be built on land to the east of Swindon. More recently this has been reduced to circa 8,000 dwellings. The identified land parcel that forms the basis of this planning application represents Phase I of the Eastern Villages and lies to the north of the A420, near South Marston. The future phases of development will complete the Eastern Villages and these will be subject to their own planning application.

1.3 The objective of the study is review the Phase I application in detail and to demonstrate that the development proposals are acceptable from a transportation and highways viewpoint. This report will also review the future phases at a high level to demonstrate that the wider growth is not prejudiced by delivery of Phase I.

1.4 The Transport Assessment, Travel Plan and all accompanying documents where applicable have been updated in response to comments raised by Swindon Borough Council (SBC), JMP Consultants and Highways England. All revisions follow collaborative dialogues and workshops with the aforementioned parties that have agreed the scope of the revised material which includes changes in land use, updated planning policies, updated traffic flows and all other relevant updates throughout the project. Consultation with the Regulatory Bodies has continued and has resulted in amendments to this documentation. This has included amendments to several sections, including:

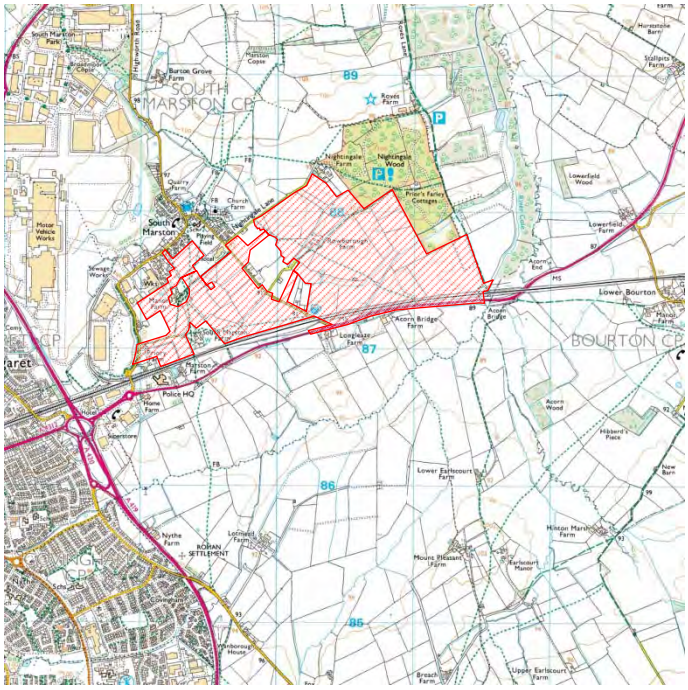
- Update traffic modelling
- Updated accident review
- Updated traffic modelling
- Greater consideration of highway interventions

2 Background Information

Location

2.1 The proposed development lies to the east of Swindon. The site is generally bound to the south by the Swindon – Paddington rail line, beyond which lies the A420. Open fields, generally in agricultural production bound the site to the east and north. The Village of South Marston lies to the north west of the site with Thornhill Road forming the western boundary of the site. Old Vicarage Lane bisects the land, forming the boundary between the South Marston and Rowborough proposals as recognised in the Local Plan documents.

2.2 The land is largely historically undeveloped. The site location and boundary are shown indicatively on Figure 2a


SOUTH MARSTON CP
BOURTON CP
South Marston
Lower Bourton
Nightingale Wood
Lowerfield Wood
Acorn Bridge
Acorn Wood
Lower Earlscourt Farm
Upper Earlscourt Farm
Hilton Manor Farm
Earlscourt Manor
Mouns Hassett Farm
Nythe Farm
ROMAN SETTLEMENT
Police HQ
Home Farm
Burton Grove Farm
Quarry Farm
Church Farm
Manor Farm
Longleaze Farm
Acorn Bridge Farm
Hibberd's Place
New Barn
Manor Farm
Stallard Farm
Horsestube Barn
Boves Farm
Prior's Farm Cottages
Motor Vehicle Works
Weigh Works
Sewage Works
Highworth Road
Marlston Copse
Lowerfield Farm
Acorn End
Branch Farm
Rowborough Moats
Rowborough House
Rowborough Farm
Lonnbeat Farm
Parsons Farm
Pierce Field
Lane
Road
Cycle Way
Footpath
Bridleway
Schools
Cemetery
Sportsfield
Allotments
Recreation Ground
Pond
Tank
Tumulus
Mast
Pylon
Tank
Sewage Works
Motor Vehicle Works
Weigh Bridge
Cattle Grid
Gate
Stile
Footbridge
Bridge
Tunnel
Level Crossing
Railway Station
Bus Station
Bus Stop
Parking
Public Toilet
Telephone
Post Box
Pillar Box
Fire Station
Hospital
Place of Worship
Cemetery
Sports Ground
Golf Course
Swimming Pool
Tennis Court
Cycle Hire
Information
Viewpoint
Picnic Site
Camping Site
Caravan Site
Youth Hostel
Hotel
Restaurant
Cafe
Public House
Shop
Post Office
Bank
Cash Point
Petrol Station
Garage
Car Park
Multi-storey Car Park
Park and Ride
Park and Ride Bus Stop
Taxi Rank
Cycle Parking

proposed to the south eastern area of Rowborough, again adjoining the main through street. The South Marston SPD sets out proposals to enhance local facilities in South Marston. Both the local centre at South Marston and the Neighbourhood centre at Rowborough, the locations as shown on the illustrative masterplan, will contain community facilities. These centres will cater for the needs of the respective neighbours with further details provided in the Design and Access Statement. At the time of writing is likely to include:

- Pub / restaurant
- Medical facilities within Neighbourhood centre
- Up to 1,500 sq. m of employment space
- Mixed use space for local retail and other services comprising up to 1,500 sq. m of Use Class A1; with provision also and in addition for A2, A3, A4, and A5 uses
- Closely associated with the primary schools
- Community and sport buildings (classes D1 and D2)

- Opportunities for strategic footpath / cycle linkages into the wider network will be exploited, connecting adjoining locations within the wider Swindon Eastern Villages to the application site.
- Legibility of the development will be ensured by developing a positive relationship between buildings, streets and spaces, with buildings fronting onto and providing opportunities for overlooking and surveillance. In addition, ensuring there is a strong definition of public (streets, opens space) and private spaces (back gardens, private driveways) making the layout legible and safe.
- A variety of landscape and development responses to the existing character of the site, including provision of a robust landscape framework including a network of formal and informal open spaces, and the creation of sports pitches located in convenient and accessible locations. At South Marston pitches are located to provide recreation opportunities for the wider community.
- The development will establish a rich and diverse green infrastructure strategy incorporating existing trees and hedgerows and connecting parks, play areas, formal sports pitches, and wildlife habitats.

2.7 The location of the various facilities and amenities on the site are in part dictated by the topography. The school is currently located on a flat area of the site that can accommodate school playing fields, which will complement the existing major hedgerow and bridleway to contribute to the green infrastructure framework for the development. The central location of the school as proposed is within the heart of Rowborough will help to establish a key focal point for the development. The network of walking and cycling routes on site will include links to the school to ensure safe connectivity.

2.8 The planning application considers a total of 2,380 new dwellings. The detailed traffic modelling, summarised in this document, assumes 2,550 dwellings, to represent a worst case (highest impact) scenario.

2.9 Future phases of the growth have been tested and are identified as Phase IA and Phase II in accordance with profiles of growth agreed during pre-application consultations and as set out in the Eastern Villages SPD.

- Phase I represents this planning application.
- Phase IA provides for Phase I plus 40,000 ha of B1/ B2 and B8 land uses to the south of the A420 together with an expansion of the Sainsbury's supermarket expansion from 7,350sqm to 10,000sqm in line with criteria agreed during consultations.
- Phase II considers the latest Local Plan allocation, in that to deliver:
  - An additional 6,000 residential dwellings south of the A420
  - retail floor space located in a network of local centres
  - Eight form entry secondary school
  - Eight forms of entry of primary school provision
  - Doctors surgery, Dentist and Pharmacy at the District Centre

2.10 Design of the transport and highway proposals is considerate to the specific concerns of the local South Marston community. The access strategy has been considered to specifically limit the increase in traffic through South Marston.

The selection of a roundabout junction on Thornhill Road provides a direct link into the development to reduce any need to travel through South Marston. Proposals for Old Vicarage Lane include the alteration of the priority such that traffic generated by the development is encouraged through the development. Access from Rowborough is proposed onto Old Vicarage Lane at two points. By this means traffic generated from within Rowborough will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road. Access under the railway is also proposed to the east.

2.11 As development progresses, a connection through South Marston and connections at Rowborough will be provided, the routes of which have been identified by the masterplan to connect to the external public highway. As such the transport strategy route provides an equitable and effective route choice for this development. The proposed link between Old Vicarage Lane and Thornhill Road is not intended to perform a southern bypass role for the South Marston area. Indeed the link has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

2.12 The potential impact of the proposals for South Marston has been fully appraised, including Pound Corner. This assessment indicates that the delivery of these proposals will not materially increase the flows through the village or impede access to the A420 for existing residents.

2.13 Furthermore in relation to the highway assessment, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included.

2.14 The impact on South Marston will be limited by:

- Discouraging the use of new highway links north of the A420 and south of South Marston being used in preference to the A420.
- Discouraging vehicles using South Marston village

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2.15 Consideration has also been given to the wider objectives in relation to transport in South Marston. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

Transport Assessment Consultations

2.16 In preparation for the planning application submission, detailed discussions have been held with both Swindon Borough Council (SBC) and representatives for Highways England (HE) over an extended period of time. These discussions have intensified over the last nine months as the planning application has been prepared. Various aspects covering the delivery of the scheme have been discussed with fundamental principles addressed and agreed. This has led to a coordinated approach to agree key aspects of scheme delivery.

2.17 This process has resulted in working agreements across various several areas of the assessment, including the following key areas:

- Use of the Swindon Wide SATURN traffic model for the high level overview
- Use of the A419 / A420 PARAMICS traffic model
- Location and the form of the Phase I access points
- Methodology to be adopted in order to review the Phase IA and 2
- Trip generation rates to be applied
- Travel Plan principles

2.18 Through the application process, discussions have been ongoing with the HE and SBC. This has led to earlier versions of this report being submitted. This amended report includes amendments / additions following observations from the regulatory bodies.

Transport Assessment Structure

2.19 The report incorporates appropriate text that reflects the agreed matters and the remainder of the report is structured in the following way:

Chapter 3: National and Local Policy Background
This chapter reviews both National and Local Policy documentation to demonstrate that this site is supported for residential use.

Chapter 4: Existing Transport Conditions
This chapter details the site location in relation to the public transport, walking, cycling networks, together with the national and local road network.

Chapter 5: Development Proposals
This Chapter reviews the development proposals and details the proposed access arrangements.

Chapter 6: Development Impact Appraisal
The chapter assesses the development in relation to Accessibility, Safety, Economy, Environment and Integration.

Chapter 7: Travel Plan
This chapter provides details on the Travel Plan that has been drafted to support this development including any measures that can be utilised to achieve the agreed modal shift targets.

Chapter 8: Development Traffic Generation
The Chapter provides details on the expected number of trips generated by this site and the methodology on how they are to be distributed within the local road network.

Chapter 9: Local Road Network Review – Journey Time
This chapter indicates the results of the assessment on journey time as predicted by the PARAMICS traffic model.

Chapter 10: Local Road Network Review – Link Flow Assessment
This chapter indicates the results of the assessment on link flows as predicted by the PARAMICS traffic model.

Chapter 11: Local Road Network Review – Junction Assessment
This chapter indicates the results of the assessment on queuing at key junctions within the road network as predicted by the PARAMICS traffic.

Chapter 12: Local Road Network Review – West of the A419
This chapter considers the potential impact of the development to the west of the A419.

Chapter 13: Limitations
This chapter defines the limitations on the above conclusions based on the accuracy of information received.

3 National and Local Policy Background

National Policy

3.1 The Government White Paper on Transport of 2004 identified the need for a transport network that can meet the challenges of a growing economy and the increasing demand for travel but can also achieve environmental objectives. This requires a transport network that provides a more reliable and freer flowing service while making walking and cycling

real alternatives for local trips. There is a clear recognition that the location and nature of development affects the amount and method of travel; and the pattern of development is itself influenced by transport infrastructure and transport policies.

3.2 **PPG13:** The Department of Communities and Local Government, Planning Policy Guidance (PPG) Note PPG13: Transport historically provided advice on how local authorities should integrate transport and land use planning. The planning policy made a contribution towards the goal of improving urban quality and vitality and to achieving a healthy rural economy and viable rural communities.

3.3 As a result, the integration of land use and transport policies became fundamental and development plan preparation led this process by aiming to reduce the need to travel, by:

- Influencing the location of different types of development relative to transport provision (and vice versa)
- Fostering forms of development which encourage walking, cycling and public transport use.

3.4 **NPPF:** PPG13 was superseded when the Department of Communities and Local Government published the National Planning Policy Framework (NPPF) in March 2012. This was produced to support reforms of the planning system and to promote sustainable growth. The NPPF sets out a framework within which Councils can produce development plans which reflect the needs and priorities of those communities.

3.5 Paragraph 30 within the NPPF encourages solutions that support reductions in greenhouse gas emissions and reduce congestion. Local planning authorities should therefore support a pattern of development which, where reasonable to do so, facilitates the use of sustainable modes of transport.

3.6 Paragraph 32 identifies that all developments generating significant amounts of movement should be supported by a Transport Statement or Transport Assessment. Plans and decisions should take account of whether:

- The opportunities for sustainable transport modes have been taken up depending on the nature and location of the site, to reduce the need for major transport infrastructure
- Safe and suitable access to the site can be achieved for all people
- Improvements can be undertaken within the transport network that cost effectively limits the significant impacts of the development. Development should only be prevented or refused on transport grounds where the residual cumulative impacts of development are severe

3.7 Paragraph 35 within NPPF identifies that plans should protect and exploit opportunities for the use of sustainable transport modes for the movement of goods or people. Therefore developments should be designed where practical to:

- Accommodate the efficient delivery of goods and supplies
- Give priority to pedestrian and cycle movements and have access to high quality public transport facilities
- Create safe and secure layouts which minimise conflicts between traffic and cyclists or pedestrians, avoiding street clutter and where appropriate establishing home zones
- Consider the needs of people with disabilities by all modes of transport

3.8 A key tool is the Travel Plan, as identified in Paragraph 36 of NPPF. All developments which generate significant amounts of movement should be required to provide a Travel Plan.

3.9 Paragraph 37 of the NNPF identifies that planning policies should aim for a balance of land uses that minimise journey lengths for employment, shopping, leisure, education and other activities. Developments should promote a mix of uses in order to provide opportunities to undertake day-to-day activities including work on site.

3.10 When setting local parking standards for residential and non-residential development, Paragraph 39 identifies that local planning authorities should take into account:

- Accessibility of the development
- The type, mix and use of development
- The availability of and opportunities for public transport
- Local car ownership levels
- An overall need to reduce the use of high-emission vehicles

3.11 **Local Transport White Paper, Creating Growth Cutting Carbon:** This document, dated January 2011, sets out the Government's vision for a sustainable local transport system that supports the economy and reduces carbon emissions. It explains how the Government is placing localism at the heart of the transport agenda, and underlines the Central Government's direct support to local authorities, including through the Local Sustainable Transport Fund.

3.12 **Manual for Streets 1 and 2 (MfS):** The UK Department for Transport (DfT) and the Department for Communities and Local Government (DCLG), with support from the Commission for Architecture and the Built Environment (CABE), commissioned WSP Group, Transport Research Laboratory (TRL), Llewelyn Davies Yeang and Phil Jones Associates to develop Manual for Streets to give guidance to a range of practitioners on effective street design.

3.13 **The Manual for Streets (March 2007)** guidance on the planning, design, provision and approval of new streets, and modifications to existing ones. It aims to increase quality of life through good design which creates more people-oriented streets. The detailed guidance applies mainly to residential streets although the overall design principles can be applied to all streets within urban areas.

3.14 A street is defined as "a highway with important public realm functions beyond the movement of motor traffic" – i.e. by its function rather than just the road hierarchy.

3.15 Manual for Streets has updated geometric guidelines for low trafficked residential streets, examines the effect of the environment on road user behaviour, and draws on practice in other countries. This research provides the evidence base upon which the revised geometric guidelines in the Manual for Streets are based, including link widths, forward visibility, visibility splays and junction spacing.

3.16 Manual for Streets 2 - Wider Application of the Principles is the result of collaborative working between the Department for Transport and the transportation industry.

3.17 The aim of the document is to extend the advantages of good design to streets and roads outside residential areas, largely covered in MfS1. By amending the way high streets and non-trunk roads are designed, the fabric of public spaces and the way people behave can be changed. It means embracing a new approach to design and breaking away from inflexible standards and traditional engineering solutions.

3.18 The new guide does not supersede Manual for Streets 1, rather it explains how the principles of the first document can be applied more widely.

3.19 **Design Manual for Roads & Bridges:** The DfT publish a large suite of documents known as the Design Manual for Roads and Bridges, which provides detailed standards and guidance on the provision of highway networks. The suite of documents provides a comprehensive manual which accommodates all current standards, advice notes and other published documents relating to the design, assessment and operation of trunk roads including motorways. The standards are routinely adopted by local highway authorities for their local highway network.

3.20 **HE Circular 02/2013:** The Strategic Road Network and the Delivery of Sustainable Development: The Circular was published in 2013 and explains how the HE will engage with the planning system and provides details on how the HE will

fulfil its remit to be a delivery partner for sustainable economic growth whilst maintaining, managing and operating a safe and efficient strategic road network.

3.21 The Circular identifies that development proposals are likely to be acceptable if they can be accommodated within the available highway capacity on the strategic road network, or they do not increase demand for use of a section that is already operating at over-capacity levels, taking account of any travel plan, traffic management and/or capacity enhancement measures that may be agreed. Furthermore it is noted that Paragraph 9 identifies that development should only be prevented or refused on transport grounds where the residual cumulative impacts of development are severe.

3.22 Paragraph 25 identifies that the overall forecast demand should be compared to the ability of the existing network to accommodate traffic over a period up to ten years after the date of registration of a planning application.

3.23 Paragraph 27 identifies that where the overall forecast demand at the time of opening of the development can be accommodated by the existing infrastructure, further capacity mitigation will not be sought.

3.24 With regard to travel plans Paragraph 29 and 30 highlights that it may be possible to free up additional capacity within the road network so that the demand generated by a proposed new development, which would otherwise be unacceptable, can be accommodated.

3.25 Paragraph 34 identifies that at locations where there insufficient capacity, the impact of the development will be mitigated to ensure that the strategic road network is able to accommodate existing and development generated traffic.

3.26 In relation to providing new access points, Paragraph 39 identifies that where appropriate, proposals for the creation of new junctions or direct means of access may be identified and developed at the Plan-making stage in circumstances where it can be established that such new infrastructure is essential for the delivery of strategic planned growth

3.27 **Swindon Borough Council Local Plan 2026:** The Swindon Borough Local Plan 2026 (formerly known as the Swindon Borough Core Strategy) is the main planning policy document for Swindon Borough, providing the planning policy framework to deliver sustainable growth up to the year 2026. The Local Plan identifies where, when, and how development will take place in Swindon. The local plan was formally adopted by Swindon Borough Council on 26th March 2015

3.28 SBC applied to the Secretary of State to save some of the policies within the Local Plan for a further period until such time they are replaced by the Swindon Core Strategy and the Site Allocations Development Plan Documents

3.29 The Local Plan includes:

- A vision and strategic objectives for the development of Swindon Borough up to 2026
- Clear policies to guide decisions on planning applications
- Broad locations of development and the allocation of strategic sites
- The infrastructure requirements required in support of the proposed Development
- The strategic context for the allocation of specific sites in either a future Council planning document and/or Neighbourhood Plans
- A key diagram, illustrating the main provisions and the strategic site allocations
- An indication of how the plan will be delivered
- Indicators to monitor the progress of the Plan

3.30 The Local Plan provides an overall spatial Vision for Swindon, as indicated below.

> "The Borough of Swindon will become a place where people choose to live, visit and invest. As an important regional centre, Swindon's appeal will stem from having an attractive and well-equipped town that has successfully blended traditional architecture with high quality contemporary buildings that incorporate sustainable design and construction principles.
>
> The achievement of a high quality public realm in the heart of the town linked to the countryside will provide a real focal point for visitors and Swindonians alike.
>
> Swindon will become one of the best business locations in the UK, offering a high quality of life, not just to its residents, but also to those from a much wider catchment area. New jobs will be created and there will be the fullest range of employment opportunities for the whole community.
>
> Swindon will be at the centre of a network of multifunctional green spaces linking the town to the wider countryside. Swindon will have responded to the needs of a growing population in a way that has protected and enhanced our natural and historic environment.
>
> People in Swindon will have the opportunity to live active, healthy and learning lifestyles."

3.31 To achieve the overall vision, 10 strategic objectives have been identified. Those relating to transport are highlighted below:

- **Strategic Objective 1:** High Quality Sustainable Development – to improve the image of Swindon, enable inclusive communities and address climate change by the provision of high quality, well designed and sustainable development.
- **Strategic Objective 2:** Infrastructure - to meet the infrastructure needs for and arising from the growth of Swindon (including health and community needs) in a timely and coordinated manner and being adequately funded.
- **Strategic Objective 7:** Transport - to support Swindon's growth through the provision of a comprehensive and sustainable transport network that is efficient, safe, affordable, accessible and easy to understand, and offers a genuine choice of modes.
- **Strategic Objective 9:** Green Infrastructure - to provide an attractive and inspirational environment to live, work, learn and play, by the provision of a far-reaching network of connected and multi-functional green spaces linked to the wider countryside.

3.32 To support the Strategic Objectives various Policies have been defined.

- **Policy SD1: Sustainable Development Principles -** To enable the delivery of sustainable development and support sustainable communities in the Borough all development proposals will:
  - Be of high quality design
  - Promote healthy, safe and inclusive communities
  - Respect, conserve, and/or enhance the natural, built and historic environments
  - Assess and address the impact of climate change through mitigation and / or adaption measures
  - Provide or contribute to the assessed local and borough wide infrastructure and service requirements
  - Contribute to the retention of jobs and growth of the local economy and complement town centre regeneration
  - Be accessible by walking, cycling and/or public transport
  - Use land and resources (such as water, energy, minerals and waste) in an efficient and effective way
- **Policy SD2: The Sustainable Development Strategy**
  Recognising its role and function in the wider area development in the Borough will be concentrated at Swindon through a combination of realizing development opportunities within Swindon's urban area; and allocated strategic sites at; Wichelstowe, Commonhead, Tadpole Farm, Kingsdown and the proposed New Eastern Villages,

Rowborough and expanded South Marston.

- **Policy SD3: Managing Development**
  When considering development proposals, the Council will take a positive approach to reflect the presumption in favour of sustainable development.

3.33 To support the delivery of the Policies, several overall 'Themes' are outlined. Theme 4 relates to Transport. This identifies specific policies to:

- Improve key transport gateways and corridors
- Provide good access to Swindon Town Centre and key destinations
- Reduce severance caused by transport corridors and the dominance of the car on the streetscene
- Minimize the environmental impact from transport (for example, vehicle emissions)
- Minimize congestion and therefore journey time, noise and air quality
- Promote healthy lifestyles and travel choices and maximize opportunities to walk and cycle
- Support good public transport provision
- Encourage innovative transport initiatives for rural areas
- **Policy TR1: Sustainable Transport Networks - The council will use its planning and transport powers to** help reduce the need to travel, and support and encourage the sustainable, safe and efficient movement of people and goods within and through the Borough.
- **Policy TR2: Transport and Development -** New development should be located and designed to reduce the need to travel and to encourage the use of sustainable transport alternatives, particularly walking and cycling, and provide the potential to maximize bus travel. Development shall be permitted where proposals provide access that is appropriate to the scale, type and location without detriment to highways safety and local amenity, and where there is an existing safe and convenient pedestrian and cycle access or provision is made for such access. Development shall be permitted where proposals will not prejudice or impede an existing or planned cycle route, or provision is made for a more satisfactory route. Development that results in the loss of existing rights of way or their disruption shall only be permitted when adequate, acceptable alternative provision or diversions be arranged. Development shall be permitted where proposals do not remove, narrow or materially impair the approved line of the Thames Path or Ridgeway National Trails, and/or public access to them. Developments shall provide appropriate mitigating measures to offset any adverse impacts on the transport network at both the construction and operational stages. Parking provision, including secure cycle and motorcycle parking, should be provided in accordance with the Council's adopted parking standards. To assess and mitigate the impact of development and to promote sustainable travel choices, the following information will be required to support planning applications:
  - A Transport Assessment: where the proposed development is likely to have significant transport and related environmental impact (in accordance with Department for Transport guidance)
  - A Travel Plan: where the proposed development is likely to have significant transport and related environmental impact (in accordance with Department for Transport guidance).

3.34 To deliver the necessary housing and employment growth identified for Swindon, specific development areas have been identified.

- **Policy NC3: New Eastern Villages - including Rowborough and South Marston Village Expansion**

  a. Land to the East of the A419, as defined on the Policies Map, is allocated for a mixed-use development. The form of the development shall comprise a series of new inter-connected distinct villages and an expanded South Marston village defined by the network of green infrastructure corridors.
  b. The development shall provide:

  - A design led approach to housing density leading to an overall average density of 40 dwellings per hectare; comprising:

i. About 6,000 dwellings at the New Eastern Villages (south of the A420);
ii. About 1,500 dwellings at Rowborough (north of the A420), and
iii. 500 dwellings at South Marston.
iv. Affordable housing shall be provided at Rowborough and south of the A420 in accordance with Policy HA2. Separate proposals are made in Policy RA3 in relation to South Marston
v. High quality public realm including outdoor civic public space;
vi. Sustainable transport links including:
vii. Walking and cycle network improvements that integrate with existing networks and provide good connectivity within the development and to the surrounding area;
viii. An express bus network through the District Centre that connects the Eastern Villages to Swindon Town Centre as part of phase 1 of the development, which includes residential development north of the A420, the District Centre and the employment allocation;
ix. Additional public transport services to connect with Swindon and internally within the development;
x. An improved gateway junction at White Hart to manage additional demand and deliver high quality public realm;
xi. Improvements to the Oxford Road/Drakes Way and Covingham Road/Dorcan Way transport corridors;
xii. A green bridge across the A419 near Covingham Drive to provide for walking, cycling and public transport;
xiii. A new road link to the Commonhead Roundabout;
xiv. A new road link under the Bristol to London railway line connecting the development north and south at Rowborough;
xv. New and/or improved accesses to the A420 for proposed residential and employment uses;
xvi. A 1000 (3ha.) space Park and Ride site; and
xvii. Measures to minimise rat-running through existing adjacent villages and east Swindon.

> An extensive green infrastructure network that maximises opportunities for habitat connectivity and enhanced biodiversity including extending the River Cole green infrastructure corridor and connecting with Nightingale Wood;
> Sports and leisure facilities, including playing pitches a leisure centre and a 25m swimming pool;
> About 40 hectares net of employment land in total (B Class Uses) to be located south and east of the A420 and A419 respectively adjacent to the White Hart Junction, in a form that complements the mixed-use District Centre, and safeguards its connection with the wider residential development in the Eastern Villages and distributed as follows:
> 2.5 hectares within B1a use-class to be located at and/or adjacent to the District Centre;
> 7.5 hectares. within B1b/c or B2 use-class; and
> 30 hectares within B8 use-class.
> About 12,000m2 (gross) of retail floorspace including a high quality District Centre with strong connectivity to the adjacent residential areas, comprising an anchor food store and complementary uses, and a network of Local Centres that offer retail provision of a scale that meets the daily shopping needs of the communities they serve, including the existing community at South Marston.
> Educational requirements comprising:

i. A minimum of 8 forms of entry of secondary provision;
ii. A minimum of 8 forms of entry of primary provision with early learning facilities;
iii. A site for a special school for children and young persons with profound, multiple and severe learning difficulties for ages 0-25;
iv. Measures to manage the demographic peak at primary and secondary schools.

> Community facilities by the means of safeguarded land and/or developer contributions, including where appropriate, flexible, multi-purpose buildings for use by the community, the public sector and for worship at the District Centre, local centres and educational requirements;
> A health care facility with GP, dentist and pharmacy at the District Centre;

> Safeguarded land for a fire station towards the southern part of the site, and
> A sewage treatment works if required.

c. The development will ensure:

> The landscape context and views to and from the North Wessex Downs AONB are respected, including potential off-site mitigation;
> The risk of flooding from the development is minimised, both within the development and at existing neighbouring communities in accordance with Policy EN6;
> Biodiversity, including the River Cole Corridor and River Cole Meadow County Wildlife Sites, is protected, integrated and enhanced; and
> The historic environment, including the Scheduled Monument, Earlscourt Manor, Great Moorleaze Farm and other Listed Buildings are protected, acknowledged and enhanced.

d. The route for the Wilts & Berks Canal as set out on the Policies Map will be safeguarded and protected from development.
e. The character and identity of Wanborough, Bishopstone and Bourton will be protected by a principle of non-coalescence between the settlements. The land between the New Eastern Villages site boundary and the existing villages shall remain part of the countryside. However, small scale development within this area, as defined on the Policies Map, will be permitted where it retains or enhances the existing character of the countryside and:

> Involves the re-use, conversion or extension of existing buildings at a scale appropriate to their location, in accordance with the criteria specified in Policy DE1; or
> Is an essential requirement directly related to the economic or social needs of the rural community.

f. The character and identity of South Marston will be protected as set out in Policy RA3.

- **Policy RA3: South Marston**

a. Development at South Marston shall be in accordance with Policies SD1 and SD2 and must contribute towards the creation of an integrated village with a distinct rural and separate identity from Swindon and other settlements. Expansion of the village will be as part of the New Eastern Villages strategic allocation and through allocated brownfield opportunities, as shown on the Proposals Map.
b. Development at South Marston shall ensure it respects the character of the existing village by:

i. Providing housing at an average of 30 dwellings per hectare; and
ii. Providing affordable housing in accordance with Policy HA2, but at a proportion of no more than 20%.

> Provide community, recreation facilities and retail provision of an appropriate scale, as part of a village centre;
> Provide green infrastructure in accordance with Policy EN1 proportionate to the scale of expansion;
> Provide an extended recreation ground to include the field to the south-west of the current recreation ground with recreational facilities of an appropriate type and scale;
> Protect historical landscape features, archaeological assets and existing green-infrastructure;
> Deliver primary school places within the village to meet the needs of an expanded South Marston; and
> Provide a new road connection between Thornhill Road and Old Vicarage Lane, within the limits of the expanded village and designed so that it does not form a bypass to South Marston;
> Provide traffic management and sustainable transport measures to minimise the volume of traffic, including traffic passing through the village, and to reduce the impact on the village, in particular at Pound Corner;
> Ensure other development/redevelopment opportunities on existing (brownfield) sites in the west and north of

the village provide strong links with the village and make the appropriate infrastructure contributions to mitigate their impact in broad accordance with the South Marston Village SPD;

> Provide mitigation measures to protect the village from flood risk; and

> Realise opportunities to facilitate main drainage and utilities supply to existing properties that currently lack connections where possible.

c. To ensure non-coalescence of South Marston with Swindon and the other proposed Eastern Villages the land between the expanded village and the railway to the south shall remain part of the countryside. Development within this area, as defined on the Proposals Map, shall only be permitted where it retains or enhances the existing character of the countryside and:

> Involves the re-use, conversion or extension of existing buildings at a scale appropriate to their location, in accordance with the criteria specified in Policy DE1; or

> Is an essential requirement directly related to the economic or social needs of the rural community.

Transport

3.35 The scale of the development facilitates the provision of an express bus network system between the new development and Swindon Town Centre and will enable longer distance bus services which will utilise the express bus route and infrastructure. Such a system should be fast, frequent, safe, reliable and cost-effective. Above all it should be attractive to those who have a choice of transport means. Opportunities to walk and cycle should be maximised, particularly along the green corridors and to integrate with existing networks.

3.36 The green bridge will provide an additional walking and cycling link across the A419 and public transport route. It will create an attractive and safe route to connect the new development and the existing communities.

3.37 The White Hart junction is a critical part of the local and strategic road network, principally providing an interchange between the A420 and the A419 Trunk Road. The White Hart Junction also has a critical role in high quality urban design and should provide a gateway to the town. Its reconfiguration should not only facilitate vehicular access to the development but should also facilitate enhanced pedestrian and cycling access. It is important that a second access to the A419 is provided to disperse traffic from the development, and this will be provided in the form of a link road to the Commonhead roundabout from within the southern part of the development.

3.38 North-south routes under the railway line will be provided to strengthen walking and cycling links. New vehicular access should be the primary route to the new Rowborough Village, and to avoid rat-running through South Marston. Transport measures should also seek to avoid rat-running through Wanborough and the existing communities of East Swindon, Stratton and Covingham including by the provision of the Commonhead link road.

3.39 In alignment with the Swindon Transport Strategy a Park and Ride should be provided as part of the development to help reduce the volume of traffic entering Swindon.

3.40 Impacts on the A419 and Junction 15 of the M4 must be considered in terms of journey time reliability and safety.

3.41 **Infrastructure Delivery Plan:** SBC is required to demonstrate that the policies in the Local Plan will be delivered in a sustainable way. To this end, there is increased emphasis on the need to identify infrastructure required to support future growth, resulting from additional housing and employment during the Plan period to 2026. The IDP focuses on those infrastructure items which will require capital expenditure

3.42 The IDP helps to identify:

- Types and costs of infrastructure

- Delivery timetable
- Gaps in funding
- Critical pieces of local and strategic infrastructure.

3.43 In relation to transport infrastructure the **WSP Swindon Transport Strategy (2009)** sets out Swindon's transport aspirations to 2026.

- **Rapid Transit**
  The Northern Rapid Transit link will deliver the first arm of Swindon's Rapid Transit scheme connecting Tadpole Farm and the Northern Development Area with Swindon town centre. The rapid transit routes from Wichelstowe and the Eastern Villages will provide fast and reliable journey times for new communities to Swindon Town Centre and complete the rapid transit network. The rapid transit network will complement the Park and Ride Strategy – additional sites are proposed at Wichelstowe, Eastern Villages and Commonhead.
- **Town Centre**
  The new Town Centre bus exchange and improved bus stops will support economic growth by providing a viable and attractive alternative to the private car for everyday journeys.
- **Eastern Villages including Rowborough and South Marston**
  The Eastern Villages will deliver a new district centre, not just to serve the new community but also the existing residential areas at Covingham, Dorcan and Parks. The main route from the New Eastern Villages into Swindon's Central Area will be via Oxford Road (A420) and White Hart Roundabout. The intention is to deliver an improved junction at White Hart that establishes strong linkages between the Eastern Villages and the rest of Swindon, prioritises access into Swindon Town Centre and improves the public realm. A Green Bridge is proposed that will provide access across the A419 – part of the Strategic Road network and a significant physical barrier – for public transport, walking and cycling. The Commonhead Link (and associated Junction 15 improvements) will provide direct access to the Strategic Road Network (A419 and M4). Junction improvements will also be required on the A420.

3.44 **Eastern Villages Draft Supplementary Planning Document 2013** (not being taken forward beyond draft stage but referred to in draft form within the Adopted Local Plan)**:** It identified a set of development principles, as detailed below:

- Sets out the overall vision and principles
- Seeks to ensure that the Vision for the development is delivered in accordance with the key principles
- Provides a detailed planning framework
- Provides a catalyst to express aspirations for the development
- Offers detail and clarity on the expected outcomes from the development

3.45 The SPD identified the overall Vision for the Eastern Village Development. The Eastern Villages development will create sustainable well-designed places where people want to live, work and spend time. The development will:

- Be a high quality, sustainable development with strategic infrastructure that benefits Swindon as a whole
- Sensitively and positively respond to the existing landscape context, natural and historic assets and the character and identity of the surrounding villages as well as enhancing biodiversity and green infrastructure;
- comprise new distinct villages with individual identities and characters linked together by green spaces that integrate with the existing urban area and wider landscape setting;
- provide the facilities and services required for the new communities and opportunities for existing communities to benefit from the development
- Form a new eastern gateway to Swindon that improves the image of the Town and maximises benefits to the wider economy including connections to the Town Centre.

3.46 The key principles were to :

- Create vibrant, attractive places that are functional, durable and capable of adapting to accommodate changing lifestyles in line with sustainable communities
- Ensure the timely delivery of necessary facilities to support these sustainable communities
- Conserve and enhance natural systems, watercourses, biodiversity and landscape settings, including integration with green infrastructure to help mitigate and facilitate adaptation to climate change
- Conserve and enhance Swindon's historic assets and cultural heritage, allowing their use and enjoyment without harming them for future generations
- Ensure the development becomes part of the wider town and supports the physical, social and economic regeneration of East Swindon. Respect and support the existing character and setting of surrounding villages;
- Contribute to and not harm the Town Centre regeneration and support the broader Swindon economy with new employment and investment opportunities;
- Contribute to an improved strategic transport network for Swindon and facilitate public transport priority for journeys into the Town Centre;
- Encourage walking and cycling, within the development and ensure high levels of accessibility and connectivity within and beyond the new Eastern Villages
- Enhance the image of Swindon by maximising place-making opportunities particularly around key nodes, gateways and frontages.

3.47 The SPD identified that the development will deliver approximately 1,500 dwellings will be located at Rowborough, 500 new homes at an extension to South Marston and approximately 6,000 dwellings south of the A420.

3.48 In relation to Transport the SPD identified the interventions identified with policy NC3 of the Local Plan, viz:

- Improved White Hart junction
- Improvements to the existing network including A420
- A green bridge
- Park and Ride site
- Rapid transit link and additional public transport services
- Measures to minimise rat running through existing communities
- New road link to Commonhead

3.49 The SPD set out a preferred long term solution for the Improvement of the White Hart roundabout which is described as the diamond interchange. Further detailed design and the securing of funding is required. The SPD recognises nevertheless that the Eastern Villages may proceed by means of the "phased implementation of improvements".

3.50 Paragraph 8.10 of the SPD stated that "The EV SPD requires that development is planned with the strategic infrastructure secured, and provided, to support the initial phases of development whilst ensuring that the overall development, and its total infrastructure requirements, is fully taken into account." Therefore the TA considers the wider strategic implications of the development.

3.51 Paragraph 8.24 (n) which stated the development should provide "contributions to wider infrastructure requirements of the EV development, including off-site highway/transportation improvements".

3.52 With regard to the A420, the improvements required are, unsurprisingly, consistent with those set out in the Local Plan and South Marston SPD (South Marston SPD not being taken forward at this stage but still referred to in draft form in the Adopted Local Plan) (see below) and included:

- A new eastern access to Rowborough (north);
- An eastern access to the Park and Ride site in the south
- Improvements to the junction at Old Vicarage Lane;

- A new access to serve the new employment area (to the south);
- Improvements to the Police Station junction and improvements to Gablecross junction.

3.53 **South Marston Draft Supplementary Planning Document June 2013** (not being taken forward beyond draft stage but referred to in draft form within the Adopted Local Plan) **and Background Transport Assessments:** The purpose of the SPD was to provide a planning framework setting out the expectations and requirements for the sustainable development of the village of South Marston. In preparation of the SPD, SBC commissioned JMP to investigate the existing challenges and to review the potential impact of the Eastern Villages.

3.54 The JMP initially identified a range of potential issues with in the transport system together with possible interventions, as detailed below.

Description	Constraint	Intervention
Pedestrian and vehicular conflict around Pound Corner	Insufficient carriageway and footway available	Realignment of the carriageway at Pound Corner to provide footway on the northern side of Old Vicarage Lane
Gaps in footway provision	Thornhill Road close to the industrial estate. Vicinity of South Marston Hotel	Continuous footway from Pound Corner to Keypoint roundabout and from Pound Corner to South Marston Hotel
Accessibility improvements	Lack of Dropped Kerbs / Tactile paving	To provide dropped kerbs / tactile paving in locations where these are not provided
High Traffic Speeds/ Through Traffic	Little evidence to support view in terms of speeds/accidents or traffic volumes.	20mph zone for village centre & 30 mph between village centre and A420 for consistency. Some additional traffic calming measures

Figure 3a: Existing constraints

3.55 The JMP study then investigated the potential impact of the delivery of the Eastern Village concentrating on the impact on the A420 and within South Marston. This report notes that increase in traffic arising from development are from a very low base. The report concluded that "with the exception of the junctions along the A420, junctions within South Marston are expected to operate with minimal queuing. This assumes that a package of measures is in place for the A420 and A419 which deters vehicles from rat running through the village" (JMP Technical Note: "*South Marston Supplementary Planning Document Part Three Transport Assessment",* June 2012". Rat running is therefore avoided, on JMPs analysis, providing journey times along the A419 and A420 continue to make these routes attractive to use. The junctions with the A420 are identified as needing interventions to ensure that this is the case.

3.56 A key conclusion of the JMP technical note, reinforcing the conclusion regarding transport impacts in South Marston itself was that a roundabout junction on Old Vicarage Lane – providing access from Rowborough would be "unlikely to result in a significant increase in traffic volumes through South Marston village"(para 5.6) since most traffic is likely to travel south.

3.57 The study also identified options for improvements at Pound corner involving one or two way traffic and seeking to widen the existing footways noting that there were pros and cons of each alternative.

3.58 The purpose of the South Marston Supplementary Planning Document (SPD) was to provide a detailed planning framework for South Marston, setting out the expectations and requirements of development within the village. It addresses the existing and the expanded village. The SPD described a vision that: "*South Marston should be a high quality, integrated village with an enhanced sense of community, activity and safety within a distinctive form that builds on the existing character and features in a sustainable manner*". In order to deliver the vision a number of issues will need to

be addressed including, in relation to transport: "*improvement of existing, and provision of new highway infrastructure, including resolving issues of through-traffic, Pound Corner, the absence of road-side paths, access onto the A420 and off-site provisions to ensure free movement along the A420*"(para 3.3).

3.59 The Development Principles in respect of Access and Movement are set out in para 6.28 to 6.46. The SPD:

- Requires a Transport Assessment (TA), to be undertaken by the developer(s), to determine the number of additional trips generated and how these will be accommodated on the existing highway network; and
- Expects a sustainable Travel Plan to accompany the TA to promote alternatives to single occupancy car use.

3.60 The SPD promoted:

- A 20mph speed restriction throughout the village (30mph outer limit) with psychological traffic calming and new gateway features.
- The retention of Rowborough Lane and Nightingale Lane as public rights of way/bridleways, with no new development being served by either.
- Junction improvements on A420 (including at Gablecross and at the Police Station access, with the introduction of a new eastern access near Acorn Bridge and capacity improvements along the A420
- Improvements to Pound Corner including for instance the provision of a raised footway on the northern side of Pound Corner;
- An all movement junction to link Rowbrough to Old Vicarage Lane is proposed consistent with the advice of JMP;
- The re-alignment and signalisation of the junction of Old Vicarage Lane and the A420 in order to provide a safe egress onto the A420, particularly for those vehicles travelling west towards White Hart Junction.
- The design of highways as a series of streets with important public realm functions, rather than as roads whose primary purpose is the movement of traffic.
- Enhanced footpath and cycle connections.

3.61 The highway principles and technical work determine that there is no need for an eastern bypass for South Marston and that connections between Old Vicarage Lane and Thornhill Road should not be considered a "southern bypass" for the village

3.62 **SBC Local Transport Plan 3 (LTP3):** The Local Transport Plan 3 was adopted in March 2011 and covers the period from 2011- 2026. It sets out an overarching strategy for the delivery of the Council's transport programme. The mission statement for the LTP3 is:

> *"a safe, effective and fit for purpose transport network that supports Swindon's ambitions for town centre regeneration and economic growth whilst protecting and enhancing quality of life and the environment for the benefit of local residents, visitors and businesses"*

3.63 While LTP3 covers maintenance and small schemes, which are not identified within the IDP, larger more strategic transport schemes are included.

4 Existing Transport Conditions

Existing Travel Behaviour Overview

4.1 A review of 2011 Census data has identified the existing mode share for residents within all of the existing Wards. The modal split is indicated in Figure 4a with the distance travelled to work is indicated in Figure 4b. The site is largely located in the Blunsdon Ward and neighbouring the Wards of St Margaret's, St Phillip and Covingham & Nythe.

4.2 Blunsdon Ward contains a large geographically area but only includes 1,323 dwellings accordingly to 2011 Census data. As a result of the limited number of dwellings, the Blunsdon ward is not deemed representative of the proposed development characteristics. The neighbouring St Margaret's Ward is located in a similar position to the highway network and would access similar sustainable facilities. In regard to the character of the St Margaret's ward, there are 4,321 households of which 3,034 are identified as family households. The household type is primarily 3 bed units supported by 2 and 4 bed units. It is considered that the St Margaret's ward is representative of the proposed development.

4.3 Based on the 2011 data, the most dominant mode of travel is by car, resulting in 69.5% of all trips within the St Margaret's ward. Travel by foot is the second most dominant mode of travel.

4.4 The 2001 Census data also provides an indication of the distance travelled to work. This indicates that 7.6% of working people work home. This percentage is largely representative of the majority of wards in the area and is consistent with the average for all of Swindon. 75.3% of the work force travel less than 10km. Therefore a large proportion of the workforce remain within the Swindon hinterland.

Mode	Percentage
Train	0.9%
Bus, minibus or coach	5.7%
Driving a car or van	69.5%
Passenger in a car or van	5.6%
Motorcycle, scooter or moped	1.3%
Taxi	0.4%
Bicycle	5.7%
On foot	7.0%
Work mainly at or from home	3.4%
Other method of travel to work	0.5%

Figure 4a: Mode Share

Mode	Percentage
Working at or from home	7.6%
Less than 2km	27.0%
2km to less than 5km	40.7%
5km to less than 10km	10.1%
10km to less than 20 km	2.2%
20km to less than 40km	3.1%
40km to less than 60km	2.4%
60km and over	3.3%
Other	3.5%

Figure 4b: Distance travelled to work

Existing Highway Network

4.5 The location of the site in relation to the local road network is indicated in Figure 4c.

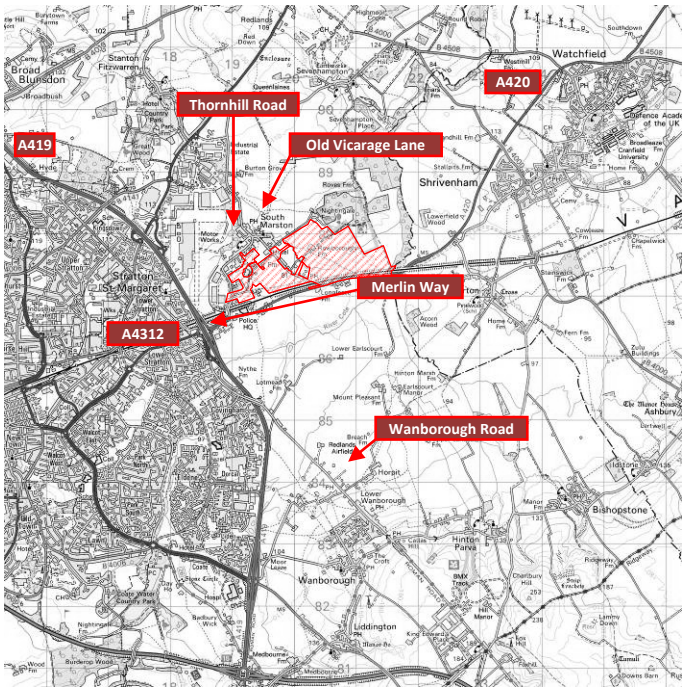

Figure 4c: Site Location in relation to the local road network

4.6 The majority of the road network adjacent to the site is classified as part of the Local Road Network (LRN), managed by SBC, with the A419 and the M4 Motorway managed by the HE. The roads managed by the HE are typically identified as being a part of the Strategic Road Network (SRN).

4.7 The A420 is located to the south of the site and forms an important corridor within the LRN. The A420 commences from Swindon and heads generally in a north easterly direction extending into Oxford. The A420 is predominantly single carriageway road which is subject to 50mph speed limit along much of the length. There are sections of dual carriageway and the section approaching Swindon is subject to a 50mph speed limit.

4.8 There are a several junctions along the length of the A420 which are predominantly at-grade. The junctions along the A420 that are within the hinterland to the site are:

- A420 / Old Vicarage Lane
- A420 / Police HQ
- A420 / Gablecross
- A420 / A419

4.9 Old Vicarage Lane forms a priority T-junction with the A420 with a ghost island right turn bay for vehicles travelling along the A420. From the junction with the A420, Old Vicarage Lane continues in a north westerly direction towards the village of South Marston. Old Vicarage Lane is rural in character and does not provide any dedicated facilities for pedestrian or cyclists. Circa 100m north of the A420, Old Vicarage Lane passes under the Swindon to Paddington rail line, as indicated below. The carriageway beneath the underbridge is of reduced cross sectional width.

Figure 4d: Southerly view along Old Vicarage Lane

4.10 To the north, Old Vicarage Lane forms simple priority junctions with Nightingale Lane and Rowborough Lane. These are minor roads which provide access to a small number of existing residential properties.

4.11 Circa 1.4km to the west of the junction with Old Vicarage Lane, the A420 provides a signalised access junction to the Wiltshire Police headquarters. There is no facility to undertake a right turn from the A420 into the police headquarters, so vehicles needing to access the facility from an easterly direction must continue through this junction toward the Gablecross roundabout and undertake a U-turn. Heading in a westerly direction from this junction, there is an off road segregated walking and cycling route.

4.12 The Gablecross roundabout lies circa 200m to the west of the Police HQ signals and is a five arm roundabout. The arms of the roundabout serve Sainsbury's supermarket to the south, Thornhill Road and the St Margaret's Retail Park to the north with the remaining arm being exit only from the retail park. The off-road pedestrian/cycle route continues westerly through Gablecross junction. There is an unsignalised pedestrian crossing across the Thornhill Road arm with a signalised pedestrian crossing to the west of the junction to provide a safe route into the supermarket.

4.13 Thornhill Road heads in a generally northerly direction to access the South Marston village. Thornhill Road is rural in nature and does not provide for dedicated facilities for pedestrian or cyclists within the highway over much of the length, as indicated below. There is however a pedestrian and cycle route alongside Thornhill Road between the Gablecross roundabout and the Keypoint roundabout and there is a segregated pedestrian and cycle route to the west of Thornhill Road between the Keypoint roundabout and the start of south Marston village. There is traffic calming (single file traffic) at the entry and exit of the village which attempts to control vehicle speeds into South Marston

4.14 Thornhill Road continues into the village of South Marston where there are sections of a discontinuous footway. In the centre of South Marston, Thornhill Road forms a priority T junction with Old Vicarage Lane, with this junction known locally as Pound Corner. The village is accessed by three roads (Highworth Road, Thornhill Road and Old Vicarage Lane) that link to the north with the A361 and to the south with the A420 and onward to the A419 trunk road and M4. These roads are considered to be potential "rat runs" between the strategic roads and to the major employment areas adjacent to the village. The alignment and relative narrowness of these roads create points of potential pedestrian and vehicular conflict, including Pound Corner.

Figure 4e: Northerly view along Thornhill Road

4.15 Circa 300m to the west of the Gablecross junction lies a grade separated junction between the A419 and the A420, known as the White Hart interchange. The interchange has two parts that are connected by part of Merlin Way. The northern part of the interchange is a grade separated roundabout with the A419 main carriageway passing over the roundabout. The roundabout connects the A420, the A419 southbound slip roads on the east side of the roundabout together with Oxford Road and Ermine Street to the west and Merlin Way to the south. The northbound A419 on and off slips have a roundabout junction with Merlin Way that is located 400m to the south of the White Hart roundabout junction and form the southern part of the interchange . Any vehicles going to the north on the A419 from the A420 have to travel southwards along Merlin Way before travelling north along the A419 and as such the present arrangement is counter intuitive. Oxford Road continues from the White Hart interchange westerly towards Swindon town centre.

4.16 The White Hart interchange incorporates a pedestrians and cyclist bridge linking the A420 movements east-west either side of the A419.

4.17 A Toys-R-Us store is located to the south east of the A419. Inbound access to the store is taken direct from the A419 southbound on slip. Outbound trips joining the A420.

4.18 Merlin Way continues south from the White Hart interchange and becomes Wanborough Road which crosses over the A419 via a single carriageway road bridge 1.5km south of White Hart Roundabout. Wanborough Road links with Wanborough and Upper Wanborough, and has one lane per direction.

4.19 Continuing from the White Hart interchange, Oxford Road (A4312) provides a westerly link into Swindon town centre. Oxford Road is a single carriageway road with a footway to the north and a footway/cycle way to the south. Oxford Road is a residential road allowing direct access to individual properties. Oxford Road has a signalised pedestrian crossing near Wanborough Road, a signalised junction with Nythe Road and a roundabout junction with Dorcan Way, with this junction being identified as the Green Bridge roundabout, being adjacent to the Green Bridge Retail Park.

4.20 From the White Hart interchange the A419 heads in north westerly and southerly directions. To the north west the road heads toward Cirencester passing through further junctions identified locally as the Honda and Rat Trap roundabouts. To the south, the road passes through the Commonhead junction to link with the M4 at Junction 14. The A419 takes the form of a lit dual-carriageway which is subject to a national speed limit of 70mph. There are no pedestrian or cyclist facilities along the A419.

4.21 The M4 motorway caters for east – west movements and provides the major ink between South West England / Wales to the west and Reading / London to the east. The grade separated junction with the M4, junction 15, is signalised. The roundabout is lit and there are no pedestrian or cyclist facilities at the junction.

Existing Sustainable Facilities and Services

Pedestrians and Cyclists

4.22 At the present day, the development land does not have any significant traffic generators within the boundary. As such, there has been no material requirement for footway / cycleway provision on site or to access the site. The nearby primary roads that do provide footways have been identified earlier in this section.

4.23 Swindon has a cycling network consisting of a mixture of both on and off road routes that also caters for walking trips. As indicated earlier, an off road cycling route commences from the police headquarters and continues westerly towards the town centre. A cycle way extends northwards from the junction with Gablecross to link South Marston. At the A420, the White Hart Interchange provides a bridge for onward travel towards the town centre. There are additional routes to the east of the site.

4.24 The figure below identifies the location of the cycleways closest to the site.

4.25 The desire line from the site is predicted to be westerly towards the supermarket and the town centre, where existing provision is available.

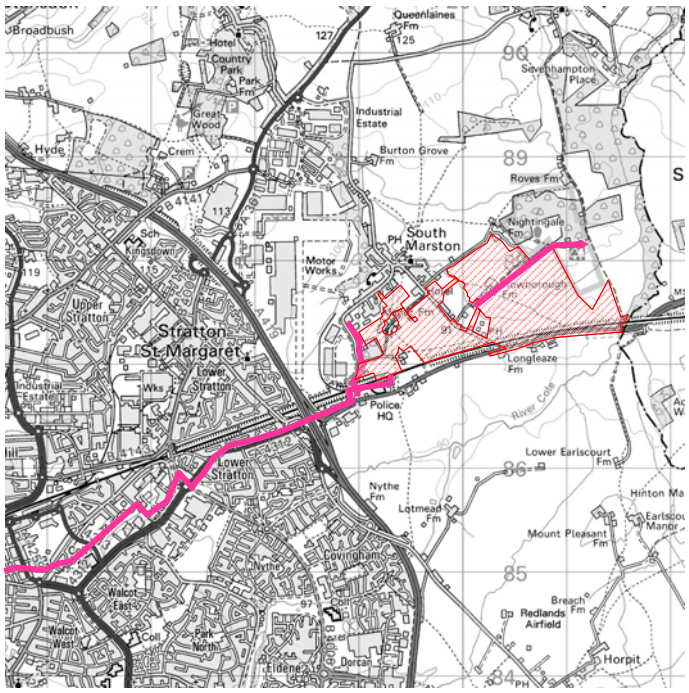

Figure 4f: Cycleways closest to site

Public Rights of Way

4.26 Public Rights of Way (PRoW) are classified as highways and as such are protected routes. The 1949 National Parks and Access to the Countryside Act placed a duty on every County Council in England and Wales to draw up and publish a definitive map and statement of PRoW in their area.

4.27 The Definitive Map is the legal record of the location and status of PRoW. The statement is a description of the PRoW shown on the definitive map.

4.28 There are four classifications of PRoW:

- Footpaths - by foot only
- Bridleways - by foot, horse or bike
- Restricted byways - by any form of transport that doesn't have a motor
- Byways open to all traffic - let you travel by any form of transport, including cars

4.29 The figure below highlights the PRoW that are closest to the site. This illustrates that there is a network of bridleways and footpaths that cross the site and connect with bridleways that penetrate other roads in the vicinity of the site. In particular there are existing strong links between South Marston and the A420 for pedestrians and cyclists to Gablecross and for pedestrians across the railway at underbridges further to the east near the Police HQ and to the west and east of Old Vicarage Lane.

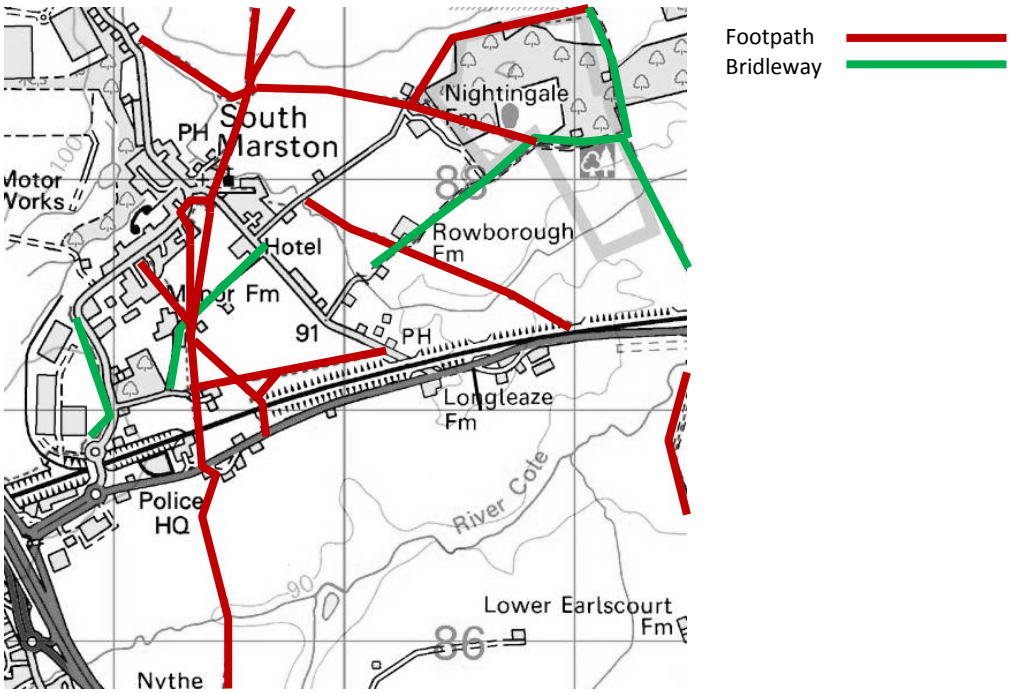

Figure 4g: Public Right of Way

Public Transport - Road

4.30 Bus services 65 and 66 operate along the routes close to the site as indicated below.

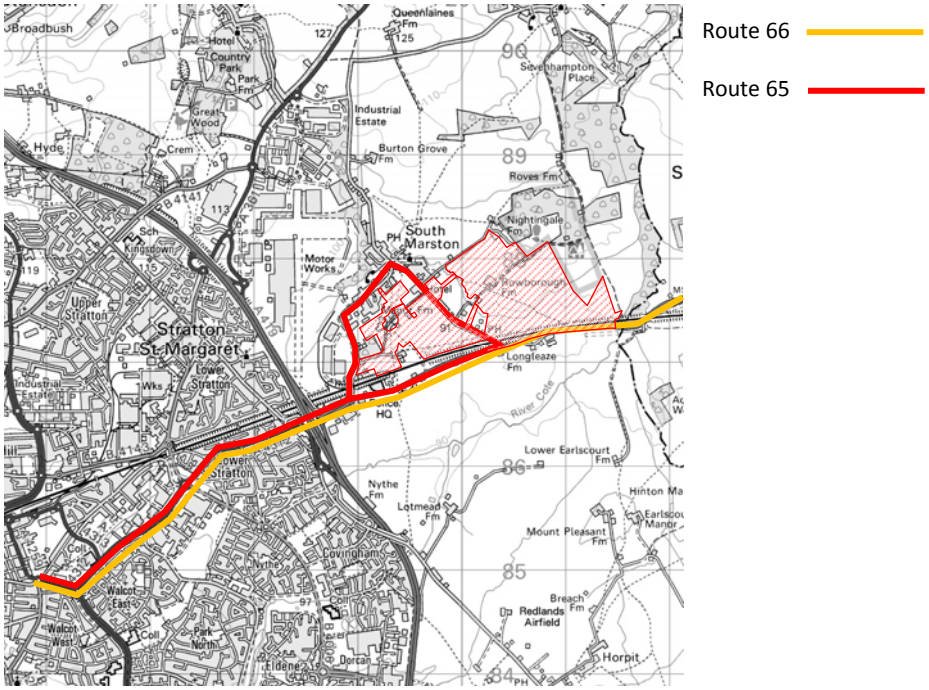

Figure 4h: Bus Routes operating close to the EV

4.31 The bus service 65 is a route managed by Stagecoach and operates between Swindon and Faringdon, Monday to Saturday. The first bus leaves Swindon Bus Station at 10:50 calling at South Marston at 11:03. For trips towards Swindon, the first bus calls at South Marston at 10:26 with a journey time of 15 minutes. This service operates typically four times daily.

4.32 The Route 66 is managed by Stagecoach and operates along the A420 between Swindon and Faringdon, Monday to Sunday. The first bus leaves Swindon at 05:40 and typically operates on a half hourly frequency. The journey time to Oxford is 1 hour and 15 minutes.

4.33 The bus station provides waiting areas, toilets and is located circa 750 metres to the south of Swindon train station.

Public Transport - Rail

4.34 The closest train station is located in central Swindon, circa 6km to the west of the proposed site. The train station, presently managed by the franchise operator, First Great Western, provides a range of facilities including:

- 578 space car park open 24 hours
- Ticket office open 7 days of the week
- Self-service ticket office
- Manned help desk
- Cash machine
- Pay phones
- Post box
- Public WiFi
- Refreshments
- Shops
- Toilets
- Waiting rooms

4.35 Swindon rail station provides three routes per hour to Didcot with a journey time of circa 17 minutes. From Didcot, onward travel to Oxford is possible. Additional routes to London Paddington are provided three times per hour with a

journey time of 63 minutes.

4.36 Trips from Swindon westwards to Bristol are supported, with two trips per hour each to Temple Meads (45 minute journey time) and to Bristol Parkway (Journey time 27 minutes).

Accident Review

4.37 Data have been obtained from Wiltshire Police for all recorded personal injury road accidents (PIAs) occurring during a five year period up to the end of February 2013 on the A419, A420, A4312 and A361 and other roads in the vicinity of the site.

4.38 A total of 418 accidents are recorded, which are summarised below, resulting in 597 casualties. Two accidents resulted in fatal injury, both to a motorcyclist. A further 51 accidents resulted in serious injury.

	Number of PIAs				Casualties
	Slight	Serious	Fatal	Total	
Year 1 to end Feb'09	86	13	0	99	133
Year 2 to end Feb'10	75	14	0	89	121
Year 3 to end Feb'11	62	6	1	69	99
Year 4 to end Feb'12	71	11	0	82	119
Year 5 to end Feb'13	71	1	1	79	125
5 year period total	365	45	2	412	597

Figure 4i: Total number of PIAs by year and severity, with casualties

4.39 A notable statistical outlier relates to the high proportion of rear end shunts, making up 44% of all accidents. Also, 21% of the accidents involved vehicles skidding.

4.40 Other notable factors are that 129 of the accidents (representing 31%) involved at least one young driver or rider aged 23 years or under and that 30% of the accidents involved vulnerable road users i.e. pedestrians and the riders of 2-wheeled vehicles. The numbers of such accidents are listed below.

	Number of PIAs				
	Under 17	17-18	19-20	21-23	TOTAL
Pedal cyclist	2	4	2	2	10
Motor cyclist	6	3	5	5	19
Car/LGV driver	0	25	33	41	99
HGV 7.5t	0	0	0	1	1
5 year period total	8	32	40	49	129

Figure 4j: PIAs involving young drivers/riders

	Number of PIAs			
	Pedestrian	Pedal cyclist	Motor Cyclist	Total
Year 1 to end Feb'09	2	10	13	25
Year 2 to end Feb'10	4	10	17	31
Year 3 to end Feb'11	2	6	6	14
Year 4 to end Feb'12	9	8	12	29
Year 5 to end Feb'13	6	7	9	22
5 year period total	23	41	57	121

Figure 4k: PIAs involving at least one vulnerable road user

4.41 For the purpose of more detailed analysis, the accidents have been split by area and shown in the table below. Significant clusters within each area have then been identified and studied in greater detail. These have been compared against expected accident rates from COBA for that particular type of road layout and volume of traffic.

	Slight	Serious	Fatal	Total
Area 1: M4 Junction 15	32	4	0	36
Area 2: A419, M4 to White Hart	27	1	1	29
Area 3: White Hart Roundabout	41	2	0	43
Area 4: A419 North of White Hart	29	7	0	36
Area 5: A420 E of A419	18	2	0	20
Area 6: A4312 W of A419	65	4	0	69
Area 7: A361 E of A419	27	0	0	27
Area 8: Other - main roads	78	20	1	99
Area 9: Other - unclassified	48	11	0	59
5 year period total	365	51	2	418

Figure 4l: All PIAs by area

Area 1/Cluster 1 – M4 Junction 15 Interchange with A419 and A346

4.42 This area of the road network covers the whole of the M4 Intersection with the A419 and A346, including the immediate junction approaches and is shown as "Area 1" on Figure 1.

4.43 A total of 36 PIAs are recorded at this location during the 5-year study period, equivalent to 7.2 PIAs per annum. Of these, none resulted in fatal injury but four resulted in serious injury, three to motorcyclists. These serious accidents do, however, appear to be on a falling trend with two occurring during the first 12 month period and one in each of the two subsequent years and none in either of the most recent two years.

4.44 There were a total of 52 casualties, averaging 1.4 per PIA.

4.45 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 7.3 PIAs per annum which compares with an observed rate of 7.2. This suggests that the junction is performing marginally better than might be expected and that overall the results reported give no particular cause for concern.

4.46 The only notable feature of the accidents is that 25 or 67% of the accidents involved rear end shunts. Thirteen of these occurring on the A419 southbound approach and entry to the roundabout.

Area 2 – A419 between M4 Junction 15 and White Hart Roundabout

4.47 This section of the road network covers the whole of the A419 between the M4 motorway to the south and White Hart Roundabout to the north.

4.48 A total of 29 PIAs were reported at this location during the 5-year study period, equivalent to 5.8 PIAs per annum. Of these one resulted in fatal injury and eight resulted in serious injury. The accidents do, however, appear to be on an overall falling trend with 6, 7, 10, 2 and 4 occurring in each of the 12-month periods studied.

4.49 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.141 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177, which compares with an observed rate of 0.067. The overall accident record, therefore, would appear to be relatively good.

4.50 Notable factors amongst the 29 reported accidents are that 55% of them involved rear end shunts and 52% involved at least one vehicle skidding. 10% involved HGVs.

4.51 One fatal accident occurred in April 2010 just north of Commonhead Roundabout when a pedal cycle travelling south with no lights was hit by a heavy goods vehicle (HGV), displacing the rider who was then run over by the HGV and three following vehicles. The fact that the rider was wearing dark clothing and displaying no lights is mentioned as a contributory factor in the accident although lighting conditions were not given as being "dark".

Area 3 – A419 junction with A420 White Hart Roundabout

4.52 At this Intersection 43 PIAs were reported including two that resulted in serious injury.

4.53 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 4.6 PIAs per annum which compares with an observed rate of 8.8. This suggests that the junction is performing worse than might be expected in terms of safety.

4.54 By far the most notable pattern amongst these accidents is that 88% of them involved rear end shunts, mainly at the three major entries to the roundabout:

- From A419 southbound off-slip – 15 PIAs, all involving rear end shunts,
- From A420 westbound – 16 PIAs, all but one involving rear end shunts,
- From A420 Merlin Way – 9 PIAs, all but one involving rear end shunts.

4.55 The two principal causation factors given for these accidents are, "failed to look properly" and "failed to judge other persons path or speed". There is clearly an issue at this junction concerning drivers not slowing sufficiently on the approaches and/or judging correctly when it is safe to proceed.

4.56 Proposals outlined elsewhere in this report to signalise the entries onto White Hart Interchange, where this problem currently exists, is likely to have safety benefits in that with a more formalised junction control drivers' decisions are made easier and there is less room for misjudgement to occur. The detailed design process will consider the accident history of this junction and will include any necessary features.

Area 4 – A419 between White Hart Roundabout and Blunsden

4.57 This section of the road network covers the A419 between White Hart Roundabout and Blunsden to the north.

4.58 A total of 36 PIAs were reported at this location during the 5-year study period, equivalent to 7.2 PIAs per annum. Of these seven resulted in serious injury. Although spread over the length of road in question, five of these accidents did occur during the hours of darkness and three involved HGVs. The accidents do, however, appear to be on an overall falling trend with 12, 9, 5, 7 and 3 occurring in each of the 12-month periods studied.

4.59 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.141 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177, which compares with an observed rate of 0.076. The overall accident record, therefore, would appear to be relatively good.

4.60 Notable factors amongst the 36 reported accidents are that 56% of them involved rear end shunts and 42% involved at least one vehicle skidding. 22% involved HGVs.

Area 5 – A420 east of White Hart Roundabout

4.61 This section of the road network covers 3.0km of the A420 between the A419 White Hart Interchange and the county boundary to the east.

4.62 A total of 20 PIAs were reported during the 5-year study period, equivalent to 4.0 PIAs per annum. Of these none resulted in fatal injury but two resulted in serious injury.

4.63 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.309 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177 which compares with an observed rate of 0.168. The overall accident record, therefore, would appear to relatively good.

4.64 Notable factors amongst the 20 accidents are that 35% of them involved rear end shunts and 35% occurred on a wet road surface. Seven of the accidents involved vulnerable road users: one pedestrian, four pedal cycles and two motorcycles.

4.65 Three small clusters have been identified on this section of road. They are small but in view of their proximity to the development site they have been analyses separately in order to identify any addressable issues. These are shown on Figure in Appendix B as Clusters 1, 2 and 3.

4.66 **Cluster 1 –A420 Gablecross Roundabout:** Four accidents were reported at Gablecross Roundabout, one of which resulted in serious injury to a pedal cyclist when a car entered the junction from Thornhill Road, failing to give way to the pedal cycle, which was already on the roundabout. Another of the accidents involved a pedal cyclist travelling west who failed to give way to a car already on the roundabout. Another accident involved a motorcyclist travelling east who hit a car whilst changing lane on the exit from the roundabout. The fourth one involved and was caused by an unmarked police car also changing lane on the eastbound exit. There is a cluster of accidents at this junction that involved vulnerable users. This junction has been identified to be signalised that will include controlled crossing points that will mitigate any increased risk of accidents.

4.67 Although three of the four accidents involved vulnerable road users there are no clear common factors to suggest any particular problem.

4.68 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 1.4 PIAs per annum which compares with an observed rate of 0.8 which is considerable lower than this.

4.69 **Cluster 2 –A420 in vicinity of Esso Service Station:** Five slight injury accidents were reported in the vicinity of the Esso service station between Gablecross Roundabout and the priority junction with Vicarage Lane. One of the accidents involved a single vehicle losing control on an icy road surface. Three of the others involved a rear end shunt in slow moving traffic, two eastbound and one westbound. The fifth accident involved a vehicle overtaking a slow moving vehicle and hitting a tractor-with-trailer turning right ahead of it.

4.70 None of the accidents appear to have resulted directly from activity associated with the filling station but all are a by-product of traffic slowing down in the general vicinity.

4.71 **Cluster 3 –A420 junction with Vicarage Lane:** Three slight injury accidents were reported at the junction of Old Vicarage Lane with the A420. One of the accidents involved a rear end shunt on the eastbound approach to the junction. The other two both involved single vehicles, one losing control turning left into Old Vicarage Lane on an icy road surface and the other whilst swerving to avoid an animal in the road. These latter two accidents resulted from unusual conditions and there does not appear to be a problem specifically related to this junction.

4.72 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 0.7 PIAs per annum, which compares with an observed rate of 0.6, indicating that the junction is performing within acceptable parameters.

Area 6 – A4312 and A4259 west of A420

4.73 This 3.5km section of the road network covers the A4312 between the A419 in the east and Drakes Way in the west, and the A4259 just south of the A4312.

4.74 A total of 69 PIAs were reported at this location during the 5-year study period, equivalent to 13.8 PIAs per annum. Of these none resulted in fatal injury but four resulted in serious injury. These serious accidents do, however, appear to be on a falling trend with one, three, zero, zero and zero in each of the 12-month periods.

4.75 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.717 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.537 which compares with an observed rate of 0.530. The overall accident record, therefore, would appear to relatively good.

4.76 Notable factors amongst the 69 reported accidents are that 46% of them involved rear end shunts and 35% involved drivers/riders aged 23 years of age or under. 29% of the accidents involved vulnerable road users: two pedestrians, six pedal cycles and 12 motor cycles. Further analysis highlights that 11 of the accidents involved both at least one young driver/rider and at least one vulnerable road user. The most notable common factor was 7 accidents involving motor cyclists aged 23 or under.

4.77 There are, however, no notable clusters of accidents along the route or other common characteristics indicating specific problem areas or addressable contributory factors.

Area 7 – A361 east of the A419

4.78 This section of the road network covers the A361 between the A419 and 3.0km to the east.

4.79 A total of 27 PIAs were reported at this location during the 5-year study period, equivalent to 5.4 PIAs per annum. Of these none resulted in either fatal or serious injury.

4.80 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.309 PIA/mvkm or based on "Road Casualties GB 2012 Table 10003" would be 0.177 which compares with an observed rate of 0.250. The overall accident record, therefore, would appear to fall somewhere between the two and does not give any particular cause for concern.

4.81 The only notable factor amongst the 27 reported accidents are that 37% of them involved vulnerable road users: zero pedestrians, four pedal cycles and six motor cycles. There are no notable clusters of accidents along the route.

Area 8 – Other main roads

4.82 This area covers all main roads of the study area that are not included in the sections above.

4.83 A total of 99 PIAs were reported on these roads during the 5-year study period, equivalent to 19.8 PIAs per annum. One accident resulted in fatal injury and 20 in serious injury.

4.84 Notable factors amongst the 99 reported accidents are that 44% of them involved vulnerable road users: nine pedestrians, 14 pedal cycles and 17 motorcycles.

4.85 One fatal accident occurred in July 2012 on B4141 Kingsdown Road when a van travelling north west from the college entrance, turned left and then across the main road into the path of a motorcyclist travelling south west, displacing the rider who sustained fatal injuries.

4.86 There are no notable clusters of accidents on these roads and they are not likely to be subjected to any significant increase in traffic as a result of the development.

Area 9– Other unclassified roads

4.87 This area covers all other roads of the study area that are not included in the sections above.

4.88 A total of 59 PIAs were reported on unclassified roads during the 5-year study period, equivalent to 11.8 PIAs per annum. No fatal accidents were reported but eleven resulted in serious injury, eight of these occurring between 8pm and 3am.

4.89 Other notable factors amongst the 59 reported accidents are that 51% of them involved vulnerable road users: nine pedestrians, 12 pedal cycles and nine motorcycles. This high percentage of accidents involving vulnerable road users is indicative of the mixed use of these roads by different types of road user, for a multitude of local and longer distance activities.

4.90 There are no notable clusters of accidents on these roads and they are not likely to be subjected to any significant increase in traffic as a result of the development.

Injury Accidents Involving Vulnerable Road Users

4.91 The initial analysis of road injury accident data that was carried out for the study highlighted the fact that there were several accidents that involved vulnerable road users. This section reviews in more detail those accidents, to determine whether any significant common factors exist. This may help identify any particular remedial intervention that might ameliorate the situation.

Pedestrian Accidents

4.92 23 accidents resulting in injury to pedestrians were reported during the five-year study period (6% of the total). None of the accidents resulted in fatal injury but ten of the 23 (43%) resulted in serious injury. The locations of the accidents are shown in the Appendix.

4.93 The number of pedestrian accidents per year shows a generally upward trend, with 2, 4, 2, 9 and 6 in each of the five years respectively. The most notable feature of the accidents is that 7 (30%) involved children, 4 aged under 11 years and 3 aged 12-15 years. The only other notable feature is that 8 (35%) of the accidents occurred during the evening, between 7pm and 11pm.

4.94 One notable observation is that four of the accidents occurred after a driver had exited their vehicle (and therefore become a pedestrian) following another incident.

4.95 The locations of the accidents are shown in the Appendix.

4.96 The only significant cluster is five accidents in the vicinity of Kingsdown School in Upper Stratton. Four of these involved children, aged 10, 12, 13 and 15 crossing the road and failing to give way to oncoming traffic, with three resulting in serious injury. The locations, again, are fairly disperse and no notable site-specific problems are indicated by the data.

Pedal Cycle Accidents

4.97 41 accidents resulting in injury to pedal cyclists were reported during the five-year study period, 10% of the total. One of the accidents resulted in fatal injury when a pedal cyclist without lights travelling south in the dark on the A419 just north of Commonhead was hit by a lorry. Another six accidents resulted in serious injury. The locations of all of the accidents are shown in the Appendix.

4.98 The accidents appear to be slightly downward trending but there no obvious common or significant factors. There do, however, appear to be a number of small clusters, each of which is discussed in more detail below.

4.99 **Cluster 1 : Four accidents on B4006 in vicinity of Kingsdown School** - This cluster is not particularly significant other than it is at the same location as the clusters of pedestrian accidents that has already been identified. Four accidents were reported, one of which resulted in serious injury. Only one of the cyclists was of school age i.e. 16. There are no clear common factors amongst the reported characteristics of the accidents. The analysis indicates that the majority of the accidents did not include school age children and there was less than one accident per year. The delivery of the development is unlikely to alter the characteristics of the traffic flow such that an increase in accidents is likely.

4.100 **Cluster 2 : Five accidents in vicinity of A4312 Drakes Way junction with Queens Drive -** A further two accidents which initially appear to be part of this cluster are in fact on a parallel side road, and have been excluded from this analysis. Four of the remaining accidents occurred in fairly close proximity but with no other common characteristics. It is noted, however, that 2 of the cyclists were cycling on the pavement. No children (aged 16 and under) were involved in the accidents reported at this location.

4.101 **Cluster 3 : Five accidents on A4312 Oxford Road south west of White Hart Roundabout** - The five accidents occurred at different locations under different circumstances. It is noted, however, that 3 of the cyclists were cycling on the pavement. No children (aged 16 and under) were involved.

4.102 **Cluster 4 : Four accidents on A420 east of White Hart Roundabout -** The grid references given for two of these accidents place them at White Hart Roundabout but in fact the descriptions all mention Gablecross. No young cyclists were involved and in two of the accidents the cyclist was at fault. In the other two, including the serious injury accident, a vehicle entering the roundabout failed to give way to a cyclist already on the circulating carriageway. Three of the accidents occurred on a Monday although it is difficult to attach any significance to this. Three occurred between 1630 and 1800 in the evening but only two of these on weekdays, one on a Saturday. Again, there are no clear patterns indicating specific problems.

4.103 **Cluster 5 : Four accidents at the Merlin Way roundabout with Covingham Drive** - All four accidents here occurred as a result of a vehicle pulling out of Covingham Drive and failing to give way to a cyclist entering the roundabout from Merlin Way south. Three of these occurred during a weekday morning peak hour when drivers may have been looking into the sun (one accident mentions dazzling as a possible contributory factor). Otherwise, visibility is good and generally free of obstructions.

Vulnerable Road User Summary

4.104 A more detailed analysis of road injury accidents involving pedestrians and pedal cyclists has identified a number of clusters of accidents but has failed to determine any particular common factors that indicate any specific problems that might be susceptible to remedial intervention.

4.105 There have, however, been a total of 9 accidents in 5 years involving vulnerable road users in the vicinity of Kingsdown School in Upper Stratton. Five children (aged between 10 and 16 years), averaging one per year, were injured on the roads surrounding the school during the study period.

Accident Summary

4.106 418 personal injury accidents were reported to have occurred within the study area during the most recent 5-year period for which information is available at the time of writing. This included two fatal accidents, both involving riders of two-wheeled vehicles. Overall there does appear to be a quite high proportion of rear shunt type accidents, and in several cases, accidents involving vulnerable road users, although this is not necessarily atypical of urban roads serving a mixture of purposes in a living environment.

4.107 The area has been split into several sections for further analysis and the numbers of accidents reported has been compared with statistics indicating expected rates for each type of road or junction given observed traffic levels. Overall the accident record appears to be quite good with observed figures generally below the equivalent expected rate.

4.108 The one notable exception to this is at White Hart Roundabout where a disproportionate number of rear shunt type accidents have occurred on three of the approaches to the junction. Proposals outlined elsewhere in this report to improve the capacity of this junction are likely to largely address this problem in the future by introducing signal control on two of the entries. These proposals will formalise the operation of junction through the introduction of traffic signals and therefore remove the decisions associated with drivers having to give way, simplifying their thought processes and thereby reducing errors of judgement.

4.109 To summarise, there are no notable safety issues that will remain unaddressed by the current development proposals. Although the development will add traffic to the network there is no reason to suppose that his will significantly compromise the relatively safe performance of the existing road system.

5 Development Proposals

Development Proposals

5.1 Full details of the development proposals are outlined in Section 2 of this document.

Development Timescales

5.2 The timescales for development delivery are dependent on many factors, including the planning process and future market demand for housing. This quantum of development will be constructed over a number of years, potentially circa 10 - 15 years. Therefore the development is likely to be built in several phases with points of access delivered progressively, as necessary.

5.3 The phasing of the development is linked to the timings of the access strategy, this is discussed below.

Transport Strategy

5.4 To create a sustainable development it is fundamental to the process that the assessment considers how the future residents will access the Eastern Villages through all modes of transport. A sequential approach is to be followed, as detailed below:

- **Encouraging environmental sustainability:** Reducing the need to travel, especially by car
- **Managing the existing network:** Making best possible use of existing transport infrastructure
- **Mitigating residual impacts:** Initially through improvements to the local public transport network, and walking and cycling facilities, and then through provision of new or expanded roads.

Development Phasing and Access Strategy

5.5 As the development will be progressed over some time, it will be necessary to phase infrastructure delivery to meet the requirements of the growing community. The development will be built in several phases, with points of access onto the existing highway network being delivered as necessary. The principal points of access are indicated illustratively below and shown in greater detail in the later sections of this report.

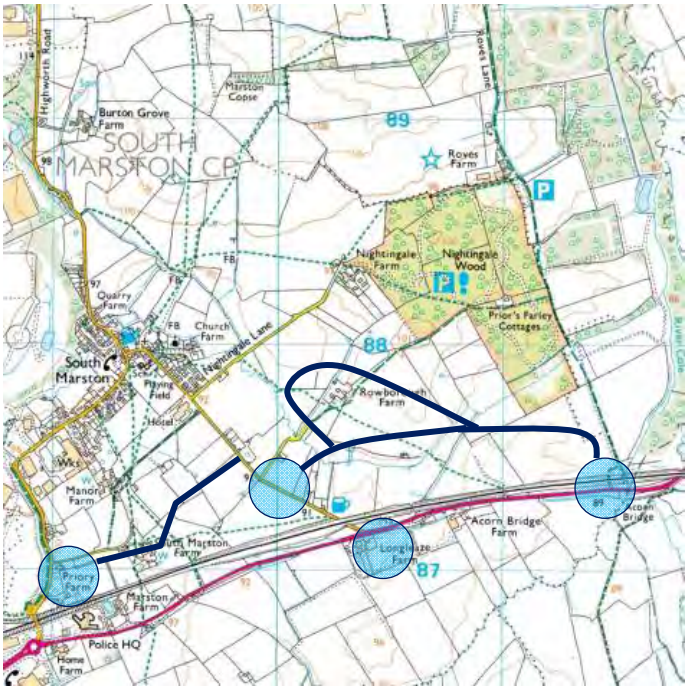

Figure 5a: Phase 1 access junction locations

5.6 This development has been identified as Phase 1 of the wider EV development, with Phase 1A and Phase 2 considering the latter development stages. It is typical that larger developments are often phased over several years. The following paragraphs identify a series of indicative infrastructure phases that are proposed to support Phase 1 of the proposed development.

Phase 1i

5.7 At the time of writing, it is envisaged that the development will commence from the western boundary of the site with access taken from Thornhill Road via a new compact roundabout. It is envisaged that this could serve the first 200 – 300 dwellings at South Marston.

Phase 1ii

5.8 The horizontal alignment of Old Vicarage Lane will be amended to provide an access into the western half of the site. This will result in Old Vicarage Lane giving way to the main line flow which will be towards the development. This will have the benefit of reducing any desire line towards South Marston.

5.9 As development progresses, the connection of the access points onto Old Vicarage Lane and Thornhill Road will allow completion of the 500 dwellings identified to the south of South Marston by providing a second access point for each new home in the expanded South Marston.

Phase 1iii

5.10 Access to the Rowborough phases of development will be taken from the eastern side of Old Vicarage Lane via a roundabout identified as part of Phase 1ii, and as described in the work completed by JMP to support the South Marston SPD, together with a new priority junction further to the south. The horizontal alignment of Old Vicarage Lane will be amended to result in Old Vicarage Lane to the north giving way to the main line flow, which will be towards the development.

5.11 Access to Rowborough is also to be delivered to the east of the development, on the A420, via a signal controlled priority junction. The phasing of the connections back to the A420, including the timing of the delivery of the rail under bridge at the eastern end of the site, will be a matter for discussion with SBC.

5.12 It is typical that larger developments are often phased over several years. The following paragraphs identify a series of indicative infrastructure phases that are proposed to support the proposed development.

South Marston Phase

5.13 At the time of writing, it is envisaged that the development will commence from the western boundary of the site with access taken from Thornhill Road via a new roundabout. It is envisaged that this could serve the first 200 – 300 dwellings at South Marston.

5.14 A further link will be provided from Thornhill Road through to Old Vicarage Lane which will then allow completion of the 500 dwellings identified to the south of South Marston. This will deliver two points of access into the South Marston development. This access arrangement will provide equitable route choice as recommended in Manual for Streets. This will distribute trips to the most appropriate routes to the A420, and provide a cohesive access strategy for servicing and emergency purposes.

Rowborough Phase

5.15 To provide access to the Rowborough area, the horizontal alignment of Old Vicarage Lane towards its current junction with the A420 will be amended to provide direct priority access into the western half of the site. This will result in the old route of Old Vicarage Lane giving way to the main line flow, which will now be towards the Rowborough development in the east. This will have the benefit of reducing any desire line towards South Marston and will give access the initial housing at Rowborough and reinforcing the primary desire immediately to the A420.

5.16 A further access to the Rowborough phases of development off Old Vicarage Lane is required to support the progressive development of the proposals from the west of Rowborough. The final access strategy includes for another additional access to the east of the development forming a new access under the railway. This access strategy for Rowborough was identified through the initial transport scoping discussions with Swindon Borough Council and Highways England.

5.17 The access strategy is indicated and described below.

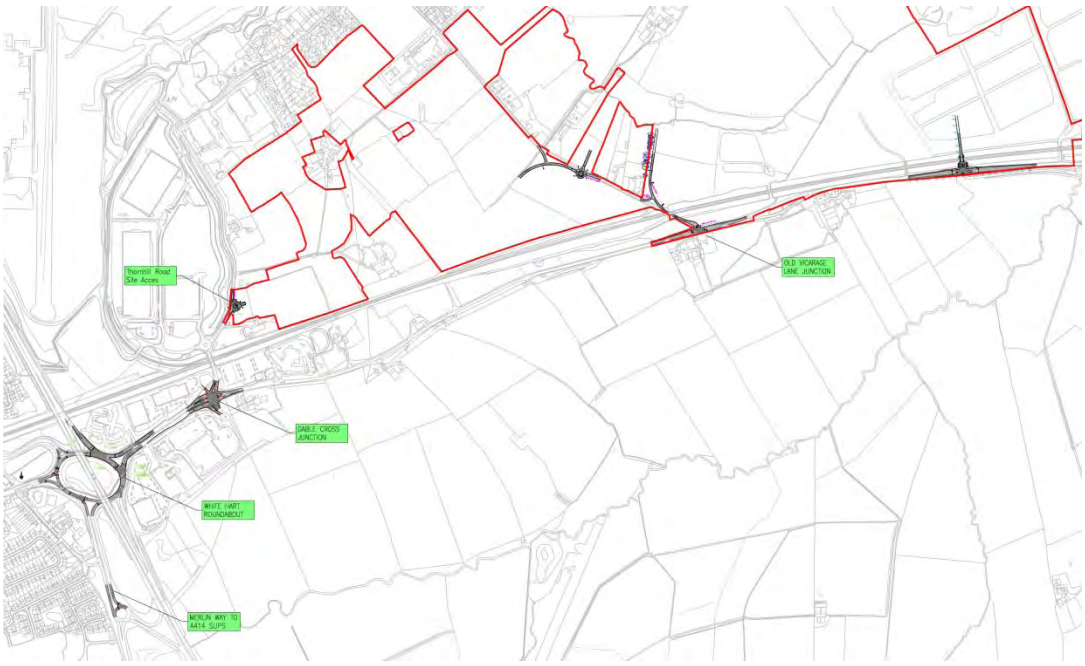

Figure 5b: Development wide access strategy

5.18 The second Rowborough access is therefore taken from the eastern side of Old Vicarage Lane via a roundabout. This will allow:

i. Timely Progression of development
ii. Opportunities for enhanced public transport
iii. Opportunities for enhanced walking and cycling on lesser trafficked roads.

5.19 Within the historic JMP Part 2 report and Technical note of 2012, concerns are raised regarding the potential for rat running from two points of access on Old Vicarage Lane and rat running along the new development between Thornhill Rd and Old Vicarage Lane. This Report and Technical Note have been outdated by several amendments to the underlying Paramics traffic model to which they relate and rely. The latest models post 2012 (latest adjusted modelling being through the summer of 2014) clearly demonstrate that the amended junction at Gable Cross, enables the primary route along the A420 in peak times to be significantly quicker and therefore more attractive route. In particular and to answer the conclusions of the note of 2012:

'2.7 ….., the provision of a link road to the south of an expanded village and connecting Thornhill Road with Old Vicarage Lane was also previously suggested in order to reduce traffic volumes in the village. Modelling indicated that if this link road was extended to Rowborough then the resultant delays and queues experienced at the Keypoint and Gablecross roundabouts would increase significantly.

5.20 In the latest modelling of 2014 this is simply not the case. It must be understood that the modelling of 2012 was thoroughly investigated both by Brookbanks and by an independently engaged Transport Consultant Arups. The model was found to contain many flaws and errors and was amended substantially through the summer of 2014. Errors/flaws were found to be but not limited to:

- Poor signal optimisation
- Irregular coding on junction features
- Vehicles not using lanes available to them
- Junction priorities not set correctly leading to poor junction use
- Over conservatism on junction calibration

5.21 With the proposed infrastructure improvements at Gablecross and the surrounding area, the queuing in the "do something" (with connections into Rowborough) scenario compared with the "do nothing" scenario shows significant reductions.

2.8. In addition, the queues and delays experienced on Old Vicarage Lane would also increase because of the increase in opposing flows where Old Vicarage Lane met the new road. Part two concluded that from a road hierarchy point of view, the A420 was considered to be the more suitable route for Rowborough movements rather than the provision of a new link road skirting the southern edge of the expanded village.

5.22 As discussed previously the latest adjusted modelling of 2014 demonstrates that the A420 to Gable Cross has significantly better journey times than the original model on which JMP have based their findings, when compared to routes either through the new development road to the south of South Marston or South Marston itself. The main access from Rowborough along the A420 will become the main route and have priority to the junction of Old Vicarage Lane giving direct access to the A420. The second access from Rowborough on to the roundabout with a development road would offer no benefits over traffic using the primary route onto Old Vicarage Lane down to the A420 route, which has been established as the quicker journey time. It can also be demonstrated that the second route gives more sustainable Public Transport routes to Rowborough. This would be through a "true loop" route through the new development road, then to the northern access point of Rowborough by the roundabout, then looping out on the southern access with priority to the

A420. This avoids having to travel along the same roads twice in short succession which only one access to Rowborough would give. Therefore two access points maximises potential for uptake.

2.9. A further disbenefit was that a link road to the south of the new development would create a physical barrier for pedestrians and cyclists and would result in single-sided development along an artificially lit highway edge. Hence it was concluded that this would adversely impact on the rural character to the south of the village and could potentially make the village more vulnerable to further encroachment to the south'.

5.23 With an appropriate network of footways and footpaths with properly demarcated crossing points offering pedestrian dominance there is no reason why the development road would form any barrier as described above. Also to be considered here is the potential for the 20mph target speed as discussed in section 3 of this technical note would further reduce the dominance of the motor vehicle.

5.24 Within this updated Transport Assessment (TA) work, further detail is offered on the results of the new modelling which demonstrate that two points of access will not be an issue. In particular through an assessment of Pound Corner the assessment indicates that…"

Two of the locations identified report a GEH of greater than 5, which result from the reduction in vehicles, being at Thornhill Road and Highworth Road at Pound Corner. The decrease in flows at these locations occurs due to the context of the access proposals, viz the realignment of Old Vicarage Lane which encourages trips into the development. The modelling predicts that the flows approaching Pound Corner could reduce by up to 141 vehicles.

5.25 Furthermore, this TA indicates there will not be a significant impact on any of the arms at Pound Corner.

5.26 With reference to Gablecross, this TA indicates that in the morning peak, the majority of the arms predict an improvement to the queuing, with the A420 eastbound predicting a slight increase. This is not unexpected as the delivery of signal control will contribute to this as the nature of signals (red lights) will introduce the need for traffic to stop. However, this will not lead to an increase delay. The signalisation of this junction provides safe opportunities for traffic to exit from Thornhill Road, rather than to wait for a gap in traffic. This is demonstrated by a review of journey times through Gablecross. The journey times have also been reviewed, finding that the southbound journey time improves through the signalisation of the Gablecross junction in the morning and evening peak periods, both in 2016 and 2026. The improvement in journey times within the Thornhill Road / A420 corridor clearly indicates a benefit and this trip is not prejudiced by the delivery of the development.

5.27 A further access point to Rowborough is also to be delivered to the east of the development, on the A420, via a signal controlled priority junction. This will then allow the completion of the development.

5.28 The access strategy together with the internal road layout has been developed specifically to minimise the traffic routing through the proposed development and the existing South Marston village. The intention is to maintain the A420 as the primary east-west corridor and providing direct and ready connection to the A420 at Thornhill Road, Old Vicarage Lane, and from the new access towards the eastern end of the development. The main connections through the site have been designed as public transport routes providing effective public transport connectivity between South Marston and Rowborough to enhance the attractiveness of public transport for residents of the new development but also the existing community. The link between Thornhill Road and Old Vicarage Lane has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

5.29 If not carefully designed, Old Vicarage Lane could result in additional traffic through South Marston. To reduce the attractiveness of this route, the access strategy reduces the importance of this link by altering the priorities with the development routes. This will encourage trips direct into the development rather than South Marston as was envisaged by the JMP background studies. By providing the second access to the Old Vicarage Lane, traffic generated from within

Rowborough towards Old Vicarage Lane will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road.

5.30 Consideration has also been given to the wider objectives in relation to transport in South Marston as set out in the Draft SPD. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the emerging Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

5.31 To reduce the potential attractiveness for development trips to pass through South Marston, consideration should be given towards measures to influence route choice. This could likely include a signage strategy to highlight alternative routes and a range of traffic calming measures providing both traffic control and a better environmental conditions within South Marston.

5.32 While the traffic management improvements may be considered to be beneficial to the network, highway users and the local communities, it should be noted that transportation capacity is adequate with the highway capacity improvements alone. As such, traffic management improvements are betterments rather than measures that are strictly necessary for network operation.

Internal Highway Network

5.33 Within the site, the masterplan proposes a street network having a clear hierarchy. The masterplan is indicative and will be confirmed through Reserved Matters. This is illustrated in Figure 5a and descried below:

Primary route: A main streets through South Marston and through Rowborough will connect to the external access points. The design speed for the internal street is based on a speed limit of 30mph, although the aspiration of the development is to achieve lower speeds through careful design of the streetscape and public realm.

The main links through the development will be designed to cater for public transport vehicles. At the time of writing it is considered that no specific public transport priority measures will be required. The purpose of the main link is to distribute the traffic on to the secondary routes, keeping the main link free flowing. However, at the appropriate time, ahead of implementation, the design of the main link will be reviewed.

It is envisaged that pedestrian and cycle movements will be catered for through on and off carriageway provision.

Secondary Routes: Secondary routes are designed to penetrate the individual development blocks and cater for vehicles at the reduced speeds, which will be reflected in the design and appearance of these roads.

Tertiary Routes: These will be designed to penetrate individual housing clusters and will be designed to encourage lower vehicle speeds and could incorporate shared spaces between motor vehicles, pedestrians and cyclists. The aspiration is for design speeds of 20 MPH on tertiary and secondary routes, thereby affording priority to walking and on street cycle movements as well as enhancing the public realm.

Walking and Cycling Provision

5.34 Published good practice identifies five main requirements for pedestrian routes. Wherever possible these should be followed when planning for pedestrians within the proposed development:

- Convenience – follow desire lines without any undue deviation from route,
- Connectivity – link multiple origin and destinations,
- Conviviality – be pleasant to use,
- Coherence – be made legible through paving and/or signage,
- Conspicuousness – promote security and safety allowing pedestrians to see and be seen by others

5.35 The 'Guidance for Cycle Audit and Cycle Review' (The Institution of Highways and Transportation, 1998) determines five main requirements for cycle routes. It is highly crucial that these requirements are recognised if the promotion of cycling to the site as a viable and attractive alternative to car use is to be successful:

- Coherence: continuous and to a consistent standard,
- Directness: closely follow desire lines as much as possible,
- Attractiveness: in aesthetic as well as objective terms
- Safety: designed to minimise risks for cyclists and others; and
- Comfort: well-maintained smooth dry surfaces, flush kerbs and gentle gradients

5.36 Overall consideration should be given towards the former Commission for Architecture and the Built Environment (CABE) principles of inclusive design, as highlighted below:

- Inclusive: so everyone can use it safely, easily and with dignity.
- Responsive: taking account of what people say they need and want.
- Flexible: so different people can use them in different ways.
- Convenient: so everyone can use them without too much effort or separation.
- Accommodating: for all people, regardless of their age, gender, mobility, ethnicity or circumstances.
- Welcoming: with no disabling barriers that night exclude some people.
- Realistic: offering more than one solution to help balance everyone's needs and recognising that one solution may not work for all

5.37 The masterplan for the site will include numerous walking and cycling routes within the development to provide a comprehensive route network that will comprise both on and off road paths. This would include an integrated walking / cycling route adjacent to the main link road through the development. This would deliver the main spine through the development, from which spurs would then access the wider development. Highway crossing points will be designed to cater for all types of pedestrian users with the routes lit where appropriate.

5.38 There are several Public Rights of Public Way across the site, these will be incorporated within the site masterplan. To encourage walking and cycling these routes through the site are likely to be enhanced to ensure that there are no barriers to movement. Across the site the improvements would include the provision of adequate surfacing to reflect the characteristics of the area and lighting where appropriate. In areas adjacent to housing, this could result in illuminated tarmacked routes and in less built up areas more low engineered surfacing, including bonded gravel. The surfacing to be used will be appropriate to the type and quantum and usage for any given route.

5.39 The walking and cycling paths will connect the individual housing blocks into the main route through the site that will ensure full connectively and route choice throughout the development.

5.40 The strongest desire line for walking and cycling is westbound. The distribution of trips, as indicated later in this report is 74% towards Swindon. The established peak hour car based trips of 788 can be used to determine the likely pedestrian

/ cyclist demands using census data. The 788 car based trips divided by the modal split of 0.695 provides a total of 1,134 trips. Pedestrian / cyclist trips represent 0.097 of the total trips, equating to 110 walking and cycling trip in the peak hour. This demand demonstrates the need for a strong walking and cycling network, which has been incorporated into the proposals.

5.41 The onsite network will connect into the external walking and cycling networks that includes the PRoW network. The predominant walking and cycling desire line towards Swindon is to be fully incorporated into the links from the development.

5.42 Walking and cycling trips to the west will be encouraged to travel through the development to make use of the high quality environs that will be delivered. The onsite routes will link into the Thornhill Road exit. A footpath will be provided along Thornhill Road to connect into the Gablecross Junction. This junction is to be converted into a signalised crossroads which will offer safe crossing points for vulnerable users.

5.43 Currently, there is a high quality walking and cycling link adjacent to the A420 commencing from the Police headquarters junction heading in a westerly direction through the Gablecross junction to connect with the White Hart interchange. As indicated above, the development will provide a connection into the walking and cycling link adjacent to Thornhill Road, which in turn links with the A420. This will ensure that trips into the supermarket and into Swindon can be carried out.

5.44 The walking and cycling strategy plan is attached in Appendix D.

Road Based Public Transport Provision

5.45 To maximise the opportunities to travel by public transport, it is proposed to improve the current routes that operate in Swindon. The options to deliver public transport enhancements have been discussed with SBC and proposals delivered following various discussions with the local bus operator, Stagecoach. The development is likely to commence at Thornhill Road and proceed broadly in an easterly direction. Therefore a connection through the development into Old Vicarage Lane is unlikely to be delivered at the outset.

5.46 Presently, two routes operated by Stagecoach pass adjacent to the site boundary. It has been discussed that neither of the existing routes are suitable for augmenting to serve the proposed development. A bespoke route serving the development has therefore been identified in consultation with a local operator, Stagecoach.

5.47 The main desire line for public transport patrons is likely to be towards Swindon. Through the discussions with Stagecoach, a bespoke service has been identified that delivers the level of service frequency which will be attractive to future residents and serves the key destination. The bespoke service will deliver a 20-minute frequency from the outset. The service is operated as a stand-alone service, making it a great deal easier to operate reliably. This delivers a highly marketable service to new residents, at a frequency that gives the greatest scope to affect mode shift, without giving rise to a service that could not be realistically sustained by long-term revenue potential.

5.48 The service also benefits from offering a highly competitive attractive journey time to the town centre, including to the rail station, especially when end-to-end journey times are factored in. The service will operate into the evening to maximise scope for returning rail users.

5.49 Through these discussions with Stagecoach, it is considered that the delivery of the bespoke service is viable over the long term. It is expected that this route would require financial support at the outset, until the patronage levels are sufficient to self-fund the route. Stagecoach has provided the likely contribution trajectory, as indicated overleaf.

Timing	Contribution
Year one	£126,492[1]
Year two	£235,511
Year three	£201,911
Year four	£164,279
Year five	£127,991
Year six	£91,703
Total contribution	£947,887

Figure 5c: Public transport contribution trajectory

Note 1 - assumed that service starts halfway through year

5.50 The overall public transport strategy is attached in Appendix E.

5.51 The long term viability of any public transport routes is critical if it is serve the community into the future. Therefore discussions with Stagecoach has established the likely level of revenue that could be generated by the development, which can be offset by the likely costs to understand the viability. As a result of this work, financial support is likely to be needed initially, but over time it is expected that patronage levels will be sufficient to safeguard the long term viability of the proposed public transport interventions. The information contained in Figure 5c identifies the contribution considered by Stagecoach necessary prior to the public transport route becoming viable.

5.52 To determine if the identified contribution is reasonable, a further analysis has been carried out. BCL has reviewed TEMPRO data that provides details on the likely trip generation by mode for geographical areas over weekly period. TEMPRO data have been interrogated, with the results indicated below.

Time period	Number of trips - Origin	Number of trips - Destination
Weekday am 07:00-10:00	9,603	10,600
Interpeak 10:00 – 16:00	14,280	14,506
Weekday pm 16:00 -19:00	7,155	6.260
Weekday off peak 19:00 -07:00	2,983	2,849
Saturday	24,255	24,363
Sunday	7,798	7,796
Total week trips	202,158	171,965

Figure 5d: TEMPRO planning data – origin and destination

5.53 In total, the above shows amounts to 374,123 bus trips being generated from 82,915 residential dwellings. This equates to 4.5 public transport trips per week per household. Therefore it is expected that the development would generate the following trips during the build out period, together with the expected revenue and level of contribution required.

Year	Occupations	Total weekly trips	Annual Trips	Predicted income	Predicted operating cost	contribution
1	60	406	21,117	£21,117	£150,000	£128,883
2	180	1,218	63,350	£63,350	£300,000	£236,650
3	300	2,030	105,584	£105,584	£300,000	£194,416
4	420	2,843	147,817	£147,817	£300,000	£152,183
5	540	3,655	190,051	£190,051	£300,000	£109,949
6	660	4,467	232,284	£232,284	£300,000	£67,716
7	780	5,279	274,518	£274,518	£300,000	£25,482

Figure 5e: Public Transport Subsidy

5.54 The above demonstrates that the level of contribution assumed by Stagecoach, presented in Figure 5c is reasonable.

Alternative Transport Modes

5.55 Existing bus routes have been identified that currently serve the South Marston area and the commuter route along the A420. These services cannot be sensible augmented and therefore discussions have taken place with Stagecoach to provide a new 20 minute service within and serving the new development. Given that this is a new service, it is sensible to

try and achieve the most efficient route possible and a second point of access on to Old Vicarage Lane allows for a "true loop" from a Public Transport providers perspective. A "true loop" is considered a loop which minimises the use of the same roads and takes the shortest journey through the area which it serves. With the second access a "true loop" connecting both the South Marston and Rowborough sections of development to the proposed early service in green in figure 5f and then the later services in red can be achieved after all phases of development.

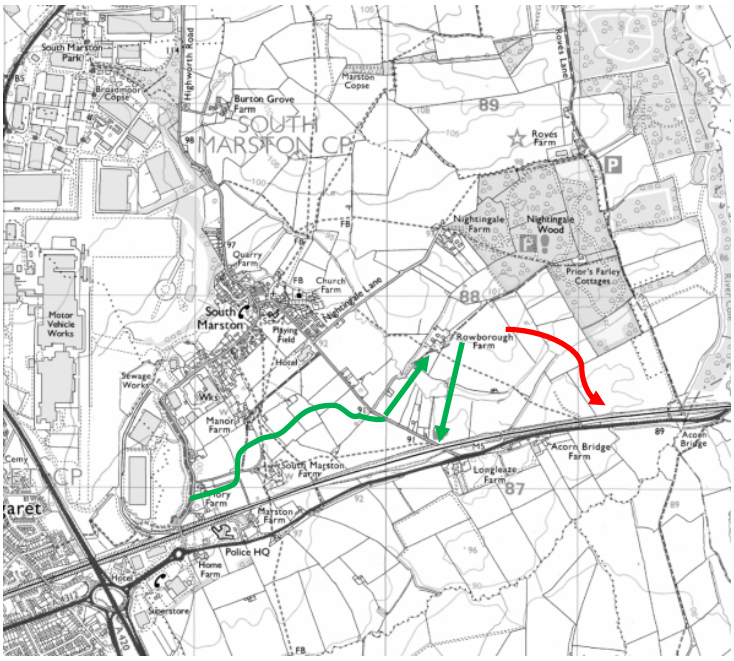

Figure 5f: Potential bus routes through the development.

5.56 Cycling and walking routes between the two areas would also be enhanced by this second route and connection giving more direct access between the two development areas and offering an alternative to the more attractive southern route for motorised vehicles to the A420.

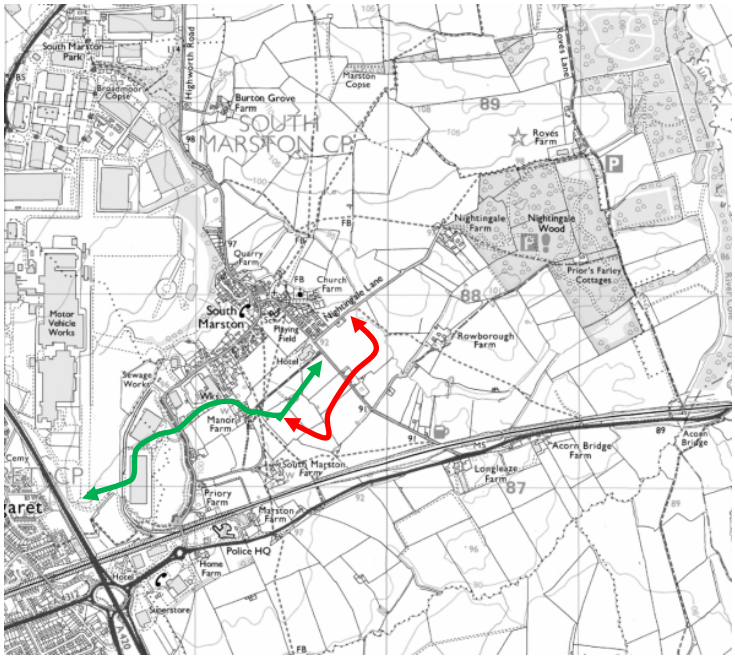

Figure 5g: Cycling and walking routes, green depicts two points of access and red with one.

Offsite Interventions

5.57 Through historical modelling of the transport impacts, various off site junctions were provisionally identified as requiring

mitigation. Further assessments have been completed to support the production of this Transportation Assessment, which supports and refines the requirements for off-site infrastructure interventions. These are identified below:

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road.
- **New Access onto A420:** New signal controlled junction at the east end of phase one to provide access from the A420.
- **Police HQ Signals:** Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire
- **Gablecross Roundabout:** Conversion to a signalled crossroads (this is a revised intervention from the original proposed intervention which was localised widening on approach arms)
- **White Hart interchange:** Signalisation off several arms of the existing interchange layout.
- **Merlin Way / A419 slip road roundabout:** Signalisation off several arms of the existing interchange layout (subsequent traffic modelling of the revised development layout together with anticipated traffic growth for the Swindon Borough raised that mitigation measures were necessary at this junction location).

5.58 This represents a comprehensive package of interventions delivered through, and arising from, the development. The detailed traffic modelling completed for this TA has also considered future development resulting from the wider growth proposals to the east of Swindon, as identified in the Local Plan. As would be expected, this work has identified further interventions that will be necessary to deliver the wider growth. However, the assessments demonstrates that delivery of this Phase I element of the growth does not prejudice the delivery of the subsequent phases. Such further interventions that have been anticipated in the Eastern Villages SPD for instance and include for diamond interchange at White Hart and the signalisation of the Greenbridge roundabout. The SPD envisages the phased implementation of wider transport improvements in accordance with the phased implementation of the EV development. The delivery of the identified interventions for Phase I does not prejudice the delivery of any longer term schemes.

Parking Provision

5.59 SBC has published guidance on parking standards, in which dwellings having four or less bedrooms shall have two spaces with all other dwellings having three spaces. This guidance was before the publication of NPPF, which at Paragraph 39 identifies that local planning authorities should take into account:

- Accessibility of the development
- The type, mix and use of development
- The availability of and opportunities for public transport
- Local car ownership levels
- An overall need to reduce the use of high-emission vehicles

5.60 The application is submitted in outline and parking numbers can be refined / defined at reserved matters stage. The parking strategy will however, reflect the primary focus of the transport strategy, which primarily seeks to target mode shift away from private car use. A flexible approach to parking design and provision will be adopted, through a provision of on-street, on-plot, courts and individual garages. At the time of writing it is envisaged that the parking provision will be delivered through a mix of allocated and unallocated parking with the SBC guidance on parking provision treated as maximum standards.

5.61 It is anticipated that every dwelling will have access to safe, secure cycle parking. It is assumed that garages, where provided, will provide suitable cycling parking and for the dwellings without garages, it is anticipated that secure facilities will be provided either within the building or in rear gardens.

6 Development Impact Appraisal

Impact Appraisal

6.1 The TA has considered the impact of new development using the principles set out in the New Approach to Appraisal (NATA). The impact of proposals are assessed in terms of the five NATA objectives for transport:

- Accessibility
- Safety
- Economy
- Environment
- Integration

Accessibility

6.2 The accessibility of the development is achieved through the successful forming of transport links from the development to the external transport routes such that a permeable layout is delivered that allows the future site occupiers to access the current range of local facilities and amenities by different modes of travel.

6.3 A qualitative review of the accessibility implications of the proposed development has been conducted. The existing level of access for cyclists and pedestrians between the proposed development and the surrounding transport system is described in Chapter 4.

Employment Facilities

6.4 Various employment opportunities are located in close proximity to the site, including South Marston Park, that comprises a site area of 93.35 Ha and Keypoint at South Marston, comprising a site of 24.54Ha. Further afield, other key employment destinations include the Great Western Hospital, Green Bridge Retail Park, Swindon Town Centre together with a range of industrial / trading estates. These offer a wide range of employment opportunities for the future residents.

6.5 The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6a, with the location indicated in Figure 6b.

Employment	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
1. South Marston Park	0.5km	✓	6	✓	2
2. Keypoint	0.5km	✓	6	✓	2
3. Great Western Hospital	5.7km		68		20
4. St Margaret's Retail Park	0.4km	✓	5	✓	1
5. Green Bridge Retail Park	2.4km		29	✓	8
6. Swindon Town Centre	5.0km		60	✓	18

Figure 6a: Distance to Employment Destinations

Figure 6b: Employment Destinations

Health Facilities

6.6 A range of health facilities are available within Swindon. The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6c, with the location indicated in Figure 6d.

Health	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
1. Elm Tree Surgery	6.0km		72		21
2. Eldene Surgery & Pharmacy	5.4km		65		19
3. Penhill Surgery	5.9km		71		21
4. Hawthorn Medical centre	5.0km		60	?	18
5. Kingswood Surgery	5.3lm		64		19
6. Park South Dental Practice	5.6km		67		20
7. Great Western Hospital	5.7km		68		20

Figure 6c: Distance to Healthcare Destinations

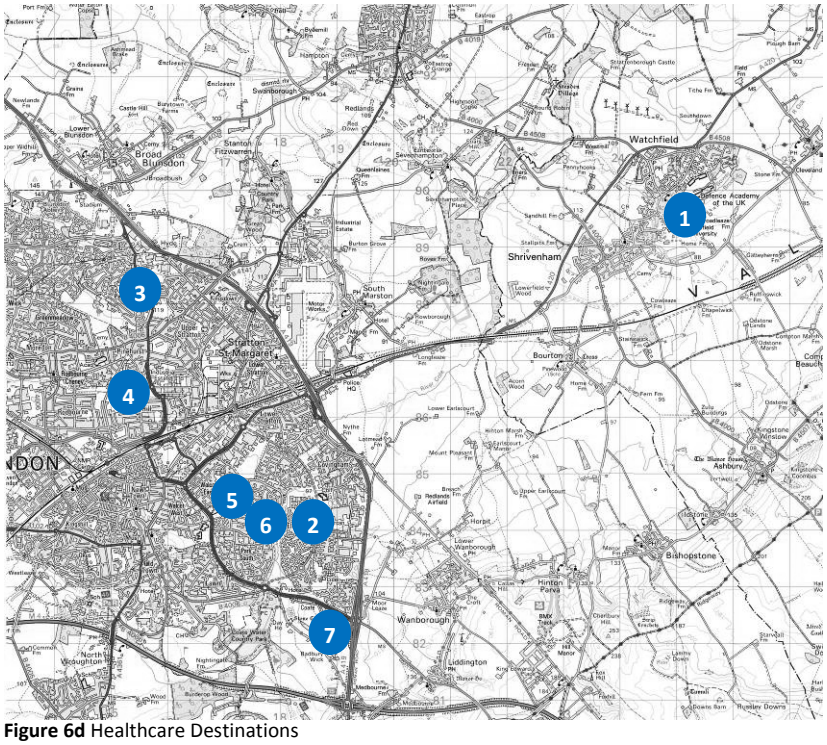

Figure 6d Healthcare Destinations

Education Facilities

6.7 There are a range of education facilities within Swindon. The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6e, with the location indicated in Figure 6f.

Schools	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
Primary					
1. Beechcroft	5.0km		60	✓	18
2. Colebrook	2.2km		26	✓	8
3. Covingham	2.4km		29	✓	8
4. Eldene	5.4km		65		19
5. Goddard Park	4.3km		52	✓	15
6. Grange	1.8 km	✓	22	✓	6
7. Liden	5.0km		60	✓	18
8. Nythe	2.3m		28	✓	8
9. South Marston	1.8km	✓	22	✓	6
Secondary Schools / Further Education					
10. Kingsdown	4.0km		48	✓	14
11. Dorcan College	3.3km		40	✓	12
12. Swindon New College	4.3km		52	✓	15
13. Swindon College	5.0kim		60	✓	18

Figure 6e: Distance to Education Destinations

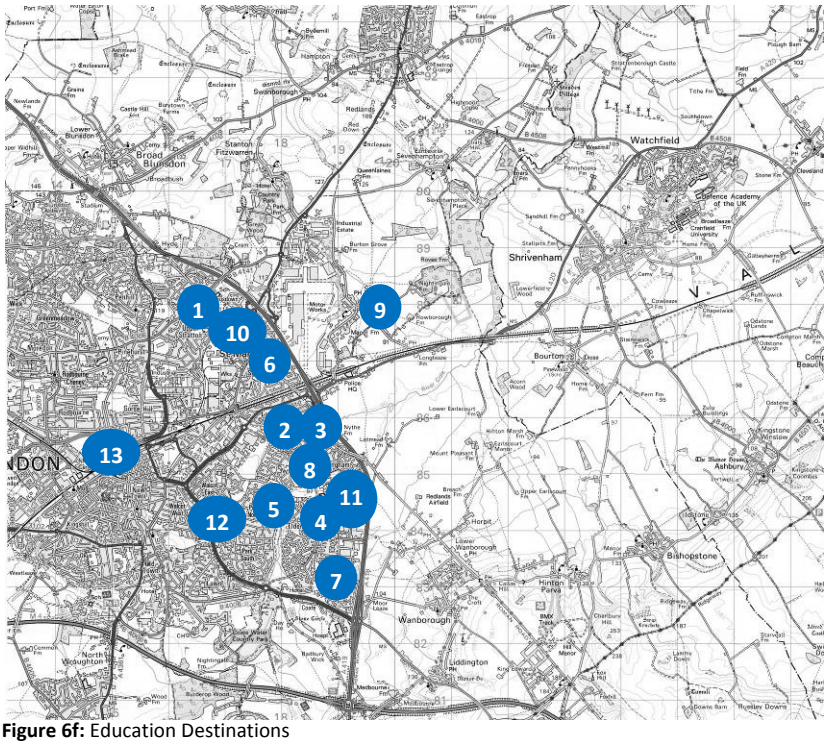

Figure 6f: Education Destinations

Miscellaneous Amenities

6.8 A full range of leisure and complementary amenities existing in the area of the development.

6.9 South Marston Village Hall provides a venue for Parish meetings and local community events and is owned and managed by the Parish Council and a local charity. The primary school also provides a playgroup facility for parents and their babies/toddlers. There is a youth club gardening/allotment club, walking club and business club. South Marston also has a church, two public houses and hotel and conference venue. A post office is available nearby as are a range of food and non-food retail outlets.

6.10 South Marston has a recreation area (2.34 Ha) located on the south eastern edge of the village, adjacent to the primary school. The village allotments are located at the northern edge of the village, on the Highworth Road.

6.11 Nightingale Wood, to the east of the site is a general recreation area and County Wildlife Site.

6.12 The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6g, with the location indicated in Figure 6h.

Shops & General Amenity	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
1. Train Station	5.3km		64		19
2. Bank	5.0km		60	✓	18
3. Greenbridge retail park	2.3km		28	✓	8
4. Supermarket	0.5km	✓	6	✓	2
5. Swindon Town centre	5.0km		60	✓	18
6. Post Office	2.3km		28	✓	8
7. South Marston	1.8km	✓	22	✓	6
8. Nightingale Wood	2.0km	✓	24	✓	7

Figure 6g: Distance to Key Local Amenities

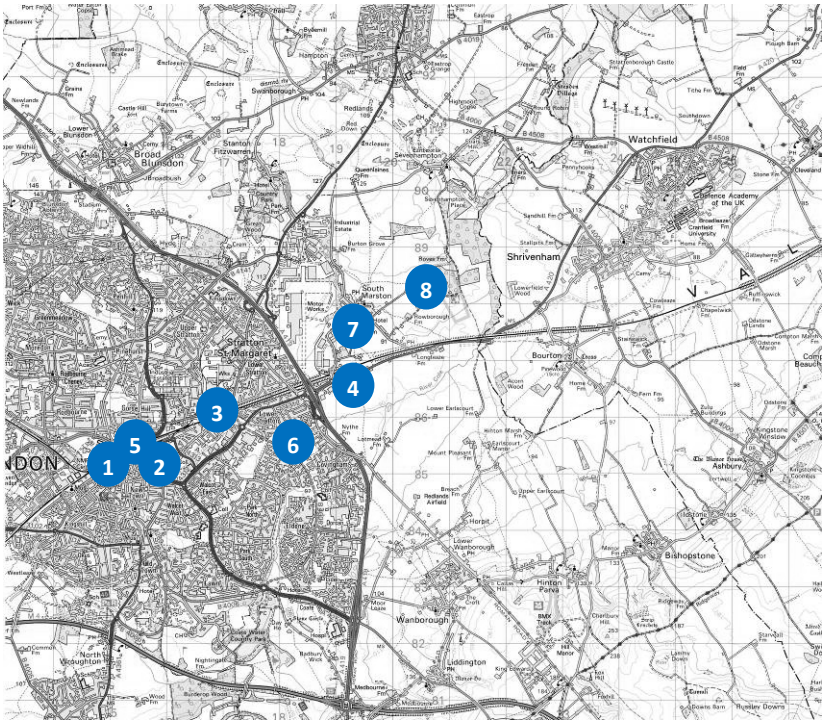

Figure 6h Education Destinations

6.13 It may be concluded that the development will have very good accessibility to a wide range of local amenities that will support the new and existing community. Figure 6i below provides a graphical representation of the 2km walking and 5km cycling isochrones, which the range of local amenities exist. The proposed development will not create any new accessibility barriers within the surrounding area. The range of facilities and services, including the provision made for education will also significantly improve as a result of the application proposals. A higher level of services and facilities and employment again will be provided within the wider Eastern Villages development most notably the District centre, employment and secondary education provision to the south of the A420 which would be within a 2 km walking isochrones of the site.

Figure 6i: Walking and Cycling Isochrones

Safety

6.14 With new developments comes the potential for increased risk of accidents in the immediate area, due to increased multi-modal traffic. The safety of the development is therefore achieved firstly by identifying the existing accident records and making changes as necessary to the highway network to mitigate any problems. Thereafter, the proposals must be designed to appropriate standards with safety reviews being conducted as necessary during the process. In respect of these requirements:

1. A review of the historical accidents has been completed that confirms there is no accident trend or risk that might materially be increased through the delivery of the development.
2. The proposals have been developed in line with recognised standards in the form of the Design Manual for Roads and Bridges and Manual for Streets. A Stage One Road Safety Audit will be completed with regards to the site access to ensure compliance with the relevant applied design standards.

Economy

6.15 The transport economic efficiency of the development is achieved in part through the successful delivery of a comprehensive transport access strategy that considers all modes of transport, to ensure journey reliability.

6.16 A key transport economic objective will be to minimise any detrimental impact upon private road users' journey time, reliability and travel costs, and to maintain or reduce public transport and non-motorised journey times. The former will be achieved through assessment of trip generation and its likely impact on the local road network, and the latter through a considered choice of mitigating measures; both will be discussed in a later section of this Transport Assessment.

6.17 It should be noted that the proposed development will deliver high quality housing in an area of Swindon that will increase the work force to maximise employment opportunities.

Environment

6.18 The transport environmental benefits of the development are achieved in part through the delivery of a sustainable transport strategy that encourages travel by walking, cycling and public transport and reduces the reliance of the single occupancy vehicle trip.

6.19 As a result of development proposals, local traffic generally increase and appropriate mitigation may be implemented to accommodate this effect. Later Chapters in this report highlight that in this case no such mitigation is required. No significant issues are apparent in relation to the environmental issues.

6.20 The development proposal is in line with transport planning policy. The Transport Assessment underlines areas in which the proposal supports local, regional and national planning transport policies as detailed in Chapter 3.

6.21 There will be no exacerbation of social exclusion resulting from the residential development since no existing travel movements will be cut off or hampered.

Summary of Site Accessibility

6.22 This section of the TA demonstrates that the proposed development site has a wide range of locational advantages in terms of site accessibility.

6.23 The site is in close proximity to Swindon and South Marston, which provide a range of amenities to serve daily needs both in relation to food retail and educational needs. The development will reinforce these services as necessary with the provision of a local centre, schools and health facilities. A higher level of services and facilities and employment again

will be provided within the wider Eastern Villages development most notably the District centre, employment and secondary education provision to the south of the A420 which would be within a 2 km walking isochrones of the site.

6.24 The development site will have excellent walking and cycling links into Swindon. Future residents will readily be able to access both road and rail public transport networks.

7 Travel Plan

Travel Plan Benefits

7.1 A Travel Plan (TP) is a management tool designed to enable the users of any site to make more informed decisions about their travel while minimising the adverse impacts of the development on the environment. This is achieved by setting out a strategy for eliminating the barriers keeping users of the site from using sustainable modes and managing single-occupancy car use.

7.2 This Transport Assessment is accompanied by a standalone Framework Travel Plan (FTP). The Transport Assessment should be read in conjunction with the FTP to fully understand the overall transport strategy for the site. The FTP has formed the basis for the sustainable transport strategy and FTP for the proposed development. The FTP will establish the overarching principles to be applied to ensure that the final TP will maximise modal shift. The TP was produced in discussion with the HE and SBC.

7.3 A summary of the Travel Plan is provided below.

7.4 If well-designed and properly managed, the implementation of a Travel Plan can lead to a decrease in the proportion of site users using private cars and an increase in the proportion using sustainable modes of public transport, including walking and cycling.

7.5 Travel Plans can also:

- Improve the environmental credentials of the proposed development
- Reduce the traffic impact on the local highway network
- Improve the health and well-being of all the site occupiers
- Reduce adverse impacts on local residents and businesses

Travel Plan Objectives

7.6 The Travel Plan for the development has several key objectives, as listed below:

- Reduction in the car based trips
- Reduction of Single Occupancy Vehicle (SOV)
- Reduction in congestion and pollution through reduced car use
- Improve the modal split of trips made by walking
- Improve the modal split of trips made by cycling
- Improve the modal split of trips made by public transport

Travel Plan Targets

7.7 The TP will need to establish mode share targets. These targets should be based on challenging, but achievable non-car and Single Occupancy Vehicle (SOV) mode share targets. The targets should be based upon current practice in the

hinterland around, and the location of, the site. The target will take account of the local geography and existing transport provision.

7.8 Based on the census information, the nature of the development and the expected travel characteristics, the target for the site will be to reduce the SOV by an ambitious 10% modal shift.

Travel Plan Coordinator

7.9 Research has shown that Travel Plans need to be managed by a Travel Plan Coordinator, who has a clear brief with dedicated resources to manage the Plan to ensure that's its objectives are met.

7.10 The Travel Coordinator role could be required for a period of 5 years.

7.11 The key responsibilities undertaken by the Coordinator as set out below will be reviewed and amended on a regular basis:

- Leading on the delivery of the TP once approved
- Coordinating the necessary data collection required to develop the 'Household Travel Pack' and 'Employee Travel Packs'
- Representing the 'human face' of the TP including liaison with residents' steering group or management committee
- Promoting the individual measures and packages
- Liaising with the relevant Council Public Transport Team
- Liaising with the Local Highway Authority over monitoring and reviews of the TP
- Assessing progress towards achieving mode-shift away from car use

Travel Plan Measures

7.12 The key to a successful TP is identifying the correct measures that will suit the future residents. It is unlikely that there will be sufficient attraction to a single measure and hence a combination of measures is the most appropriate approach to take.

7.13 In order to maximise the uptake of sustainable transport measures of the development, sustainable transport modes will be available and will be promoted during all stages of the development process from the design, construction and initial marketing of the development through to initial occupation and then on to full occupation of the site.

7.14 The key stages of the Travel Plan process integrate with the key stages of the development process as set out below:

- Before occupation of any dwellings - pre occupation
- During the period when dwellings are being occupied – during occupation
- After dwellings have been occupied – post occupation

8 Development Traffic Generation

Introduction

8.1 In the context of the Eastern Villages proposals, the primary objective of transport network modelling is to provide the tool to assess the effects of growth on the transport network and help inform the need for interventions to ensure the network operates satisfactorily into the future. Scoping discussions with representatives of the HE and SBC concluded that the impact of the development should be considered and appraised using the existing traffic network models for Swindon. These include the SATURN model identified as the Swindon Multi-Model Transport Model (SMMTM) and the A419/A420 PARAMICS strategic model.

8.2 The SMMTM is a validated and calibrated model for a base year of 2006 and covers the geographical area of Swindon. This is a SATURN model, which is suitable for high level strategic modelling. The A419 / A420 PARAMICS model is a micro-simulation model has been calibrated and validated to a more current base year.

8.3 It was agreed with representatives of the HE and SBC that the A419 / A420 PARAMICS provides the most appropriate model to test the effects of the development. However, as the SMMTM model covered a wider geographical area, it was also agreed initially to model development impacts using SATURN, with the results then being used to facilitate the PARAMICS model. This methodology provided the following advantages:

- The PARAMICS model can inherit all growth in traffic demand and committed development traffic including trip distributions across the network.
- The SATURN model enables the understanding of the wider reassignment impacts of background traffic that might arise from a large development such as the EV. These are then inherited by the PARAMICS model.
- The SATURN model enables the understanding of the wider impacts on the local road network, west of the A419, which the PARAMICS model does not cover.
- Where appropriate, the SATURN model could be used to establish trip distribution from the site for EV trips, which is then inherited by the PARAMICS model.
- This methodology retains consistency with all previous studies undertaken regarding the EV/EDA development by JMP using the PARAMICS model.

Assessment Scenarios

8.4 The following development / network scenarios were agreed with representatives of SBC and the HA:

- 2006 base year
- 2016 opening year plus committed developments
- 2016 opening year plus committed developments plus Phase I*
- 2026 future year plus committed developments
- 2026 future year plus committed developments plus Phase I*
- 2026 future year plus committed developments plus Phase I plus Phase IA*
- 2026 future year plus committed developments plus Phase I plus Phase IA Plus Phase II*

**** Note:*** The Phase I, IA and II development scenarios are outlined in Section 2 of this report.

Committed developments

8.5 Committed development assumptions to be included to create the 2026 future year from the 2006 base model have been agreed with representatives of SBC and the HA. The assumptions have been entered into the SMMTM variable demand model. The proposed development of 140 houses at the Crown Timber site in South Marston has also been added to the model to the Do Minimum SATURN highway model.

8.6 The introduction of the committed developments resulted in the inclusion of 18,147 houses, 10,938 jobs and 127,500 sqm of retail floor space.

8.7 The proposed development of 140 houses at the Crown Timber site in South Marston has also been added to the model to the Do Minimum SATURN highway model.

8.8 The list of committed developments is included within the Halcrow modelling report, attached in Appendix F.

Committed Infrastructure Schemes

8.9 The following network changes have been made to the SATURN highway model for the forecast year, in accordance with advice from SBC, to create the Do Minimum highway scenario:

- Blunsdon Bypass.
- Commonhead Flyover.
- Schemes associated with Wichelstowe, including the proposed new layout at M4 Junction 16, the signalisation of Croft Road/Wichelstowe Access and Croft Road/Pipers Way and the internal Wichelstowe road network.
- Changes associated with the Union Square development.
- Transfer Bridges and Bruce Street Bridges Great Western Way changes.
- The consented highway improvements associated with the Commonhead mixed development, including the proposed changes at M4 Junction 15, A419 and development access.
- Removal of the following car parks prior to 2016:
  - Cheltenham Street (102)
  - College (98)
  - Regents Close (12)
  - Princes Street (119)
  - Carlton (590)
  - Catherine Street (6)
  - Queen Street (8)
  - Queenstown (145)
- The incorporation of the following cars parks:
  - Union Square: 850 MSCP (short and long stay) and 694 residential and office use
  - Regent's Circus: 238 for food store and 212 for office and leisure
  - 1,000 spaces north of the railway line (long stay).
- The provision of bus services to serve the Wichelstowe development;
- Coding bus service 15A to the Northern Development Area (NDA) and extension into the Tadpole Farm development.
- Removal of the P&R bus service to Groundwell P&R.

Trip Generation

8.10 Trip generation for the proposed development has been agreed between BCL and representatives of SBC / HE by way of a Technical Note reference 203064 contained in Appendix F. The following tables summarise the agreed development trips. For Phase I, trips are shown in Figure 8a.

Trips	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Phase I trips	226	788	833	477

Figure 8a: Trip generation – Phase I

8.11 Trips for the agreed development profile for Phase IA have been developed and agreed with representatives of SBC / HE using TRICS. For Phase IA, trips are shown in Figure 8b.

Trips	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Phase IA trips	460	189	157	415

Figure 8b: Trip generation – Phase IA

8.12 In relation to Phase II, a similar methodology as agreed for Phase I has been applied, with the resultant trips indicated below.

Trips	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Phase II trips	281	1534	1678	898

Figure 8c: Trip generation – Phase II

Distribution

8.13 The trip distribution has been determined, employing the same methodology as the historical work carried out on the EV development by Halcrow and JMP, by using model zone 68, which is a residential area in Dorcan. The resultant trip distribution is shown below.

Destination	Morning Peak		Evening Peak	
	IN	OUT	IN	OUT
Swindon	74%	65%	65%	74%
A420 (E)	11%	9%	9%	5%
M4 (W)	8%	16%	13%	11%
M4 (E)	4%	8%	12%	6%
A419 (N)	3%	2%	2%	4%

Figure 8d: Trip distribution Phase I

PARAMICS Modelling

8.14 JMP has previously developed a PARAMICS microsimulation model of the A419 corridor covering the section between M4 J15 and the Blunsdon Bypass, and also part of the A420, including the junctions with Thornhill Rd and Old Vicarage Lane.

8.15 The SMMTM flows from SATURN were used as input data for the PARAMICS modelling, which then included the following stages.

Task 1: Base Year Model Results

8.16 Outputs from the existing base year PARAMICS model (2008) were extracted in the form of link flows, queuing at critical junctions and journey times.

Task 2: Future Year Model – 2026 Do Nothing

8.17 The SATURN model results were reviewed and any necessary changes in terms of committed development or committed network schemes were made to the model.

8.18 The PARAMICS model was then reviewed in terms of network structure to reflect any changes in committed network schemes. Demand matrices for PARAMICS were derived from the 2026 Do Nothing (DN) SATURN model. Traffic signals were re-optimised to ensure the smooth movement of traffic in the network and to reduce queuing.

Task 3: Future Year Model – 2026 Do Something – Phase I

8.19 The details of the Phase 1 EV were incorporated into PARAMICS. The PARAMICS demand matrices for this option were derived from the SATURN model and traffic signals will be re-optimised.

Task 4: Future Year Model – 2026 Do Something – Phase IA

8.20 Sensitivity tests were performed to analyse the impact of the potential extension to the Sainsbury's site and the delivery

of the employment land to the south of the A420.

Task 5: Future Year Model – 2026 Do Something Phase II

8.21 The details of the Phase II EV development (including quantum of development, location and internal network structure) as well as proposed transport interventions were incorporated into the PARAMICS model. As before, the PARAMICS demand matrices for this option were derived from the SATURN model and traffic signals will be re-optimised.

Task 6: Opening year assessment

8.22 During the consultation period the HE circular was published. This requires an assessment year to reflect the opening year, assumed to be 2016. As before, the PARAMICS demand matrices for this option were derived from the SATURN model.

9 Local Road Network Review – Journey Times

Journey Time Assessment Introduction

9.1 The PARAMICS model has been used to provide estimation of journey times across key routes through the A419 and A420 corridors. The impact on average journey times can be compared between the 'with' and 'without' development scenario tests to identify the impacts of development. The agreed routes that are applicable to this development are indicated below:

- **Route 1:** A419 southbound from Blunsdon Bypass to south of M4 J15
- **Route 2:** A419 northbound from south of M4 J15 to Rattrap Junction
- **Route 3:** A419 northbound from south of M4 J15 to Gablecross A420
- **Route 4:** A419 southbound from Gablecross A420 to south of M4 J15
- **Route 5:** A419 southbound from Blunsdon Bypass to White Hart Off-slip
- **Route 6:** A419 northbound from Merlin Way Junction to Blunsdon Bypass
- **Route 7:** A420 eastbound from White Hart to east of Old Vicarage Lane
- **Route 8:** A420 westbound from east of Old Vicarage Lane to White Hart Junction

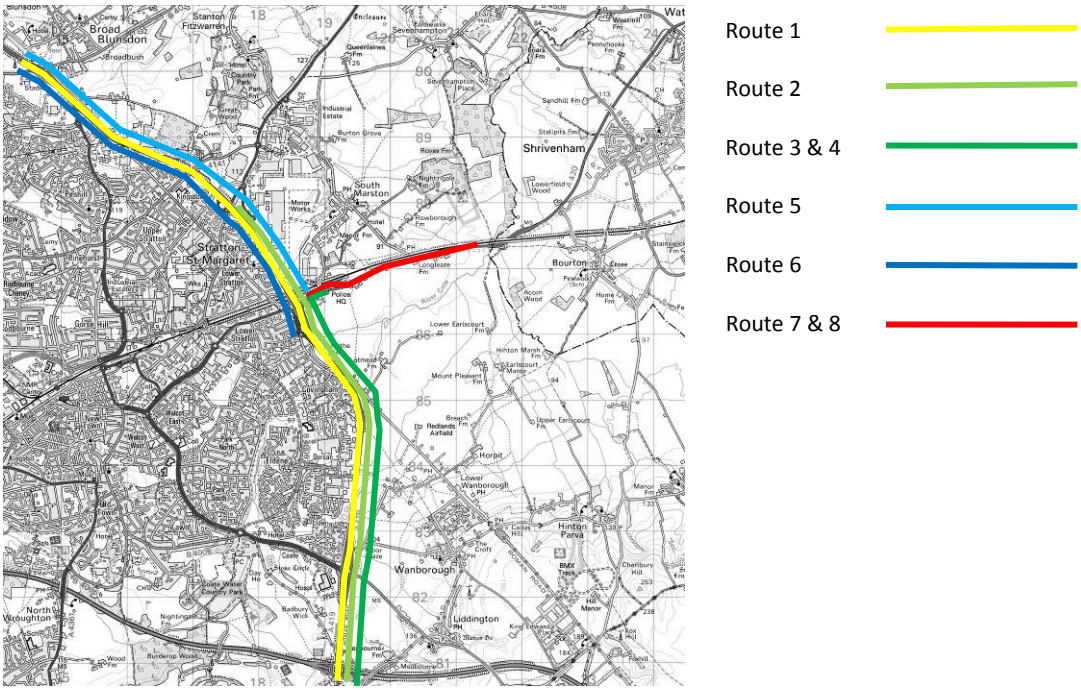

Figure 9a: Journey Time results

9.2 To assess the significance of the difference in flows and journey times, the Design Manual for Roads and Bridges at Volume 12, Section 2 outlines a methodology known as GEH Statistic, which are frequently employed to compare to sets of model data. The formula for the GEH statistic is indicated below:

$$GEH = \sqrt{\frac{2(M-C)^2}{M+C}}$$

Where M and C represents the two sets of data

9.3 As an initial appraisal, the GEH statistics are used to appraise the difference in journey times as a result of the development generated trips. The result of this analysis can reflect both an increase and a decrease. Where this assessment shows a GEH of less than 5, this indicates no material impact.

9.4 Further to the initial GEH screening, when considering the scenarios with and without the development, the difference both in relation to actual and percentage is reported, together with a consideration of journey speed. In relation to a significant difference, it is considered a difference of greater than 60 seconds could be noticeable from day to day variations.

Phase I Journey Time Results

9.5 The results of the journey time assessment are indicated in the figures below. This includes both the GEH statistics and a review of the likely average speed of vehicles along the route, in both 2016 and 2026

Route	2016 Base	2016 with Phase I	Difference	GEH	Percentage	KPH 2016 Base	KPH 2016 with Phase I	Difference
Route 1 –AM	451	461	10	0.5	2.2%	93.4	91.4	-2.0
Route 1 - PM	443	453	10	0.5	2.2%	95.0	93.0	-2.0
Route 2 –AM	426	429	3	0.2	0.8%	66.8	66.2	-0.6
Route 2 - PM	446	451	5	0.2	1.0%	63.7	63.1	-0.6
Route 3 - AM	503	516	13	0.6	2.6%	47.9	46.7	-1.2
Route 3 - PM	410	494	84	4.0	20.6%	58.9	48.8	-10.1
Route 4 - AM	345	352	7	0.4	1.9%	66.8	65.5	-1.3
Route 4 - PM	345	348	3	0.2	0.8%	66.8	66.2	-0.6
Route 5 - AM	157	159	2	0.1	1.2%	128.3	126.8	-1.5
Route 5 - PM	147	148	1	0.0	0.4%	136.8	136.3	-0.5
Route 6 - AM	177	177	0	0.0	0.1%	126.4	126.3	-0.1
Route 6 - PM	188	188	0	0.0	0.0%	118.9	118.8	-0.1
Route 7 - AM	260	343	83	4.8	31.9%	26.3	19.9	-6.4
Route 7 - PM	258	362	104	5.9	40.4%	26.5	18.9	-7.6
Route 8 - AM	251	420	169	9.2	67.3%	27.2	16.3	-10.9
Route 8 - PM	280	324	44	2.5	15.7%	24.4	21.1	-3.3

Figure 9b: 2016 Phase I Journey Time Results, [seconds]

9.6 The results demonstrate that Routes 3, 7 and 8 may potentially be affected by the delivery of the scheme in the opening year.

9.7 Route 3 A419 northbound from south of M4 J15 to Gablecross A420 is identified to have an increase in journey time by more than 60 seconds in the PM peak. The AM peak does not indicate an issue. The increase reported is 84 seconds and considering the length of this journey it is considered that this would be within a daily variance of the trips and typically would not be noticeable. The total length of this route is circa 6.3km, resulting in a delay of only 13 seconds per km. This is not significant. The reliability of the route is a factor of the predicted aggregate queuing and this information is presented in the following chapter of this report. The aggregate queuing information indicates that queuing is not predicted to increase significantly. Therefore, this does not represent a severe impact.

9.8 Route 7 and 8 relate to the A420 between the White Hart and Old Vicarage Lane junctions. This section of highway delivers the principle points of access and therefore it is not unexpected that there would potentially be an impact at these locations. The mitigation strategy has been developed to minimise the impact of the development, including signalisation of White Hart and Gablecross.

9.9 The results for the compassion between 2026 for Phase I are presented below.

Route	2026	2026 with Phase I	Difference	GEH	Percentage	KPH 2026	KPH 2026 with Phase I	Difference
Route 1 –AM	643	482	-161	6.8	-25.0%	65.5	87.4	21.9
Route 1 - PM	435	453	18	0.9	4.1%	96.8	93.0	-3.8
Route 2 –AM	550	710	160	6.4	29.1%	51.7	40.1	-11.6
Route 2 - PM	911	451	-460	17.6	-50.5%	31.2	63.1	31.9
Route 3 - AM	741	581	-160	6.2	-21.6%	32.6	41.5	8.9
Route 3 - PM	914	494	-420	15.8	-46.0%	26.4	48.8	22.4
Route 4 - AM	464	353	-111	5.5	-23.9%	49.7	65.3	15.6
Route 4 - PM	393	348	-45	2.3	-11.5%	58.6	66.2	7.6
Route 5 - AM	357	172	-185	11.4	-51.8%	56.5	117.2	60.7
Route 5 - PM	149	148	-1	0.1	-0.7%	135.3	136.2	0.9
Route 6 - AM	254	351	97	5.7	38.2%	87.9	63.6	-24.3
Route 6 - PM	334	188	-146	9.0	-43.7%	66.8	118.7	51.9
Route 7 - AM	257	328	71	4.2	27.6%	26.6	20.9	-5.8
Route 7 - PM	483	362	-121	5.9	-25.1%	14.2	18.9	4.7
Route 8 - AM	479	329	-150	7.5	-31.3%	14.3	20.8	6.5
Route 8 - PM	330	324	-6	0.3	-1.8%	20.7	21.1	0.4

Figure 9c: 2026 Phase I Journey Time Results, [seconds]

9.10 This comparison of journey times indicates that the majority of the results predict a GEH of greater than of 5. However, several of these are due to an improvement in journey times, indicating that the interventions improve the operation of the road network.

9.11 There is a GEH in excess of 5 due to an increase on the following routes:

- Route 2 in the morning peak
- Route 6 in the morning peak

9.12 Routes 2 and 6 relate to the northbound flow along the A419 and therefore both report the same delay. Route 6 indicates an increased delay of 97 seconds, which would also be reflected in the Route 2 delay. Therefore the delay in route 2 would be circa 63 seconds between the M4 and the A420 / Merlin Way slip roads. This delay over this length of road would likely be within daily variation of this trip.

9.13 The impact on Route 6 is a result of the interventions releasing the traffic from White Hart and the Merlin Way roundabout. This is through the removal of the Merlin Way southbound queue which increases the flow towards Rattrap and Blunsdon junctions.

9.14 In relation to the likely impact on vehicle speeds, it is identified that vehicle speeds largely increase within the majority of the identified corridors.

9.15 Overall, it may be concluded that delivery of Phase I will not significantly affect the operation of the road network.

Phase IA Journey Time Results

9.16 The results of journey time assessments are indicated figures below. This comparison assumes that Phase IA would be approved subsequent to Phase I, therefore becoming a consented scheme forming the base transport environment for comparative reasons. The Phase I, transport interventions as outlined in this report have been tested in this scenario.

Route	2026 with Phase I	2026 with Phase IA	Difference	GEH	Percentage	KPH 2026	KPH with Phase IA	Difference
Route 1 –AM	482	482	0	0.0	0.0%	87.4	87.4	0.0
Route 1 - PM	453	440	-13	0.6	-2.9%	93.0	95.7	2.7
Route 2 –AM	710	725	15	0.6	2.1%	40.1	39.2	-0.9
Route 2 - PM	451	929	478	18.2	106.0%	63.1	30.6	-32.5
Route 3 - AM	581	720	139	5.4	23.9%	41.5	33.5	-8.0
Route 3 - PM	494	847	353	13.6	71.5%	48.8	28.5	-20.3
Route 4 - AM	353	353	0	0.0	0.0%	65.3	65.3	0.0
Route 4 - PM	348	337	-11	0.6	-3.2%	66.2	68.4	2.2
Route 5 - AM	172	172	0	0.0	0.0%	117.2	117.2	0.0
Route 5 - PM	148	148	0	0.0	0.0%	136.2	136.2	0.0
Route 6 - AM	351	312	-39	2.1	-11.1%	63.6	71.5	7.9
Route 6 - PM	188	336	148	9.1	78.7%	118.7	66.4	-52.3
Route 7 - AM	328	544	216	10.3	65.9%	20.9	12.6	-8.3
Route 7 - PM	362	333	-29	1.6	-8.0%	18.9	20.5	1.6
Route 8 - AM	329	639	310	14.1	94.2%	20.8	10.7	-10.1
Route 8 - PM	324	462	138	7.0	42.6%	21.1	14.8	-6.3

Figure 9d: 2026 Phase IA Journey Time Results, [seconds]

9.17 The results of the sensitivity test indicates that six routes have predicted increases in journey time that give rise to a GEH in excess of five, viz:

- Route 2: A419 northbound from south of M4 J15 to Rattrap Junction
- Route 3: A419 northbound from south of M4 J15 to Gablecross A420
- Route 6: A419 northbound from Merlin Way Junction to Blunsdon Bypass
- Route 7: A420 eastbound from White Hart to east of Old Vicarage Lane

- Route 8: A420 westbound from east of Old Vicarage Lane to White Hart Junction
- Route 8 - A420 westbound from east of Old Vicarage Lane to White Hart Junction

9.18 These results indicate there will be increases in journey times for traffic travelling in both directions along the A419 and along the A420. This would indicate that additional mitigation may be required for the impact of the Phase IA development.

Phase II Journey Time Results

9.19 The results of journey time assessments are indicated below. This comparison assumes that Phase II would be approved subsequent to Phase IA, therefore becoming a consented scheme and will form the base transport environment. The Phase II, wider EV interventions as outlined in the SBC transport strategy have been tested in this scenario. Two separate tables are provided for the two options for the layout of the White Hart junction; firstly as a signalled roundabout and secondly as a diamond interchange.

Route	2026 with Phase IA	2026 with Phase IIR	Difference	GEH	Percentage	KPH 2026 with Phase IA	KPH Phase IIR	Difference
Route 1 –AM	482	492	10	0.5	2.1%	87.4	85.6	-1.8
Route 1 - PM	440	722	282	11.7	64.1%	95.7	58.3	-37.4
Route 2 –AM	725	516	-209	8.4	-28.8%	39.2	55.1	15.9
Route 2 - PM	929	576	-353	12.9	-38.0%	30.6	49.4	18.8
Route 3 - AM	720	521	-199	8.0	-27.7%	33.5	46.3	12.8
Route 3 - PM	847	545	-302	11.5	-35.7%	28.5	44.3	15.8
Route 4 - AM	353	415	62	3.2	17.6%	65.3	55.5	-9.8
Route 4 - PM	337	670	333	14.8	98.8%	68.4	34.4	-34.0
Route 5 - AM	172	152	-20	1.6	-11.6%	117.2	132.6	15.4
Route 5 - PM	148	215	67	5.0	45.3%	136.2	93.8	-42.4
Route 6 - AM	312	178	-134	8.6	-42.9%	71.5	125.4	53.9
Route 6 - PM	336	266	-70	4.0	-20.8%	66.4	83.9	17.5
Route 7 - AM	544	595	51	2.1	9.4%	12.6	11.5	-1.1
Route 7 - PM	333	559	226	10.7	67.9%	20.5	12.2	-8.3
Route 8 - AM	639	607	-32	1.3	-5.0%	10.7	11.3	0.6
Route 8 - PM	462	724	262	10.8	56.7%	14.8	9.4	-5.4

Figure 9e: 2026 Phase IIR Journey Time Results, [seconds] – White Hart Signalised Roundabout

Route	2026 with Phase IA	2026 with Phase IID	Difference	GEH	Percentage	KPH 2026 with Phase IA	KPH Phase IID	Difference
Route 1 –AM	482	635	153	6.5	31.7%	87.4	66.4	-21.0
Route 1 - PM	440	892	452	17.5	102.7%	95.7	47.2	-48.5
Route 2 –AM	725	518	-208	8.3	-28.6%	39.2	55.0	15.8
Route 2 - PM	929	646	-283	10.1	-30.4%	30.6	44.0	13.4
Route 3 - AM	720	695	-25	0.9	-3.4%	33.5	34.7	1.2
Route 3 - PM	847	664	-183	6.7	-21.6%	28.5	36.3	7.8

Route	2026 with Phase IA	2026 with Phase IID	Difference	GEH	Percentage	KPH 2026 with Phase IA	KPH Phase IID	Difference
Route 4 - AM	353	378	24	1.3	6.9%	65.3	61.0	-4.3
Route 4 - PM	337	629	292	13.3	86.7%	68.4	36.6	-31.8
Route 5 - AM	172	369	197	12.0	114.4%	117.2	54.7	-62.5
Route 5 - PM	148	490	342	19.1	231.0%	136.2	41.2	-95.0
Route 6 - AM	312	214	-98	6.0	-31.4%	71.5	104.2	32.7
Route 6 - PM	336	331	-5	0.3	-1.4%	66.4	67.4	1.0
Route 7 - AM	544	545	1	0.0	0.1%	12.6	12.6	0.0
Route 7 - PM	333	519	186	9.0	56.0%	20.5	13.2	-7.3
Route 8 - AM	639	473	-167	7.1	-26.1%	10.7	14.5	3.8
Route 8 - PM	462	477	15	0.7	3.3%	14.8	14.3	-0.5

Figure 9f: 2026 Phase IID Journey Time Results, [seconds] - White Hart Diamond Interchange

9.20 These tests indicates that the majority of the routes except are predicted to have GEH in excess of five. However, several of these are based on beneficial reduction in journey times.

9.21 This would indicate that additional mitigation may be required to mitigate the impact of the Phase II development.

9.22 The development of the Phase I proposals, being the subject of this proposed application, is not predicated on the successful delivery of the Phase II wider EV proposals and interventions forming the SBC strategy for the area as a whole. However, it is important that the Phase I proposals do not compromise the wider growth by the implementation of interventions that might fetter the opportunity for delivery of the intended future infrastructure. This matter is considered in greater detail in Section 10 and 11 of this document.

Thornhill Road / Old Vicarage Lane Link Road Journey Times

9.23 To serve the new community at South Marston, a connection between Thornhill Road and Old Vicarage Lane will be delivered. This link will be designed to further encourage motorists to use the A420 as the primary and substantially preferred route. A close to 20mph target speed is envisaged with careful design and treatment and would accord with "Manual for Streets".

9.24 To check which could be the preferred route between the site and Swindon, the journey times for each has been reviewed within the Paramics traffic model. In reality the design of the link road will result in a less attractive route, still.

9.25 The potential results are provided below.

- Route 1 - Traffic travelling southbound along Thornhill Road
- Route 2 - Traffic travelling westbound along the A420
- Route 3 - Traffic travelling westbound through the development

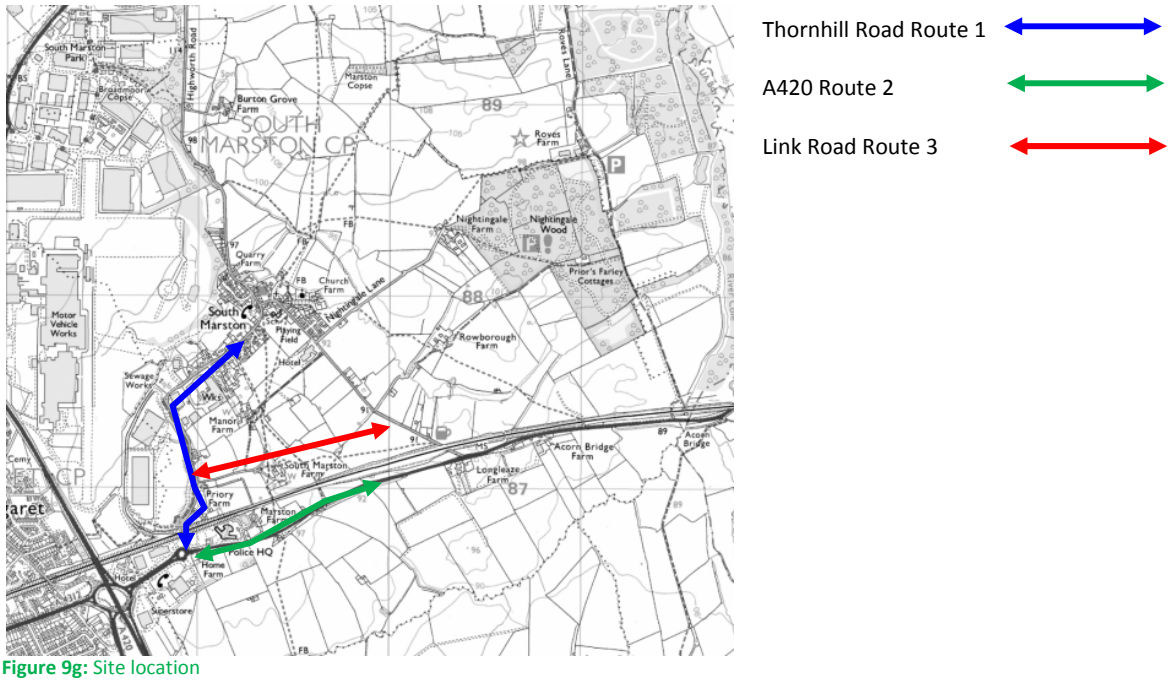

Figure 9g: Site location

9.26 The results of the journey time assessment are indicated in the figures below.

Link	2016 with development	2026 with development
Route 1	145	138
Route 2	60	59
Route 3	80	80

Figure 9h: Journey Time Results, [seconds] movements – Morning Peak

Link	2016 with development	2026 with development
Route 1	138	134
Route 2	60	61
Route 3	80	80

Figure 9i: Journey Time Results, [seconds] movements – Evening Peak

9.27 The figure above demonstrates that Route 2, along the A420 is clearly the preferred route. Manual for Streets suggests that the developments should deliver equitable route choice. The access points identified delivers a cohesive strategy that disperses vehicles to the most appropriate routes.

9.28 The Transport Assessment (TA) drafted to support the planning application assesses the impact of the development and the access strategy. The TA demonstrates that through a package of highway interventions, the impact of the development has been mitigated. The TA forms part of the Neighbourhood Plan evidence base but has been updated significantly since then culminating in several new runs of the adjusted model (the previous Swindon model found to be flawed) throughout the summer of 2014 in discussion with SBC and HE. The Data described and referenced in this note is therefore collated from the latest information provided to SBC throughout 2015, subsequent to the 2014 model runs.

Journey Time Summary

9.29 Journey time assessments indicate no significant impact on the highway network when the Phase I development scenario is tested alongside the proposed highway interventions. As would be expected, the assessment also indicates that additional mitigation may needed to support the subsequent Phase IA and Phase II development scenarios. The

mitigation suggested for Phase I will not prejudice the later phases and this is demonstrated in greater detail in Sections 10 and 11 of this report.

10 Local Road Network Review – Link Flow Assessment

Phase 1 Link Flow Assessment

10.1 The PARAMICS model has been employed to predict the volume of traffic on the road links within Swindon in the base case and with development scenarios. As an initial appraisal, the GEH statistics are used to appraise the increase in flows as a result of the development generated trips. Where this assessment shows a GEH of less than 5, this indicates no material impact, whereas a GEH of greater than 5 would then require a more detailed assessment on link capacity. The tables below also show the change in the number of vehicles and the percentage change on each link.

10.2 The link flows are assessed at specific locations as indicated in Figure 10a. The links that connect into Pound Corner are Old Vicarage Lane south of Highworth Road (the southern arm), Thornhill Road west of Highworth Road (western arm) and Highworth Road north of Old Vicarage Lane (north-east arm).

Location	Morning Peak					Evening Peak				
	2016	Phase I	Change	% Change	GEH	2016	Phase I	Change	% Change	GEH
M4 Motorway	6444	6469	25	0.4%	0.3	6745	6783	38	0.6%	0.5
A419, J15 to Commonhead	5531	5700	169	3.1%	2.3	5399	5525	126	2.3%	1.7
A4259 at Commonhead junction	2825	2916	90	3.2%	1.7	2966	3046	80	2.7%	1.5
Purley Rd at Commonhead junction	403	410	6	1.5%	0.3	600	611	11	1.9%	0.5
Pack Hill at Commonhead junction	251	251	0	0.0%	0.0	309	308	-1	-0.2%	0.0
A419, Commonhead to White hart	5084	5318	235	4.6%	3.3	4665	4832	167	3.6%	2.4
A420 at White Hart	2753	3725	972	35.3%	17.1	3218	4225	1007	31.3%	16.5
Ermin Street at White Hart	533	654	122	22.8%	5.0	674	844	170	25.2%	6.2
Oxford Road at White Hart	1410	1716	306	21.7%	7.7	1309	1811	501	38.3%	12.7
Merlin Way at White Hart	1400	1576	176	12.6%	4.6	1458	1588	130	8.9%	3.3
A419 north of White Hart	5350	5458	109	2.0%	1.5	5501	5596	94	1.7%	1.3
Thornhill Road north of Gablecross	413	772	358	86.8%	14.7	405	817	413	102.1%	16.7
A420 east of Gablecross	2064	2765	700	33.9%	14.3	2055	2839	784	38.1%	15.8
Supermarket/access S of Gablecross	479	526	48	9.9%	2.1	745	915	170	22.8%	5.9
A420 west of Old Vicarage Lane	2163	2874	711	32.9%	14.2	2165	2950	785	36.2%	15.5
A420 east of Old Vicarage Lane	1969	2180	211	10.7%	4.6	1916	2198	282	14.7%	6.2
Old Vicarage Lane north A420	157	382	225	143.1%	13.7	132	409	277	210.6%	16.9
Old Vicarage Lane S of Highworth Rd	124	208	85	68.6%	6.6	94	204	110	116.5%	9.0
Thornhill Road west of Highworth Rd	280	286	6	2.0%	0.3	209	230	21	10.0%	1.4
Highworth Rd N of Old Vicarage Lane	282	363	81	28.9%	4.5	209	284	75	35.9%	4.8
A420 west of New Eastern Access	1969	2199	230	11.7%	5.0	1916	2199	283	14.8%	6.2

A420 east of New Eastern Access	1969	1986	17	0.9%	0.4	1916	1986	70	3.6%	1.6
New Eastern Access Road	0	256	256	n/a	n/a	0	256	256	n/a	n/a
Link between Honda RB and Hyde Rd	4595	4654	59	1.3%	0.9	4536	4629	92	2.0%	1.4
Link between Hyde Road and Bypass	4585	4642	57	1.2%	0.8	4484	4554	69	1.5%	1.0

Figure 10a: Two way flow link analysis – 2016 Phase I

10.3 The review of the As demonstrated in the Figure 10a above, several links have a predicted GEH in excess of 5, viz:

- A420 at White Hart in the morning and evening peak
- Ermin Street in the morning and evening peak
- Oxford Road at White Hart in the evening peak
- Merlin Way at White Hart in the morning peak
- Thornhill Road north of Gablecross in the morning and evening peak
- A420 east of Gablecross in the morning and evening peak
- A420 west of Gablecross in the morning and evening peak
- Supermarket south of Gablecross in the morning and evening peak
- A420 west of New Eastern Access

10.4 The proposed development will deliver access points within the Thornhill Road, Old Vicarage Lane and A420 corridors and therefore it is expected that increases in traffic flows and the GEH Statistic will be concentrated at these locations. An increase in traffic flow does not necessary equate to an increase in delay and / or congestion.

10.5 The potential to increase delay and congestion relates to not only the link, but the junctions within any link. The potential impact at junctions is considered later in this report.

10.6 The theoretical highway capacity refers to the maximum level of traffic that can be accommodated; this being a function of the design and width of the road. Traffic flow along a link has the potential to create congestion if the theoretical highway capacity of that link is exceeded.

10.7 To assess the potential impact of the increase in flows within those links identified above, an assessment has been carried out of the predicted traffic flows against the theoretical highway capacity as indicated in DMRB TA79/99, Capacity of Urban Roads. This document considers the capacity of single lanes of traffic, based on vehicle flow. This is presented

Location	Link Type	Theoretical Capacity	M Maximum Peak	
			Phase I	%
A420 at White Hart	UAP1	3180	2295	72.2%
Ermine Street at White Hart	UAP3	1100	463	42.1%
Oxford Road at White Hart	UAP2	1470	1038	70.6%
Merlin Way at White Hart	UAP2	2940	1269	43.2%
Thornhill Road north of Gablecross	UAP4	900	431	47.9%
A420 east of Gablecross	UAP1	1590	1518	95.5%
A420 west of Gablecross	UAP	3180	2208	69.4%
A420 east of Old Vicarage Lane	UAP2	1470	1126	76.6%
Old Vicarage Lane north A420	UAP4	900	621	69.0%
Old Vicarage Lane S of Highworth Rd	UAP4	900	208	23.1%
A420 west of New Eastern Access	UAP2	1470	1135	77.2%

Figure 10b: Two way flow link analysis – 2016 Phase I

10.8 The assessment of the two way link flow indicates that the majority of the links operate well within theoretical highway capacity limits. One section is highlighted which includes the A420 between the Gablecross junction and the Old Vicarage Lane junction. These links are approaching capacity. However, the flow along this link will be controlled by signalled controlled junctions that will create platoons of traffic, rather than a steady stream. This is likely to allow the traffic to free flow.

10.9 A review of the traffic levels indicates that the predicted traffic levels do not exceed the theoretical highway capacity.

10.10 A review of the 2026 is presented below.

	Morning Peak					Evening Peak				
Location	2026	Phase I	Change	% Change	GEH	2026	Phase I	Change	% Change	GEH
M4 Motorway	7374	7019	-355	-4.8%	4.2	7813	7746	-67	-0.9%	0.8
A419, J15 to Commonhead	4159	4187	28	0.7%	0.4	3837	3817	-20	-0.5%	0.3
A4259 at Commonhead junction	3339	3448	109	3.3%	1.9	3200	3248	48	1.5%	0.8
Purley Rd at Commonhead junction	595	601	6	1.0%	0.2	480	492	12	2.5%	0.5
Pack Hill at Commonhead junction	210	226	16	7.6%	1.1	117	120	3	2.6%	0.3
A419, Commonhead to White Hart	5715	5921	206	3.6%	2.7	5393	5404	11	0.2%	0.1
A420 at White Hart	2930	3680	750	25.6%	13.0	3051	3974	923	30.3%	15.6
Ermin Street at White Hart	703	747	44	6.3%	1.6	835	860	25	3.0%	0.9
Oxford Road at White Hart	1440	1566	126	8.7%	3.3	1277	1580	303	23.7%	8.0
Merlin Way at White Hart	1850	2090	240	13.0%	5.4	2114	2329	215	10.2%	4.6
A419 north of White Hart	6277	6293	16	0.3%	0.2	5845	5785	-60	-1.0%	0.8
Thornhill Road north of Gablecross	530	787	257	48.5%	10.0	645	937	292	45.3%	10.4
A420 east of Gablecross	2136	2723	587	27.5%	11.9	2072	2662	590	28.5%	12.1
A420 west of Gablecross	2737	3508	771	28.2%	13.8	2868	3749	881	30.7%	15.3
Supermarket/access S of Gablecross	541	584	43	7.9%	1.8	806	957	151	18.7%	5.1
A420 east of Old Vicarage Lane	2081	2251	170	8.2%	3.7	2026	2386	360	17.8%	7.7
Old Vicarage Lane north A420	106	502	396	373.6%	22.7	66	513	447	677.3%	26.3
Old Vicarage Lane S of Highworth Rd	101	96	-5	-5.0%	0.5	66	144	78	118.2%	7.6
Thornhill Road west of Highworth Rd	500	359	-141	-28.2%	6.8	391	282	-109	-27.9%	5.9
A420 west of New Eastern Access	1986	2115	129	6.5%	2.8	2011	2229	218	10.8%	4.7
A420 east of New Eastern Access	1984	1982	-2	-0.1%	0.0	2007	2035	28	1.4%	0.6
New Eastern Access Road	0	155	155	n/a	n/a	0	236	236	n/a	n/a
Link between Honda RB and Hyde Rd	5633	5593	-40	-0.7%	0.5	5041	5018	-23	-0.5%	0.3
Link between Hyde Road and Bypass	5546	5504	-42	-0.8%	0.6	4927	4935	8	0.2%	0.1

Figure 10c: Two way flow link analysis – 2026 Phase I

10.11 As demonstrated in the Figure 10b above, several links have a predicted GEH in excess of 5, viz:

- A420 at White Hart in the morning and evening peak
- Oxford Road at White Hart in the evening peak
- Merlin Way at White Hart in the morning peak
- Thornhill Road north of Gablecross in the morning and evening peak
- A420 east of Gablecross in the morning and evening peak
- A420 west of Gablecross in the morning and evening peak
- A420 west of Old Vicarage Lane in the morning and evening peak
- A420 east of Old Vicarage Lane in the evening peak

- Old Vicarage Lane north A420 in the morning and evening peak
- Old Vicarage Lane south of Highworth Road in the evening peak
- Thornhill Road west of Highworth Road in the morning and evening peak (decrease in flows)
- Highworth Road north of Old Vicarage Lane in the morning peak (decrease in flows)

10.12 The proposed development will deliver access points within the Thornhill Road, Old Vicarage Lane and A420 corridors and therefore it is expected that increases in traffic flows and the GEH Statistic will be concentrated at these locations. However, the increases in traffic along these links also have to be considered in the context of the proposed intervention strategy along the A420, which are intended to mitigate the effects of the development. In this sense, the journey time assessments are more representative of the whole network operation.

10.13 Two of the locations identified report a GEH of greater than 5, which result from the reduction in vehicles, being at Thornhill Road and Highworth Road at Pound Corner. The decrease in flows at these locations occurs due to the context of the access proposals, viz the realignment of Old Vicarage Lane which encourages trips into the development. Table 10c predicts that the flows approaching Pound Corner could reduce by up to 141 vehicles.

10.14 In summary therefore whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner.

10.15 Outwith the primary corridors providing access to the site, only three corridors report a GEH in excess of 5, viz:

- Oxford Road at White Hart in the evening peak
- Merlin Way at White Hart in the morning peak
- Old Vicarage Lane south of Highworth Road in the evening peak

10.16 The GEH statistic is predicted to be 8.0 within the Oxford Road corridor in the evening peak period. Notwithstanding the GEH statistic, the two way flow is predicted to be 1,580, which is within the theoretical capacity of the highway when assessed in accordance with the DMRB and therefore no specific mitigation is necessary.

10.17 The GEH statistic is predicted to be 5.4 within the Merlin Way corridor in the morning peak period. Notwithstanding the GEH statistic that only marginally exceeds the threshold, the two way flow is predicted to be 2,090, which is within the theoretical capacity of the highway when assessed in accordance with the DMRB and therefore no specific mitigation is necessary.

10.18 The GEH statistic is predicted to be 7.6 within Old Vicarage Lane south of Highworth Road in the evening peak period. Notwithstanding the GEH statistic, the increase is due to the low base levels of traffic, with the two way flow is predicted to be 144, which is well within the theoretical capacity of the highway when assessed in accordance with the DMRB. Therefore, no specific mitigation is necessary.

10.19 To further assess the implications of the increase in flows, the theoretical capacities of those links exceeding a GEH of 5 have been assessed against the theoretical highway capacity. These are illustrated below.

Location	Link Type	Theoretical Capacity	Maximum Peak	
			Phase I	%
A420 at White Hart	UAP1	3,180	2142	67.4%
Oxford Road at White Hart	UAP2	1,470	940	63.9%
Merlin Way at White Hart	UAP2	2,940	1296	44.1%
Thornhill Road north of Gablecross	UAP4	900	495	55.0%
A420 east of Gablecross	UAP1	1,590	1746	109.8%
A420 west of Gablecross	UAP1	3,180	2101	66.1%
A420 east of Old Vicarage Lane	UAP2	1,470	1084	73.7%
Old Vicarage Lane north A420	UAP4	900	454	50.4%
Old Vicarage Lane S of Highworth Rd	UAP4	900	96	10.7%

Figure 10d: Two way flow link analysis – 2026 Phase I

10.20 The results of this assessment indicate the majority of the links predicted to experience the highest increases will remain within the thresholds of theoretical highway capacity. The A420 east of Gablecross is predicted to exceed the theoretical capacity. However, TA79/99 indicates that theoretical capacity could be 10% higher than that indicated, meaning that the capacity threshold would not be exceeded. Furthermore, it is noted that in 2016, the capacity threshold is not exceeded, demonstrating that the rise in background traffic growth is the resultant cause of the threshold being exceeded.

10.21 The operation of this link has been assessed within the traffic model. The Journey times for the A420 indicate a reduction in vehicle speeds, but not to a significant degree. The capacity indicated is theoretical, with the operation of junctions often being the controlling feature. The junctions within this corridor have been assessed, identifying that they will operate within capacity.

10.22 The potential to widen the A420 has been identified for the subsequent Phases, this would increase the highway capacity such the thresholds would not be exceeded.

10.23 Based on the above, it is considered that there are no highway capacity concerns.

Phase 1A Link Flow Assessment

10.24 The link flows for the Phase IA scenario are assessed at specific locations as indicated in Figure 10b. This comparison assumes that Phase IA would approved subsequent to Phase I, therefore becoming a consented scheme forming the base line transport environment.

Location	Morning Peak					Evening Peak				
	Phase I	Phase IA	Change	% Change	GEH	Phase I	Phase IA	Change	% Change	GEH
M4 Motorway	7019	7034	15	0.2%	0.2	7746	7692	-54	-0.7%	0.6
A419, J15 to Commonhead	4187	4196	9	0.2%	0.1	3817	3800	-17	-0.4%	0.3
A4259 at Commonhead junction	3448	3504	56	1.6%	0.9	3248	3136	-112	-3.4%	2.0
Purley Rd at Commonhead junction	601	591	-10	-1.7%	0.4	492	444	-48	-9.8%	2.2
Pack Hill at Commonhead junction	226	226	0	0.0%	0.0	120	115	-5	-4.2%	0.5
A419, Commonhead to White Hart	5921	5967	46	0.8%	0.6	5404	5412	8	0.1%	0.1
A420 at White Hart	3680	3923	243	6.6%	3.9	3974	4137	163	4.1%	2.6
Ermin Street at White Hart	747	747	0	0.0%	0.0	860	839	-21	-2.4%	0.7
Oxford Road at White Hart	1566	1707	141	9.0%	3.5	1580	1602	22	1.4%	0.6
Merlin Way at White Hart	2090	2130	40	1.9%	0.9	2329	2402	73	3.1%	1.5
A419 north of White Hart	6293	6266	-27	-0.4%	0.3	5785	5698	-87	-1.5%	1.1
Thornhill Road north of Gablecross	787	740	-47	-6.0%	1.7	937	886	-51	-5.4%	1.7
A420 east of Gablecross	2723	2973	250	9.2%	4.7	2662	2754	92	3.5%	1.8

Location	Morning Peak					Evening Peak				
	Phase I	Phase IA	Change	% Change	GEH	Phase I	Phase IA	Change	% Change	GEH
A420 west of Gablecross	3508	3831	323	9.2%	5.3	3749	3865	116	3.1%	1.9
Supermarket/access S of Gablecross	584	689	105	18.0%	4.2	957	929	-28	-2.9%	0.9
A420 east of Old Vicarage Lane	2251	2157	-94	-4.2%	2.0	2386	2297	-89	-3.7%	1.8
Old Vicarage Lane north A420	502	451	-51	-10.2%	2.3	513	479	-34	-6.6%	1.5
Old Vicarage Lane S of Highworth Rd	96	134	38	39.6%	3.5	144	126	-18	-12.5%	1.5
Thornhill Road west of Highworth Rd	359	361	2	0.6%	0.1	282	270	-12	-4.3%	0.7
Highworth Rd N of Old Vicarage Lane	376	383	7	1.9%	0.4	343	315	-28	-8.2%	1.5
A420 west of New Eastern Access	2115	1985	-130	-6.1%	2.9	2229	2206	-23	-1.0%	0.5
A420 east of New Eastern Access	1982	1909	-73	-3.7%	1.7	2035	2021	-14	-0.7%	0.3
New Eastern Access Road	155	159	4	2.6%	0.3	236	226	-10	-4.2%	0.7
Link between Honda RB and Hyde Rd	5593	5575	-18	-0.3%	0.2	5018	5089	71	1.4%	1.0
Link between Hyde Road and Bypass	5504	5479	-25	-0.5%	0.3	4935	4980	45	0.9%	0.6

Figure 10e: Two way flow link analysis – Phase IA

10.25 Only one link has a predicted GEH in excess of 5, A420 west of Gablecross in the morning peak.

10.26 The GEH statistic is predicted to be 5.3 within A420 corridor. Notwithstanding the GEH statistic is marginally above the threshold, the two way flow is predicted to be 3,831, which is within the theoretical capacity of the highway when assessed in accordance with the DMRB and therefore no specific mitigation is necessary.

10.27 As similarly predicted for the Phase I results, there is a reduction in flows on approach to Pound Corner.

Phase II Link Flow Assessment

10.28 The link flows for the Phase II scenario are assessed at specific locations as indicated in Figure 10c. This comparison assumes that Phase II would approved subsequent Phase IA, therefore becoming a consented scheme and will form the base line transport environment. This assessment assumed a Diamond Interchange at White Hart.

Location	Morning Peak					Evening Peak				
	Phase IA	Phase II	Change	% Change	GEH	Phase IA	Phase II	Change	% Change	GEH
M4 Motorway	7034	6182	-852	-12.1%	10.5	7692	6556	-1136	-14.8%	13.5
A419, J15 to Commonhead	4196	3771	-425	-10.1%	6.7	3800	3721	-79	-2.1%	1.3
A4259 at Commonhead junction	3504	3077	-427	-12.2%	7.4	3136	2918	-218	-7.0%	4.0
Purley Rd at Commonhead junction	591	912	321	54.3%	11.7	444	982	538	121.2%	20.1
Pack Hill at Commonhead junction	226	330	104	46.0%	6.2	115	1042	927	806.1%	38.5
A419, Commonhead to White Hart	5967	6138	171	2.9%	2.2	5412	5933	521	9.6%	6.9
A420 at White Hart	3923	1755	-2168	-55.3%	40.7	4137	2606	-1531	-37.0%	26.4
Ermin Street at White Hart	747	685	-62	-8.3%	2.3	839	699	-140	-16.7%	5.0
Oxford Road at White Hart	1707	981	-726	-42.5%	19.8	1602	1199	-403	-25.2%	10.8
Merlin Way at White Hart	2130	761	-1369	-64.3%	36.0	2402	797	-1605	-66.8%	40.1
A419 north of White Hart	6266	6477	211	3.4%	2.6	5698	6586	888	15.6%	11.3
Thornhill Road north of Gablecross	740	1077	337	45.5%	11.2	886	1150	264	29.8%	8.3
A420 east of Gablecross	2973	2178	-795	-26.7%	15.7	2754	1859	-895	-32.5%	18.6
A420 west of Gablecross	3831	3434	-397	-10.4%	6.6	3865	3512	-353	-9.1%	5.8
Supermarket/access S of Gablecross	689	378	-311	-45.1%	13.5	929	475	-454	-48.9%	17.1
A420 east of Old Vicarage Lane	2157	1541	-616	-28.6%	14.3	2297	1536	-761	-33.1%	17.4
Old Vicarage Lane north A420	451	356	-95	-21.1%	4.7	479	443	-36	-7.5%	1.7
Old Vicarage Lane south of Highworth Rd	134	361	227	169.4%	14.4	126	438	312	247.6%	18.6
Thornhill Road west of Highworth Rd	361	511	150	41.6%	7.2	270	545	275	101.9%	13.6
Highworth Rd N of Old Vicarage Lane	383	382	-1	-0.3%	0.1	315	405	90	28.6%	4.7
A420 west of New Eastern Access	1985	1944	-41	-2.1%	0.9	2206	1788	-418	-18.9%	9.4

Location	Morning Peak					Evening Peak				
	Phase IA	Phase II	Change	% Change	GEH	Phase IA	Phase II	Change	% Change	GEH
A420 east of New Eastern Access	1909	1943	34	1.8%	0.8	2021	1768	-253	-12.5%	5.8
New Eastern Access Road	159	195	36	22.6%	2.7	226	197	-29	-12.8%	2.0
Link between Honda RB and Hyde Rd	5575	5609	34	0.6%	0.5	5089	5226	137	2.7%	1.9
Link between Hyde Road and Bypass	5479	6033	554	10.1%	7.3	4980	4870	-110	-2.2%	1.6

Figure 10f: Two way flow link analysis – Phase II

10.29 This demonstrates numerous links that have a predicted GEH in excess of 5, including beneficial results reflecting decreases in link flows. A review of the link flows, where the GEH Statistic is greater than 5, indicates these would be within theoretical highway capacity when assessed in accordance with the DMRB.

10.30 The results also indicate that there could potentially be an increase in traffic through South Marston. As such, appropriate further mitigation may be needed to support the Phase II scenario at that time taking the form for instance of additional traffic management measures.

Impact on Oxfordshire County Council Road Network

10.31 Through the consultation process, Oxfordshire County Council have raised a concern regarding the potential impact of the development to the east along the A420 into Oxfordshire. To identify the potential impact of the development, the link flows presented above have been reviewed to identify the change in traffic. The relevant traffic figures are illustrated below.

Location	Morning Peak				Evening Peak			
	base	With Dev	diff	%	base	With Dev	diff	%
2016 phase 1	1969	1986	17	0.9%	1916	1986	70	3.6%
2026 phase 1	1984	1982	-2	-0.1%	2007	2035	28	1.4%
2026 phase 1A	1982	1909	-73	-3.7%	2035	2021	-14	-0.7%
2026 phase 2	1909	1943	34	1.8%	2021	1768	-253	-12.5%

Figure 10g: A420 potential traffic flows - A420 east of New Eastern Access

10.32 This demonstrates that in the morning peak the impact is below 5%, with a peak two way trip generation of only 70 trips, It is considered that this increase will be indiscernible from background traffic.

10.33 Similarly in the evening peak, it is predicted that the increase in trips will be 28, equivalent to less than one trip every minute. Through natural spreading of vehicles it is considered that this level of traffic will also be indiscernible from back ground traffic and will not affect the operation of the road network or any junction therein.

10.34 In relation to traffic routing, this traffic has been distributed towards Oxford. It Is expected that this traffic will remain on the A420 until reaching the Oxfordshire hinterland. It is not considered reasonable that any of these trips are re-routing through local roads to reach the A419 to the west with potential routes taking significantly longer timer than the direct route to the west along the A420.

Link Flow Summary

10.35 The development proposals will increase traffic in the surrounding area. All links have been tested in accordance with DMRB standards to determine the magnitude of any impacts and where appropriate the highway link capacity. IN the Phase I and IA scenarios, all highways where are material impact is predicted will continue to operate within theoretical highway capacity when assessed in accordance with the DMRB

10.36 Additional mitigation may be needed for the future Phase II scenario, but this will not be prejudiced as a result of the current Phase I proposals.

11 Local Road Network Review – Junction Assessment

Junction Assessment Introduction

11.1 The PARAMICS model has been used to predict the level of vehicular queuing at the highway junctions across the road network. As an initial appraisal, the GEH statistics are used to appraise the increase in queuing as a result of the development generated trips. Where a junction assessment shows a GEH of less than five, this indicates no material impact, whereas GEH of greater than five would then suggest a more detailed assessment of that junction.

11.2 The reported queue length is the maximum queue length on a particular junction approach measured every five minutes during the peak hour, which is then averaged. The maximum queue will be taken to be the longest of the queues on a multi-lane approach, i.e. the same length of queue may not be present in all lanes. This is presented as the length of queue in metres, not vehicles.

M4 Junction 15

11.3 The current junction is a grade separated junction with the M4 running beneath the circulatory carriageway linking to the A419 and A436, as indicated in Figure 11a below.

Figure 11a: M4 Junction 15

11.4 The 2016 results of the PARAMICS queue length analysis are indicated in Figure 11b.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
A419 Southbound approach	69	81	12	16.3%	1.3	77	98	21	27.5%	2.3
M4 westbound off slip	73	72	1	-0.4%	0.0	97	104	7	6.7%	0.6
A346 northbound approach	51	61	10	18.0%	1.2	46	49	3	8.6%	0.6
M4 eastbound off slip	135	138	3	2.4%	0.3	132	132	0	-0.2%	0.0

Figure 11b: M4 Junction 15, 2016 queue length summary Phase I

11.5 In the opening year in 2016, the maximum reported increase in queuing is predicted to be 11m, representing an increase of two vehicles. This demonstrates that the junction in the morning peak is unlikely to be affected. A similar impact is reported in the evening peak, with a maximum increase of 21m, reflecting an equivalent increase in queuing of four vehicles. This is not considered significant to affect the operation of this junction.

11.6 The 2026 Phase 1 results of the PARAMICS queue length analysis are indicated in Figure 11c.

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
A419 Southbound approach	65	84	19	28.3%	2.1	80	76	-4	-4.5%	0.4
M4 westbound off slip	75	78	3	3.7%	0.3	202	251	49	24.1%	3.2
A346 northbound approach	99	213	114	114.9%	9.1	1040	1070	30	2.9%	0.9
M4 eastbound off slip	131	133	2	1.4%	0.2	134	129	-5	-3.4%	0.4

Figure 11c: M4 Junction 15, 2026 queue length summary Phase I

11.7 The results indicate there will not be a significant impact on the majority of the arms, with only the A346 northbound approach reporting a GEH in excess of five. PARAMICS predicts an increase in queuing of 116 metres. This reflects an additional 19 PCU's.

11.8 A similar comparison of the Phase IA and Phase II results is indicated below.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
A419 Southbound approach	84	91	7	8.0%	0.7	76	83	7	9.5%	0.8
M4 westbound off slip	78	80	2	1.6%	0.1	251	241	-10	-3.9%	0.6
A346 northbound approach	213	357	144	67.6%	8.5	1070	1136	66	6.2%	2.0
M4 eastbound off slip	133	145	12	8.9%	1.0	129	128	-1	-0.9%	0.1

Figure 11d: M4 Junction 15, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
A419 Southbound approach	91	152	61	67.3%	5.5	83	630	547	654.5%	28.9
M4 westbound off slip	80	88	8	10.1%	0.9	241	127	-114	-47.5%	8.4
A346 northbound approach	357	111	-246	-69.0%	16.1	1136	74	-1062	-93.5%	43.2
M4 eastbound off slip	145	102	-43	-29.5%	3.8	128	138	10	7.6%	0.8

Figure 11e: M4 Junction 15, 2026 queue length summary Phase II

11.9 These results indicate that once Phase IA is included, there is only a single link that predicts a GEH in excess of 5.0. For the Phase II test, the A419 northern approach reports an increase in queuing that results in a GEH in excess of 5.0.

11.10 Improvements to the A419 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP). The Phase 1 development is expected to make contributions toward the IDP, as will the Phase IA and Phase II scenarios, when these are progressed. This being a strategic element of infrastructure delivery, it is proposed that mitigation is supported through infrastructure funding rather than a site specific intervention.

Honda South Roundabout

11.11 This roundabout is located to the east of the A419 and west of the development. The roundabout provides access to and from the southbound A419 and the A361 and provides access to the Honda car manufacturing works. The broad layout of this junction is indicated below.

Figure 11f: Honda South Roundabout

11.12 The 2016 and 2026 results of the PARAMICS queue length analysis with Phase I are indicated in below.

	AM Peak					PM Peak				
Arm	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
Northern approach	8	9	1	7.2%	0.2	4	0	-4	-100.0%	2.7
Eastern approach	6	5	-1	-12.5%	0.3	23	22	-1	-4.9%	0.2
Western approach	8	9	1	19.5%	0.5	14	17	3	18.0%	0.6

Figure 11g: Honda South Roundabout, 2016 queue length summary Phase I

	AM Peak					PM Peak				
Arm	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
Northern approach	287	14	-273	-95.1%	22.2	3	3	0	2.8%	0.0
Eastern approach	104	6	-98	-94.5%	13.3	24	22	-2	-6.5%	0.3
Western approach	6	8	2	27.9%	0.7	16	19	3	17.3%	0.7

Figure 11h: Honda South Roundabout, 2026 queue length summary Phase I

11.13 The results indicate that in 2016, the operation of the junction is not materially affected. In the 2026 morning peak the operation of the junction is improved, with minimal effect in the evening peak.

11.14 A similar comparison of the 2026 with Phase IA and Phase II results is indicated below.

	AM Peak					PM Peak				
Arm	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
Northern approach	14	11	-3	-23.0%	0.9	3	2	-1	-30.5%	0.6
Eastern approach	6	6	0	5.8%	0.1	22	23	1	4.8%	0.2
Western approach	8	9	0	4.6%	0.1	19	20	1	2.0%	0.1

Figure 11i: Honda South Roundabout, 2026 queue length summary Phase IA

	AM Peak					PM Peak				
Arm	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
Northern approach	11	116	105	969.2%	13.2	2	160	158	6778.2%	17.5
Eastern approach	6	13	7	121.6%	2.4	23	129	106	449.9%	12.1
Western approach	9	48	39	461.9%	7.5	20	3	-17	-83.5%	4.8

Figure 11j: Honda South Roundabout, 2026 queue length summary Phase IID

11.15 These results indicate that once Phase IA is included, there are no links that are predicted to have an GEH in excess of 5.0. For the Phase II test, reports an increases in queuing that results in a GEH in excess of 5.0. This link would require investigation in support of the Phase II scenario to identify the need for appropriate mitigation.

The Rat Trap Roundabout

11.16 The current three arm roundabout is located to the west of the A419 and the proposed development. The roundabout provides access to and from the northbound A419 and the A361, which in turn provides access to the Honda car manufacturing works. The broad layout of this junction is indicted below.

Figure 11k Rat Trap Roundabout

11.17 The results of the 2016 and 2026 PARAMICS queue length analysis with Phase I are indicated below.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
Eastern approach	15	12	-3	-19.7%	0.8	40	46	6	15.6%	1.0
Southern approach	69	70	1	1.5%	0.1	55	54	-1	-0.2%	0.0
Western approach	73	65	-8	-11.3%	1.0	97	109	12	12.9%	1.2

Figure 11l: Rat Trap junction, 2016 queue length summary Phase I

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
Eastern approach	23	20	-3	-11.9%	0.6	120	136	16	13.1%	1.4
Southern approach	295	802	507	172.3%	21.7	61	66	5	8.8%	0.7
Western approach	439	680	241	54.7%	10.2	107	180	73	68.6%	6.1

Figure 11m: Rat Trap junction, 2026 queue length summary Phase I

11.18 The results indicate that in 2016, this junction would be largely unaffected.

11.19 In 2026, the results indicate that this junction could be affected by the Phase I proposals. However, the queuing indicated in the 2026 base indicates that queuing is significantly higher than in 2016, indicating that committed developments will have an impact on this junction, Therefore it is considered that a mitigation strategy to respond to the cumulative impacts may be required.

11.20 A similar comparison of the Phase IA and Phase II results is indicated below.

Arm	AM Peak					PM Peak				
	2026 with Phase 1	2026 with Phase IA	Change	% change	GEH	2026 with Phase 1	2026 with Phase IA	Change	% change	GEH
Eastern approach	20	25	5	22.9%	1.0	136	156	20	15.0%	1.7
Southern approach	802	757	-45	-5.7%	1.6	66	64	-2	-3.0%	0.2
Western approach	680	618	-62	-9.0%	2.4	180	171	-9	-5.4%	0.7

Figure 11n: Rat Trap junction, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase 1A	2026 with Phase II	Change	% change	GEH	2026 with Phase 1A	2026 with Phase II	Change	% change	GEH
Eastern Approach	25	16	-9	-36.1%	2.0	156	79	-77	-49.3%	7.1
Southern Approach	757	882	125	16.5%	4.4	64	156	92	143.0%	8.8
Western Approach	618	483	-135	-22.0%	5.8	171	179	8	5.0%	0.6

Figure 11o: Rat Trap junction, 2026 queue length summary Phase II

11.21 These results indicate that once Phase IA and more specifically when Phase II is included, most approaches report an increase in queuing that results in a GEH in excess of 5.0. This junction would require investigation at the appropriate time to identify the appropriate mitigation.

11.22 Improvements to the A419 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP). The Rat Trap junction is a recognised constraint on the highway network. The Phase 1 development is expected to make contributions toward the IDP, as will the Phase IA and Phase II scenarios, when these are

progressed. This being a strategic element of infrastructure delivery, it is proposed that mitigation is supported through infrastructure funding rather than a site specific intervention.

Commonhead Junction

11.23 Commonhead Junction lies south west of the proposed development. The current layout for the Commonhead junction is indicated below. This junction provides access to the A419 to the A4259 Marlborough Road, Pack Hill and Purley Road.

Figure 11p: Commonhead Junction layout

11.24 The results of the 2016 PARAMICS queue length analysis is indicated below.

	AM Peak					PM Peak				
Arm	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
SB off-slip	34	40	6	18.8%	1.0	25	32	7	29.5%	1.4
Pack Hill	16	18	2	14.8%	0.6	15	15	0	-1.7%	0.1
Purley Road	25	27	2	8.4%	0.4	41	41	0	0.4%	0.0
NB off-slip	2	12	10	418.6%	3.7	12	14	2	12.0%	0.4
A4259 EB	26	32	6	25.1%	1.2	66	74	8	12.7%	1.0

Figure 11q: Commonhead Junction, 2016 queue length summary Phase I

11.25 The result of the assessment indicates that there is no material impact in 2016.

11.26 The results of the 2026 PARAMICS queue length analysis is indicated below.

	AM Peak					PM Peak				
Arm	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
SB off-slip	39	55	16	42.2%	2.4	28	34	6	22.3%	1.1
Pack Hill	28	31	3	12.1%	0.6	11	9	-2	-15.4%	0.5
Purley Road	55	73	18	32.6%	2.2	27	30	3	8.1%	0.4
NB off-slip	6	9	3	50.6%	1.1	6	12	6	99.4%	2.0
A4259 EB	47	59	12	26.4%	1.7	62	85	23	35.5%	2.6

Figure 11r: Commonhead Junction, 2026 queue length summary Phase I

11.27 The results indicate there will not be a significant impact on any of the arms and therefore it is considered that this junction is unlikely to be materially affected by the Phase I scenario.

11.28 A similar comparison of the Phase IA and Phase II results is indicated below.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
SB off-slip	55	58	3	6.1%	0.4	34	36	2	5.2%	0.3
Pack Hill	31	31	0	0.5%	0.0	9	10	1	8.8%	0.3
Purley Road	73	68	-5	-7.1%	0.6	30	32	2	6.7%	0.4
NB off-slip	9	4	-5	-56.5%	2.0	12	4	-8	-67.0%	2.8
A4259 EB	59	65	6	10.5%	0.8	85	83	-2	-2.1%	0.2

Figure 11s: Commonhead Junction, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
SB off-slip	58	236	178	305.3%	14.7	36	263	227	638.3%	18.6
Pack Hill	31	44	13	39.7%	2.0	10	37	27	274.5%	5.6
Purley Road	68	56	-12	-17.2%	1.5	32	41	9	30.6%	1.6
NB off-slip	4	18	14	344.9%	4.2	4	34	30	758.5%	6.9
A4259 EB	65	73	8	12.2%	1.0	83	1364	1281	1546.5%	47.6

Figure 11t: Commonhead Junction, 2026 queue length summary Phase II

11.29 These results indicate that once Phase IA and more specifically when Phase II is included, most approaches reports an increase in queuing that results in a GEH in excess of 5.0. This junction will require investigation at the appropriate time to identify the appropriate mitigation to support the wider growth.

Blunsdon Bypass

11.30 The PARAMICS modelling has been employed to provide queue lengths for various junction along the Blunsdon Bypass. The queue length results for 2016 and 2026 with Phase I are indicated below.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
Cricklade Rd East - Left	51	52	1	0.4%	0.0	140	136	-4	-2.7%	0.3
Cricklade Rd East - Right	50	44	-6	-10.5%	0.8	121	116	-5	-4.2%	0.5
Cricklade Rd North	92	104	12	12.9%	1.2	173	215	42	24.1%	3.0
Cricklade Rd South	174	82	-92	-52.9%	8.1	118	117	-1	-1.1%	0.1
B4019 North - Left	330	382	52	15.7%	2.7	313	70	-243	-77.5%	17.5
B4019 North Ahead	330	382	52	15.7%	2.7	313	70	-242	-77.5%	17.5
B4019 South	627	241	-386	-61.5%	18.5	207	220	13	6.4%	0.9
B4019 Bridge East	60	45	-15	-25.0%	2.1	95	90	-5	-4.8%	0.5
B4019 East	75	75	0	-0.3%	0.0	73	74	1	1.6%	0.1
B4019 Bridge West	103	116	13	13.2%	1.3	108	114	6	5.9%	0.6
B4019 North - Left	249	73	-176	-70.8%	13.9	181	82	-99	-54.7%	8.6
B4019 North Ahead	249	73	-176	-70.8%	13.9	181	82	-99	-54.7%	8.6

Figure 11u: Blunsdon bypass, 2016 queue length summary Phase I

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
Cricklade Rd East - Left	412	476	64	15.5%	3.0	227	216	-11	-5.0%	0.8
Cricklade Rd East - Right	469	543	74	15.8%	3.3	260	224	-36	-13.9%	2.3
Cricklade Rd North	85	75	-10	-10.8%	1.0	134	165	31	23.8%	2.6
Cricklade Rd South	174	177	3	1.6%	0.2	144	142	-2	-1.3%	0.2
B4019 North - Left	379	381	2	0.4%	0.1	384	115	-269	-70.1%	17.0
B4019 North Ahead	379	381	1	0.4%	0.1	384	115	-269	-70.1%	17.0
B4019 South	676	366	-310	-45.8%	13.6	381	249	-132	-34.5%	7.4
B4019 Bridge East	67	55	-12	-17.7%	1.5	106	97	-9	-8.5%	0.9
B4019 East	73	73	0	0.2%	0.0	73	73	0	0.1%	0.0
B4019 Bridge West	112	119	7	6.1%	0.6	114	117	3	2.7%	0.3
B4019 North - Left	151	68	-83	-54.7%	7.9	212	80	-132	-62.2%	10.9
B4019 North Ahead	151	68	-83	-54.7%	7.9	212	80	-132	-62.2%	10.9

Figure 11v: Blunsdon bypass, 2026 queue length summary Phase I

11.31 The results indicate that in 2016 there are no links reporting an increase in excess of a GEH of 5. The highest increase reported is 52m on the B4019; however the length of queue is reported to decrease on several links within this corridor. On balance, it is considered that the delivery of the development will not materially affect the operation of the bypass.

11.32 In 2016, the modelling predicts the majority of the junction along this corridor will materially be unaffected by the proposals, with Cricklade Road reporting the highest increase in queues. It is noted that queuing is predicted to decrease on several arms when the development traffic is included, notably on the B4019 south arm. This consequence of queue reduction has been investigated within the Paramics model. This has indicated that the Paramics results are valid and that the main issue appears to be that that there is a high level of additional 'committed' traffic demands in this locality in the future, but without any additional infrastructure responding to the need for increased capacity. The 2026 results predict an increase in queuing without the development. A review of the road network in the area indicates that there are alternative routes that traffic could reasonable divert to, thus leading to the potential reduction in queuing. At the time of writing, it is therefore considered that the model in this area will require a more detailed review when assessing the future demands expected in this locality.

11.33 It is considered that these increases are unlikely to materially affect the operation of the road network. A review of the resultant queue lengths indicate that the queuing should not impact on the operation of any secondary junction.

11.34 Furthermore, it has been noted that the modelling assumes significant committed development in this area, connecting into the bypass based on an assumed access strategy. it is considered that when the committed development in this area is considered in greater detail, the level of queuing is likely to be reduced.

11.35 A similar comparison of the Phase IA and Phase II results is indicated below.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
Cricklade Rd East - Left	476	430	-46	-9.7%	2.2	216	219	3	1.3%	0.2
Cricklade Rd East - Right	543	485	-58	-10.6%	2.5	224	226	2	0.6%	0.1
Cricklade Rd North	75	77	2	1.9%	0.2	165	164	-1	-0.6%	0.1
Cricklade Rd South	177	176	-1	-0.3%	0.0	142	138	-4	-2.8%	0.3
B4019 North - Left	381	381	1	0.1%	0.0	115	119	4	3.4%	0.4
B4019 North Ahead	381	381	0	0.1%	0.0	115	119	4	3.4%	0.4
B4019 South	366	378	12	3.2%	0.6	249	276	27	10.9%	1.7
B4019 Bridge East	55	56	1	2.9%	0.2	97	103	6	5.7%	0.6
B4019 East	73	73	0	-0.1%	0.0	73	73	0	-0.1%	0.0
B4019 Bridge West	119	119	0	0.2%	0.0	117	117	0	0.1%	0.0
B4019 North - Left	68	70	2	2.2%	0.2	80	80	0	0.0%	0.0
B4019 North Ahead	68	70	2	2.2%	0.2	80	80	0	0.0%	0.0

Figure 11w: Blunsdon bypass, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
Cricklade Rd East - Left	430	335	-95	-22.3%	4.9	219	220	1	0.6%	0.1
Cricklade Rd East - Right	485	397	-88	-18.2%	4.2	226	229	3	1.6%	0.2
Cricklade Rd North	77	76	-1	-0.6%	0.0	164	119	-45	-27.7%	3.8
Cricklade Rd South	176	179	3	1.5%	0.2	138	140	2	1.0%	0.1
B4019 North - Left	381	382	1	0.3%	0.0	119	180	61	51.6%	5.0
B4019 North Ahead	381	382	1	0.3%	0.0	119	180	61	51.6%	5.0
B4019 South	378	453	75	19.9%	3.7	276	146	-130	-47.3%	9.0
B4019 Bridge East	56	70	14	23.7%	1.7	103	81	-22	-21.2%	2.3
B4019 East	73	76	3	3.6%	0.3	73	74	1	0.4%	0.0
B4019 Bridge West	119	115	-4	-4.0%	0.4	117	117	0	-0.3%	0.0
B4019 North - Left	70	74	4	5.4%	0.4	80	76	-4	-4.5%	0.4
B4019 North Ahead	70	74	4	5.4%	0.4	80	76	-4	-4.5%	0.4

Figure 11x: Blunsdon bypass, 2026 queue length summary Phase II

11.36 These results indicate that there could be several junctions affected once the Phase II is included. These junctions will require further investigation at the appropriate time to identify appropriate mitigations to support the wider growth.

White Hart Interchange

11.37 White Hart Junction lies south west of the proposed development on the A420 / A419 corridor. The junction provides access between the A420 and A419 and to Merlin Way and Ermin Street. The current layout for the White Hart junction is indicated below.

Figure 11y: White Hart Interchange

11.38 As part of the identified interventions, a scheme to signalise the interchange has been developed and tested for Phase I. The proposed junction layout is contained in the Appendix C. The queue length results are indicated below.

	AM Peak					PM Peak				
Arm	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
Eastern A420 arm	54	214	160	298.6%	13.9	60	154	94	158.1%	9.1
Merlin Way approach	125	264	139	111.4%	10.0	76	227	151	199.5%	12.3
Oxford Road approach	43	89	46	106.3%	5.7	51	157	106	210.0%	10.4
A419 southbound off slip	96	86	-10	-9.8%	1.0	84	77	-7	-9.2%	0.9
Ermin Way	41	98	57	139.0%	6.8	47	69	22	46.8%	2.9

Figure 11z: White Hart Interchange, 2016 queue length summary Phase I

	AM Peak					PM Peak				
Arm	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
Eastern A420 arm	268	101	-167	-62.1%	12.2	166	180	14	8.3%	1.0
Merlin Way approach	315	217	-98	-31.0%	6.0	347	286	-61	-17.4%	3.4
Oxford Road approach	49	69	20	41.2%	2.6	306	79	-227	-74.2%	16.4
A419 southbound off slip	556	76	-480	-86.3%	27.0	137	113	-24	-17.4%	2.1
Ermin Way	201	23	-178	-88.6%	16.8	29	29	0	0.0%	0.0

Figure 11aa: White Hart Interchange, 2026 queue length summary Phase I

11.39 The 2016 results indicate several links exceeding the 5.0 threshold. In 2026 several of these are positive reductions indicating that the impact of the development is mitigated. With reference to the operation of the SRN, the queuing on the slip roads do not block back onto the mainline. Furthermore, in 2026 the queue on the slip road reduces from 556m to 76m, a significant improvement.

11.40 A similar comparison of the Phase IA and Phase II results is indicated below. The Phase IA model includes the same intervention as Phase I, whereas Phase II will include the diamond interchange intervention.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
Eastern A420 arm	101	103	2	1.9%	0.2	180	260	80	44.9%	5.4
Merlin Way approach	217	270	53	24.1%	3.4	286	287	1	0.1%	0.0
Oxford Road approach	69	206	137	198.4%	11.7	79	84	5	6.1%	0.5
A419 southbound off slip	76	77	1	0.6%	0.1	113	101	-12	-10.5%	1.1
Ermin Way	23	465	442	1921.7%	28.3	29	115	86	296.6%	10.1

Figure 11ab: White Hart Interchange, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
Eastern A420 arm	103	117	14	13.2%	1.3	260	110	-150	-57.8%	11.1
Merlin Way approach	270	221	-49	-17.9%	3.1	287	205	-82	-28.3%	5.2
Oxford Road approach	206	434	228	111.2%	12.8	84	432	348	417.3%	21.7
A419 southbound off slip	77	645	568	742.2%	29.9	101	771	670	663.5%	32.1
Ermin Way	465	425	20	28.3%	2.1	115	299	184	160.0%	12.8

Figure 11ac: White Hart Interchange, 2026 queue length summary Phase II

11.41 The results of the Phase Ii indicate several links report a GEH of greater than 5. This indicates that at the appropriate time more consideration should be given towards the proposed future intervention at this location.

Merlin Way/A419 slip road junction

11.42 This junction lies to the south of the White Hart interchange and forms a connection to the A419 corridor. The current layout for the junction is indicated below.

Figure 11ad: Merlin Way Roundabout

11.43 As part of the identified interventions, a scheme to signalise the southbound junction approach has been developed and tested for Phase I. The proposed junction layout is contained in the Appendix C. The queue length results are indicated below. This was identified as being required in the first instance in 2026. But through discussions with the HA, it was considered appropriate to include this intervention in 2016.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
A420 Merlin Way southbound	108	226	118	109.26%	9.1	89	99	10	11.24%	1.0
A419 off-slip	48	65	17	35.42%	2.2	48	76	28	58.33%	3.5
Merlin Way northbound	57	33	-24	-42.11%	3.5	32	37	5	15.63%	0.8

Figure 11ae: Merlin Way, 2016 queue length summary Phase I

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
A420 Merlin Way southbound	236	63	-173	-73.2%	14.1	219	46	-173	-78.9%	15.0
A419 off-slip	265	86	-179	-67.4%	13.5	144	104	-40	-28.0%	3.6
Merlin Way northbound	265	201	-64	-23.9%	4.1	242	196	-46	-19.0%	3.1

Figure 11af: Merlin Way, 2026 queue length summary Phase I

11.44 The results indicate that through the introduction of the intervention, queuing at this junction decreases on all arms, in both morning and evening peak period. A review of the queuing at the internal stop line indicates that the length of queueing is consistently 5 vehicles, equivalent of circa 28m. The available distance without inhibiting the circulatory movement is 33m, demonstrating that the circulatory carriageway is not blocked. Furthermore, during the times the internal has the red signal, this will result in the southbound Merlin Way southbound would have the opposing green signal. This would result in traffic continuing south along Merlin Way which would reduce the occurrence of the circulatory flow. The initial modelling did not include this intervention, and without this the delay on the southbound Merlin Way was excessive, with the queuing extended back to the White Hart interchange, especially in the 2026.

11.45 A similar comparison of the Phase IA and Phase II results is indicated below. The Phase IA model includes the same intervention as Phase I, whereas Phase II will include the diamond interchange intervention.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
A420 Merlin Way southbound	63	46	-17	-27.3%	2.3	46	46	0	0.0%	0.0
A419 off-slip	86	81	-5	-6.7%	0.6	104	104	0	0.0%	0.0
Merlin Way northbound	201	270	69	34.3%	4.5	196	253	57	29.4%	3.8

Figure 11ag: Merlin Way, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
A420 Merlin Way southbound	46	12	-34	-75.0%	6.4	46	12	-34	-75.0%	6.4
A419 off-slip	81	0	-81	-100.0%	12.7	104	0	-104	-100.0%	14.4
Merlin Way northbound	270	161	-109	-40.4%	7.4	253	17	-236	-93.2%	20.3

Figure 11ah: Merlin Way, 2026 queue length summary Phase II

11.46 The results demonstrate that through the inclusion of the diamond interchange, specifically in the evening peak, the queuing is predicted to decrease.

Gablecross Junction

11.47 Gablecross junction lies south west of the proposed development on the A420 corridor. The junction provides access to and from the A420 from Thornhill Road and Sainsbury's supermarket. The current layout for the Gablecross junction is indicated below.

Figure 11ai: Gablecross junction

11.48 As part of the identified interventions, a scheme to convert the junction to a signalised crossroad has been identified. This layout has been included within the Phase I and Phase IA PARAMICS models, with the layout attached in Appendix C. The queue length results are indicated below.

	AM Peak					PM Peak				
Arm	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
A420 Eastbound	45	122	77	171.5%	8.4	56	169	113	203.6%	10.7
Retail Park Approach	11	5	-6	-52.2%	2.0	24	100	76	314.5%	9.6
Thornhill Rd Approach	53	108	55	105.0%	6.2	29	76	47	161.7%	6.5
A420 Westbound	39	107	68	177.3%	8.0	41	90	49	116.7%	6.0
Sainsbury's Approach	12	49	37	324.6%	6.8	20	118	98	488.2%	11.8

Figure 11aj: Gablecross, 2016 queue length summary Phase I

11.49 The results of the 2016 assessment indicate that the queue is predicted to increase on all the arms. The majority of the development traffic is predicted to pass through this junction so the increases are not unexpected. The delivery of the crossroads provides for the optimum traffic control and allows for the inclusion of pedestrian crossings points.

11.50 The provision of the signal control allows for the potential linking of the signals with the White Hart Interchange to maximise traffic control. The overall queue length is such that the queuing should not queue back into the White Hart Interchange. Were this to occur, the operation of White Hart would clearly be affected, although the results presented indicate that this is not the case.

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
A420 Eastbound	49	117	68	136.9%	7.4	64	222	158	245.7%	13.2
Retail Park Approach	10	3	-7	-73.1%	3.0	29	60	31	109.3%	4.7
Thornhill Rd Approach	116	85	-31	-26.3%	3.0	33	73	40	123.0%	5.6
A420 Westbound	100	82	-18	-18.4%	1.9	59	88	29	48.8%	3.4
Sainsbury's Approach	82	52	-30	-36.3%	3.6	42	125	83	199.5%	9.1

Figure 11ak: Gablecross, 2026 queue length summary Phase I

11.51 The results indicate that in the morning peak, the majority of the arms predict an improvement to the queuing, with on the A420 eastbound predicting an increase. The increase of 67m is equivalent to 12 PCU, which is not considered significant.

11.52 Furthermore, the delivery of signal control will add to the queuing as the nature of signals (red lights) will introduce the need for traffic to stop. However, this will not necessarily lead to an increase delay. The signalisation of this junction provides safe opportunities for traffic to exit from Thornhill Road, rather than to wait for a gap in traffic. This is demonstrated by a review of journey times through Gablecross.

11.53 To identify the potential delay in traffic, a further journey route has been considered, as indicated below.

- Traffic travelling southbound along Thornhill Road
- Traffic travelling westbound along the A420

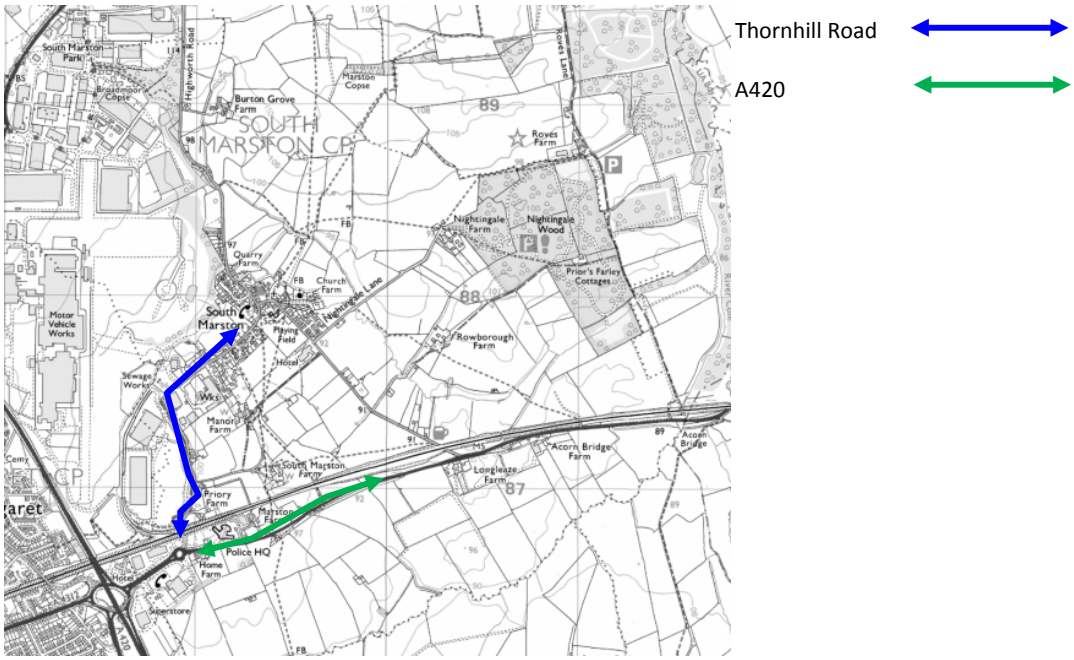

Figure 11l: Site location

11.54 The results of the journey time assessment are indicated in the figures below.

Link	2016 Base	2016 with development	2026 Base	2026 with development
Thornhill Road – Southbound	199.5	145.1	455.4	137.5
Route 1 – Westbound	101.3	59.8	182.7	58.9

Figure 11am: Journey Time Results, [seconds] movements – Morning Peak

Link	2016 Base	2016 with development	2026 Base	2026 with development
Thornhill Road – Southbound	163.9	138.1	152.4	133.9
Route 1 – Westbound	100.7	60.0	108.6	61.4

Figure 11an: Journey Time Results, [seconds] movements – Evening Peak

11.55 The figure above highlights that the southbound journey time improves through the signalisation of the Gablecross junction in the morning and evening peak periods, both in 2016 and 2026. The improvement in journey times within the Thornhill Road / A420 corridor clearly indicates a benefit and this trip are not prejudiced by the delivery of the development.

11.56 A similar queue comparison of the Phase IA and Phase II results is indicated below.

	AM Peak					PM Peak				
Arm	2026 Phase I	2026 with Phase IA	Change	% change	GEH	2026 Phase I	2026 with Phase IA	Change	% change	GEH
A420 Eastbound	117	293	176	151.3%	12.3	222	172	-50	-22.5%	3.6
Retail Park Approach	3	8	5	187.4%	2.3	60	33	-27	-45.9%	4.1
Thornhill Rd Approach	85	149	64	75.0%	5.9	73	75	2	2.2%	0.2
A420 Westbound	82	90	8	9.8%	0.9	88	105	17	18.2%	1.6
Sainsbury's Approach	52	62	10	17.9%	1.2	125	125	0	0.0%	0.0

Figure 11ao: Gablecross, 2026 queue length summary Phase IA

	AM Peak					PM Peak				
Arm	2026 Phase IA	2026 with Phase II	Change	% change	GEH	2026 Phase IA	2026 with Phase II	Change	% change	GEH
A420 Eastbound	293	284	-9	-3.1%	0.5	172	243	71	41.3%	4.9
Retail Park Approach	8	20	12	146.7%	3.2	33	134	101	311.8%	11.1
Thornhill Rd Approach	149	89	-60	-40.1%	5.5	75	88	13	17.2%	1.4
A420 Westbound	90	92	2	2.7%	0.3	105	81	-24	-22.7%	2.5
Sainsbury's Approach	62	151	89	144.6%	8.6	125	124	-1	-1.1%	0.1

Figure 11ap: Gablecross, 2026 queue length summary Phase II

11.57 The results indicate the highest level of queuing will occur on the A420 eastbound approach. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

Police HQ junction

11.58 The Police HQ signals lie due south of the development on the A420 corridor and provides access for the Police HQ to and from the A420. The current layout for the junction is indicated below.

Figure 11aq: Police HQ junction

11.59 The queue length results reported by the modelling are indicated below.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
A420 Eastbound	54	45	-9	-16.1%	1.2	46	48	1	3.0%	0.2
Police HQ	8	15	7	89.6%	2.1	19	29	10	49.9%	2.0
A420 Westbound	82	124	42	51.0%	4.1	75	90	15	18.9%	1.6

Figure 11ar: Police HQ, 2016 queue length summary Phase I

11.60 The result of the 2016 assessment indicates a maximum increase of 42m, equivalent to 7 PCU. This change is not considered material.

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
A420 Eastbound	53	39	-14	-26.7%	2.1	46	43	-3	-6.8%	0.5
Police HQ	19	28	9	50.2%	1.9	24	34	10	43.7%	1.9
A420 Westbound	146	81	-65	-44.7%	6.1	97	106	9	10.1%	1.0

Figure 11as: Police HQ, 2026 queue length summary Phase I

11.61 The results indicate that there is unlikely to be a material impact.

11.62 A similar comparison of the Phase IA and Phase II results is indicated below.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
A420 Eastbound	39	38	-1	-2.0%	0.1	43	26	-17	-39.4%	2.9
Police HQ	28	27	-1	-2.0%	0.1	34	29	-5	-14.5%	0.9
A420 Westbound	81	94	13	15.9%	1.4	106	108	2	1.3%	0.1

Figure 11at: Police HQ, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
A420 Eastbound	38	67	29	74.7%	3.9	26	72	46	174.9%	6.5
Police HQ	27	24	-4	-14.2%	0.8	29	27	-3	-6.4%	0.3
A420 Westbound	94	72	-22	-23.0%	2.4	108	79	-29	-26.7%	3.0

Figure 11au: Police HQ, 2026 queue length summary Phase II

11.63 The results indicate the introduction of Phase IA will not significantly affect the results. The results of the modelling indicate that the predicted queue lengths will not extend back to affect the operation of Gablecross. The maximum predicted queue length is 124m with 170m additional queuing length capacity available. The introduction of Phase II will result in GEH statistics greater than 5.0. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

Old Vicarage Lane

11.64 Old Vicarage lane junction lies to the south of the proposed development and provides access between the A420 and Old Vicarage Lane. The current layout for the junction is indicated below.

Figure 11av: Old Vicarage Lane junction

11.65 As part of the identified interventions, a scheme to signalise the junction has been identified. This layout has been included within the Phase I and Phase IA PARAMICS models, with the layout attached in Appendix C. The queue length results are indicated below.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
Old Vicarage Lane	26	84	58	219.8%	7.8	21	81	60	280.3%	8.4
A420 Westbound	75	154	78	103.4%	7.3	43	95	52	119.2%	6.2
A420 Eastbound	10	144	134	1343.0%	15.3	2	161	159	8761.7%	17.6

Figure 11aw: Old Vicarage Lane, 2016 queue length summary Phase I

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
Old Vicarage Lane	30	27	-3	-8.8%	0.5	13	74	61	446.1%	9.1
A420 Westbound	71	107	36	51.7%	3.9	77	85	8	10.5%	0.9
A420 Eastbound	122	126	4	3.5%	0.4	103	130	27	26.6%	2.5

Figure 11ax: Old Vicarage Lane, 2026 queue length summary Phase I

11.66 The results indicate GEH statistics in excess of 5.0 both in the morning and evening peak periods on Old Vicarage Lane and the A420. This is primarily as a result of introducing signal control in replacement to the present priority arrangement.

11.67 A similar comparison of the Phase IA and Phase II results is indicated below.

	AM Peak					PM Peak				
Arm	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
Old Vicarage Lane	27	88	61	220.8%	8.0	74	61	-13	-17.7%	1.6
A420 Westbound	107	235	128	119.6%	9.8	85	83	-2	-2.6%	0.2
A420 Eastbound	126	117	-9	-7.2%	0.8	130	118	-12	-8.9%	1.0

Figure 11ay: Old Vicarage Lane, 2026 queue length summary Phase IA

	AM Peak					PM Peak				
Arm	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
Old Vicarage Lane	88	162	74	85.2%	6.7	61	62	1	2.4%	0.2
A420 Westbound	235	67	-168	-71.4%	13.7	83	72	-11	-13.2%	1.2
A420 Eastbound	117	85	-32	-27.4%	3.2	118	99	-19	-16.7%	1.9

Figure 11az: Old Vicarage Lane, 2026 queue length summary Phase II

11.68 The results indicate the introduction of Phase IA will not significantly affect operation of the network, with the westbound approach in the evening peak increasing by 128 metres.

11.69 The introduction of Phase II will result in GEH statistics greater than 5.0. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

11.70 The existing bridge under the railway at Old Vicarage Lane is narrow, having a width of circa 4.5m. Proposals to signalise the route beneath the bridge have been developed, by providing a shuttle arrangement.

11.71 A review of the Paramics results indicate that there are no queues reported at the signals under the bridge on Old Vicarage Lane in the 2016 and 2026 scenario. This is generally the case when the queue is not greater than 2 vehicles at any time.

Eastern Access

11.72 A further connection to the A420, via a new signalised junction, is to form part of the access strategy, as indicated below.

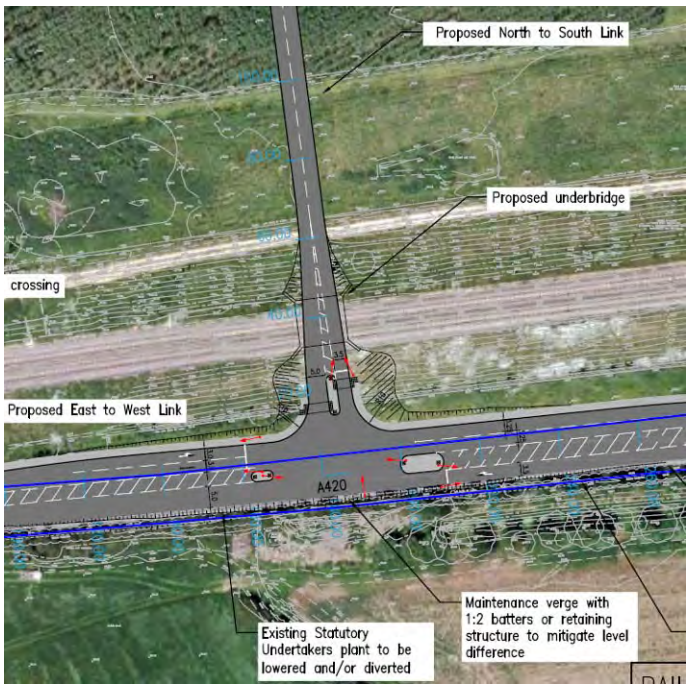

Figure 11ba: A420 Eastern Access

11.73 The queue length results reported by the modelling are indicated below. This junction is a new provision and as such there are no base results.

Arm	AM Peak					PM Peak				
	2016 Base	2016 with Phase I	Change	% change	GEH	2016 Base	2016 with Phase I	Change	% change	GEH
A420 WB	n/a	75	75	n/a	n/a	n/a	35	35	n/a	n/a
A420 EB	n/a	92	92	n/a	n/a	n/a	52	52	n/a	n/a
Dev Road N	n/a	0	0	n/a	n/a	n/a	0	0	n/a	n/a

Figure 11bb: A420 Eastern Access, 2016 queue length summary Phase I

11.74 The result of the 2016 assessment indicates a maximum queue of 92m. The review of the Paramics model indicates that this junction operates satisfactorily.

11.75 The 2026 results are presented below.

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
A420 WB	n/a	104	104	n/a	n/a	n/a	35	35	n/a	n/a
A420 EB	n/a	104	104	n/a	n/a	n/a	58	58	n/a	n/a
Dev Road N	n/a	0	0	n/a	n/a	n/a	0	0	n/a	n/a

Figure 11bc: A420 Eastern Access, 2026 queue length summary Phase I

11.76 The results indicate that there is unlikely to be a material impact.

11.77 A similar comparison of the Phase IA and Phase II results is indicated below.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
A420 WB	104	92	-12	-11.1%	1.2	35	40	5	16.7%	0.9
A420 EB	104	201	97	94.4%	7.9	58	35	-23	-40.0%	3.4
Dev Road N	0	0	0	0.0%	0.0	0	0	0	0.0%	0.0

Figure 11bd: A420 Eastern Access, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
A420 WB	92	40	-52	-56.3%	6.4	40	40	0	0.0%	0.0
Police HQ	201	121	-80	-40.0%	6.3	35	69	34	100.0%	4.8
A420 Westbound	0	23	23	n/a	6.8	0	0	0	0.0%	0.0

Figure 11be: A420 Eastern Access, 2026 queue length summary Phase II

11.78 The results indicate the introduction of Phase IA will not significantly affect the results. The introduction of Phase II will result in GEH statistics greater than 5.0. However, improvements to the A420 corridor are proposed in the SBC wider transport strategy and recognised in the Infrastructure Delivery Plan (IDP) to support the Phase II development scenario.

11.79 This junction has been subjected to a Road Safety Audit and the Designer's Response is attached in Appendix H.

Pound Corner

11.80 Pound corner lies in the centre of South Marston and presently provides a sub-standard highway layout, which is construed by the proximity of properties adjacent to the highway. The current layout for the junction is indicated below.

Figure 11bf: Pound Corner junction

11.81 The queue length results are indicated below.

Arm	AM Peak					PM Peak				
	2026 Base	2026 with Phase I	Change	% change	GEH	2026 Base	2026 with Phase I	Change	% change	GEH
Highworth Road	0	0	0	0%	n/a	0	0	0	0%	n/a
Old Vicarage Lane	12	10	-2	-16.7%	0.9	1	5	4	500%	2.1
Thornhill Road	5	5	0	0%	0.3	4	3	-1	-25%	0.5

Figure 11bg: Pound Corner, 2026 queue length summary Phase I

11.82 The results indicate there will not be a significant impact on any of the arms and therefore it is considered that this junction is unlikely to be materially affected by the Phase I scenario. The level of queuing predicted to occur is unlikely to change significantly and will be likely to be largely indiscernible.

11.83 A similar comparison of the Phase IA and Phase II results is indicated below. This indicates that the increase in queuing is again likely to be indiscernible once phase IA is taken into account.

Arm	AM Peak					PM Peak				
	2026 with Phase I	2026 with Phase IA	Change	% change	GEH	2026 with Phase I	2026 with Phase IA	Change	% change	GEH
Highworth Road	0	0	0	0.0%	n/a	0	0	0.	0.0%	n/a
Old Vicarage Lane	10	8	-2	-20%	0.6	5	7	2	40%	0.8
Thornhill Road	5	5	0	0.0%	0.0	3	4	1	33%	0.2

Figure 11bh: Pound Corner, 2026 queue length summary Phase IA

Arm	AM Peak					PM Peak				
	2026 with Phase IA	2026 with Phase II	Change	% change	GEH	2026 with Phase IA	2026 with Phase II	Change	% change	GEH
Highworth Road	0	0	0	0.0%	n/a	0	0	0	0.0%	n/a
Old Vicarage Lane	8	20	12	150%	3.2	7	29	22	314%	5.1
Thornhill Road	5	8	3	60%	1.2	4	11	7	175%	2.7

Figure 11bi: Pound Corner, 2026 queue length summary Phase II

11.84 The results indicate the introduction of Phase IA or Phase II will not significantly affect the results. As such Pound Corner will not be materially affected by the development proposals. Nonetheless, SBC has investigated a number of minor improvement options for Pound Corner, which are likely to be supported through the IDP.

Greenbridge Junction

11.85 This junction is located to the west of the development site along the A4312. The layout is indicated below.

Figure 11bj: Greenbridge junction layout

11.86 This junction is outside the extents of the PARAMICS model, therefore this junction has been assessed using turning flows extracted from the SATURN model via Arcady. The 2026 results are indicated below.

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Slade Drive	0.651	2	0.506	1
B4006 northbound	0.713	3	0.413	1
Drakes Way	0.504	1	0.638	2
Greenbridge approach	0.503	1	0.753	3
B4006 southbound	0.085	0	0.291	0

Figure 11bk: Greenbridge assessment results, 2026 +Phase I

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Slade Drive	0.630	2	0.508	1
B4006 northbound	0.731	3	0.453	1
Drakes Way	0.500	1	0.617	2
Greenbridge approach	0.629	2	0.754	3
B4006 southbound	0.091	0	0.292	0

Figure 11bl: Greenbridge assessment results, 2026 +Phase IA

Link	Morning Peak		Evening Peak	
	RFC	Max Queue	RFC	Max Queue
Slade Drive	0.611	2	0.515	1
B4006 northbound	0.764	3	0.448	1
Drakes Way	0.782	4	0.793	4
Greenbridge approach	0.809	4	0.763	3
B4006 southbound	0.116	0	0.349	1

Figure 11bm: Greenbridge assessment results, 2026 +Phase II

11.87 The results of the Greenbridge assessment indicates that this junction will operate satisfactorily in all the three development scenarios, with a peak RFC reported for Phase II at 0.809.

Site Access Point – Thornhill Road

11.88 The access strategy is identified above. The development will be accessed from Thornhill Road via a compact roundabout as illustrated below. This access point has been assessed, with the results indicated below.

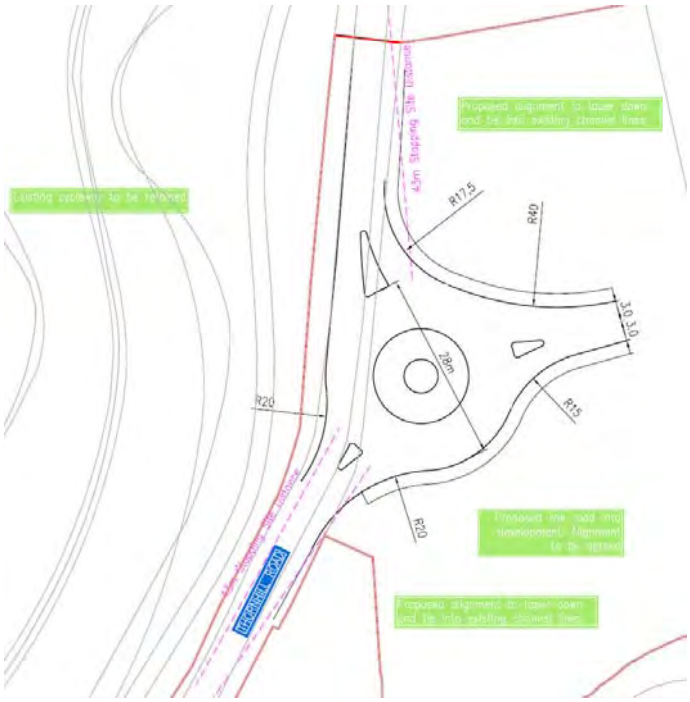

Figure 11bn: Thornhill Road Access

Phase I				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Thornhill Road north	0.297	0	0.189	0
Development access	0.275	0	0.153	0
Thornhill Road south	0.290	0	0.614	2
Phase IA				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Thornhill Road north	0.276	0	0.180	0
Development access	0.233	0	0.144	0
Thornhill Road south	0.293	0	0.606	2
Phase II				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Thornhill Road north	0.399	1	0.468	1
Development access	0.765	3	0.510	1
Thornhill Road south	0.333	1	0..787	4

Figure 11bo: Thornhill Road access 2026 assessment results

11.89 This indicates that the junction operates satisfactorily with all the development scenarios.

Site Access Point – Old Vicarage Lane South

11.90 The development will be accessed from Old Vicarage Lane via a priority junction as illustrated below. This access has been assessed with the results below.

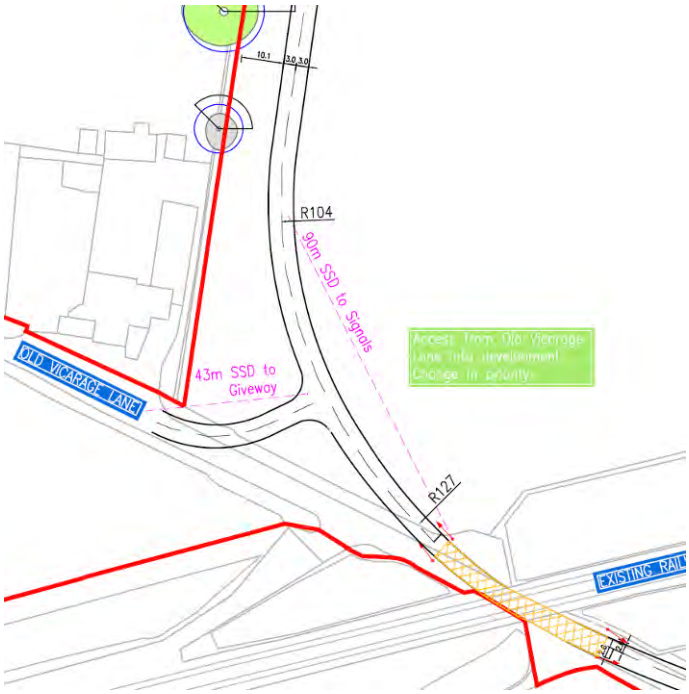

Figure 11bp: Old Vicarage Lane south access

Phase I				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Lane	0.366	1	0.294	0
Development access	0.017	0	0.005	0
Phase IA				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Lane	0.369	1	0.271	0
Development access	0.015	0	0.005	0
Phase II				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Lane	0.209	0	0.194	0
Development access	0.110	0	0.105	0

Figure 11bq: Old Vicarage Lane access 2026 assessment results

11.91 This indicates that this junction operates within acceptable limits with a maximum RFC predicted at 0.369.

Site Access Point – Old Vicarage Lane north

11.92 The development will be accessed from Old Vicarage Lane via a roundabout as illustrated below. This access has been assessed with the results below.

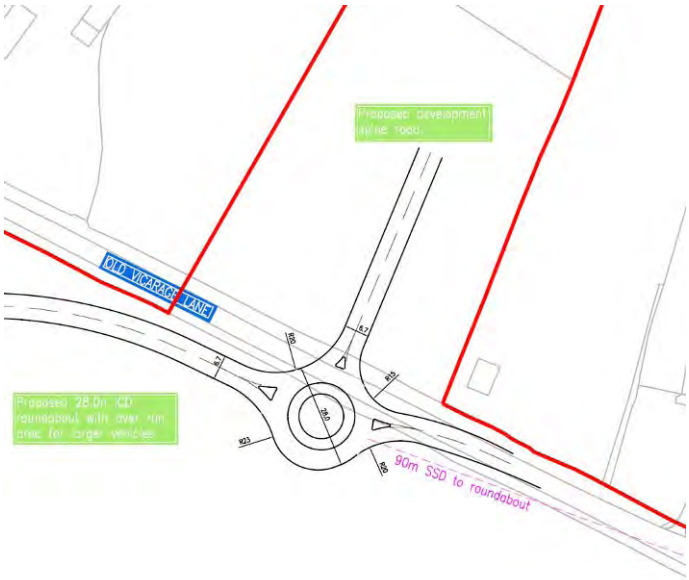

Figure 11br: Old Vicarage Lane north access

Phase I				
	Morning Peak		Evening Peak	
Links	RFC	Max Queue	RFC	Max Queue
Old Vicarage Road north	0.085	0	0.091	0
Development access east	0.172	0	0.111	0
Old Vicarage Road south	0.059	0	0.045	0
Phase IA				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Road north	0.083	0	0.083	0
Development access east	0.154	0	0.103	0
Old Vicarage Road south	0.060	0	0.046	0
Phase II				
	Morning Peak		Evening Peak	
Link	RFC	Max Queue	RFC	Max Queue
Old Vicarage Road north	0.310	0	0.249	0
Development access east	0.279	0	0.178	0
Old Vicarage Road south	0.149	0	0.171	0

Figure 11bs: Old Vicarage Lane roundabout access 2026 assessment results,

11.93 This indicates that the will operate within capacity, with a maximum RFC reported at 0.374.

Junction Summary

11.94 The TA has assessed a number of junctions across the Local Road Network and Strategic Road Network that could potentially be affected by the development proposals. Where appropriate, intervention works are promoted to support the development growth promoted in this Phase. Phase IA and Phase II scenarios have also been tested, representing the wider Eastern Villages growth, including the associated interventions.

11.95 The results indicate that the interventions identified are adequate to mitigate the development impact. The junction interventions are identified below.

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road.
- **New Access onto A420:** New signal controlled junction at the east end of phase one to provide access from the A420.
- **Police HQ Signals:** Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire
- **Gablecross Roundabout:** Conversion to a signalled crossroads
- **White Hart interchange:** Signalisation off several arms of the existing interchange layout.
- **Merlin Way / A419 slip road roundabout:** Signalisation off several arms of the existing interchange layout.

11.96 The assessment process has identified a range of possible interventions, as identified above. However, the HE and SBC have also identified potential alternative schemes at several of these locations. Through the subsequent discussions with the Regulatory Bodies, it may be that a contribution towards these alternative schemes could be agreed, as opposed to the completion of the BCL identified schemes.

11.97 In the immediate environs of South Marston, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included. Whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and evening peaks, following the Phase 1 development (as indicated in Figure 10c). This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner. Notwithstanding, the additional traffic generated by the development, with the improvements proposed to the Gablecross roundabout and the junction of Old Vicarage Lane, the level of queuing at Gablecross roundabout on the Thornhill Road arm is forecast to reduce compared to the no development and no improvement position and the queuing on Old Vicarage Lane will be limited and acceptable.

12 Local Road Network Review – West of the A419

12.1 The previous chapters reviewed the output from the Paramics model, which primarily considers the A419 and the A420. The development will have an impact beyond these corridors, especially towards the west into Swindon. This area is included in the Swindon wide Saturn model. This model has been used to extract link flow data for several key links, as presented below.